

BOUNDEDNESS OF COMPLEMENTS OF LOG CANONICAL PAIRS

ABSTRACT. Given an $\mathbb{R}$ -complementary pair with finite rational coefficients of fixed dimension, we show the existence of a monotonic N -complement for some positive integer N . As a consequence, we prove the boundedness of N -complements for any $\mathbb{R}$ -complementary pair with DCC coefficients, answering a question of Shokurov.

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1. INTRODUCTION

We work over the field of complex numbers $\mathbb{C}$.

The goal of this paper is to prove the boundedness of monotonic lc complements for any $\mathbb{R}$ -complementary pair with finite rational coefficients. This improves a result of Birkar [Bir19, Theorems 1.7 and 1.8] by removing the Fano type condition, hence resolves a conjecture of Shokurov (cf. [Sho20, Conjecture 1]). Our results rely on the recent proofs of the abundance conjecture [OAI26a] and Shokurov's global index conjecture [OAI26d].

Theorem 1.1. *Let d, p be two positive integers and $\Phi \subset [0, 1] \cap \mathbb{Q}$ a finite set. Then there exist a positive integer $p \mid N$ depending only on d, p and Φ satisfying the following.*

Assume that $(X/Z \ni z, B)$ is an $\mathbb{R}$ -complementary pair of dimension d such that $B \in \Phi$. Then $(X/Z \ni z, B)$ has a monotonic N -complement $(X/Z \ni z, B^+)$.

As a consequence, boundedness of lc complements hold for any $\mathbb{R}$ -complementary pair with DCC coefficients, improving a result of the second author with Han and Shokurov [HLS26, Theorem 1.3] by removing the Fano type condition.

Corollary 1.2. *Let d, p be two positive integers and $\Phi \subset [0, 1]$ a DCC set. Then there exist a positive integer $p \mid N$ depending only on d, p and Φ satisfying the following.*

Assume that $(X/Z \ni z, B)$ is an $\mathbb{R}$ -complementary pair of dimension d such that $B \in \Phi$. Then $(X/Z \ni z, B)$ has an N -complement $(X/Z \ni z, B^+)$.

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Structure of the paper. In Section 2 we fix notation and recall the results of [OAI26a, OAI26b, OAI26c, OAI26d] that we use. In Section 3 we reduce Theorem 1.1 to the case of a $\mathbb{Q}$ -factorial dlt pair (Y, A) such that $-(K_Y + A)$ is semi-ample over the base, and in Section 4 we prove the uniform denominator bounds for the canonical bundle formula that we need. Section 5 extends these bounds to the quotient of the base of a canonical bundle formula by a finite group. Section 6 identifies the sections of an anti-pluricanonical sheaf on the union of lc centers over a minimal lc center with the invariant sections on a source. Section 7 shows that the complements of generalized pairs of Fano type given by [FM20] can be chosen uniformly over a finite base change, and Section 8 collects the results on log canonicity under pullback which we need to extend a complement from the lc centers to the whole pair. Theorem 1.1 is proved in Section 9.

2. PRELIMINARIES AND EXTERNAL INPUTS

We will adopt the standard notation and definitions on birational geometry in [Sho92, KM98, BCHM10] and use them freely.

Definition 2.1. A *contraction* is a projective surjective morphism $f : X \rightarrow Z$ between normal quasi-projective varieties such that $f_*\mathcal{O}_X = \mathcal{O}_Z$.

Notation 2.2. Let X be a normal variety. We denote by $\mathbb{C}(X)$ the function field of X and by $\text{div}(\varphi)$ the divisor of a nonzero rational function $\varphi \in \mathbb{C}(X)$. For a prime divisor E and an $\mathbb{R}$ -divisor D on X , $\text{mult}_E D$ is the multiplicity of D along E . Let $D = \sum_i d_i D_i$ be an $\mathbb{R}$ -divisor on X , where the D_i are the irreducible components of D . For a set $I \subset \mathbb{R}$, we write $D \in I$ if $d_i \in I$ for every i . For a positive integer N , we write $NI \subset \mathbb{Z}$ if $Nb \in \mathbb{Z}$ for every $b \in I$. We define $\lfloor D \rfloor$, $\lceil D \rceil$, and $\{D\} := D - \lfloor D \rfloor$ componentwise. A set $I \subset \mathbb{R}$ satisfies the *descending chain condition* (DCC) if any decreasing sequence in I stabilizes, and the *ascending chain condition* (ACC) if any increasing sequence in I stabilizes. For a point $\mathbf{v} \in \mathbb{R}^m$, the *rational envelope* of $\mathbf{v}$ is the smallest affine subspace of $\mathbb{R}^m$ containing $\mathbf{v}$ which is defined over $\mathbb{Q}$ (cf. [HLS26, Section 5]).

Let $f : X \rightarrow Z$ be a projective morphism of normal quasi-projective varieties and D, D' two $\mathbb{R}$ -Cartier $\mathbb{R}$ -divisors on X . We write $D \sim_{\mathbb{R}, Z} D'$ (resp. $D \sim_{\mathbb{Q}, Z} D'$, $D \sim_Z D'$) if $D - D' \sim_{\mathbb{R}} f^*L$ (resp. $D - D' \sim_{\mathbb{Q}} f^*L$, $D - D' \sim f^*L$) for some $\mathbb{R}$ -Cartier $\mathbb{R}$ -divisor (resp. $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor, Cartier divisor) L on Z . Let $z \in Z$ be a (not necessarily closed) point. A property of X , f , or divisors on X holds *over a neighborhood of z* if it holds over U after replacing Z with an open neighborhood U of z and X with $f^{-1}(U)$. In particular, $D \sim_{\mathbb{R}} 0$ (resp. $D \sim_{\mathbb{Q}} 0$, $D \sim 0$) over a neighborhood of z means $D \sim_{\mathbb{R}, U} 0$ (resp. $D \sim_{\mathbb{Q}, U} 0$, $D \sim_U 0$) over some open neighborhood U of z . Since any Cartier divisor on Z is principal over a neighborhood of z , $D \sim 0$ over a neighborhood of z if and only if $D = \text{div}(\psi)$ over a neighborhood of z for some nonzero $\psi \in \mathbb{C}(X)$. We say that D is *vertical*/ Z if $f(\text{Supp } D) \neq Z$, and *horizontal*/ Z if every component of D dominates Z . We write $D = D^h + D^v$, where D^h is horizontal/ Z and D^v is vertical/ Z .

Let $n := \dim X$. A *rational top form* on X is a nonzero element of $\bigwedge^n \Omega_{\mathbb{C}(X)/\mathbb{C}}$. We fix a rational top form ω on X and let $K_X := \text{Div}(\omega)$ be its divisor. For an integer m , an *m -canonical form* on X is a nonzero rational section of $\mathcal{O}_X(mK_X)$, that is, an element $\theta = \varphi\omega^{\otimes m}$ with $\varphi \in \mathbb{C}(X)$ nonzero; for $m \geq 0$ these are the nonzero elements of $(\bigwedge^n \Omega_{\mathbb{C}(X)/\mathbb{C}})^{\otimes m}$. We define

$$\text{Div}(\theta) := \text{div}(\varphi) + mK_X,$$

which does not depend on the choice of ω . If θ is an m -canonical form, then θ^{-1} is a $(-m)$ -canonical form and $\text{Div}(\theta^{-1}) = -\text{Div}(\theta)$. For any birational map $Y \dashrightarrow X$ of normal varieties, we let $K_Y := \text{Div}(\omega)$ be the divisor of ω on Y . In this way canonical divisors are chosen compatibly on all birational models, and an m -canonical form on X is also an m -canonical form on Y . We write $\text{Div}_Y(\theta)$ and $\text{div}_Y(\varphi)$ when the variety is not clear from the context.

Definition 2.3 (Pairs, cf. [CHL24, Definition 2.7]). A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$ -divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$ -Cartier. A *pair*

$(X/Z \ni z, B)$ consists of a contraction $f : X \rightarrow Z$, a (not necessarily closed) point $z \in Z$, and an $\mathbb{R}$ -divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$ -Cartier over a neighborhood of z . If B is a $\mathbb{Q}$ -divisor, then we call (X, B) or $(X/Z \ni z, B)$ a $\mathbb{Q}$ -pair. If $(X/Z \ni z, B)$ is a pair for every point $z \in Z$, then we call $(X/Z, B)$ a pair. If Z is a point, then we omit Z and z and call (X, B) a *projective pair*. If f is the identity and $z = x \in X$, then we write $(X \ni x, B)$.

Let (X, B) be a pair. A *prime divisor over X* is a prime divisor E on a normal variety Y with a projective birational morphism $g : Y \rightarrow X$, and $\text{center}_X E := g(E)$ is the *center* of E on X . Let $K_Y + B_Y := g^*(K_X + B)$. The *log discrepancy* of E with respect to (X, B) is $a(E, X, B) := 1 - \text{mult}_E B_Y$; it does not depend on Y . We say that (X, B) is *lc* (resp. *klt*) if $a(E, X, B) \geq 0$ (resp. > 0) for every prime divisor E over X . An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$, and an *lc center* of (X, B) is the center of an lc place. A *minimal lc center* is an lc center which is minimal with respect to inclusion. We say that an lc pair (X, B) is *dlt* if there exists a log resolution $g : Y \rightarrow X$ of (X, B) such that $a(E, X, B) > 0$ for every g -exceptional prime divisor E . We say that (X, B) is *log smooth* if X is smooth and $\text{Supp } B$ is a simple normal crossing (snc) divisor. We say that $(X/Z \ni z, B)$ is *lc* (resp. *klt*) if (X, B) is lc (resp. klt) over a neighborhood of z .

Let (X, B) be an lc pair and $D \geq 0$ an $\mathbb{R}$ -Cartier $\mathbb{R}$ -divisor on X . The *lc threshold* of D with respect to (X, B) is

$$\text{lct}(X, B; D) := \sup\{t \geq 0 \mid (X, B + tD) \text{ is lc}\}.$$

If $f : X \rightarrow T$ is a contraction and P is a prime divisor on T with generic point η_P , then $\text{lct}(X, B; D)$ over the generic point of P is

$$\sup\{t \geq 0 \mid (X, B + tD) \text{ is lc over a neighborhood of } \eta_P\}.$$

Let (X, B) be a dlt $\mathbb{Q}$ -pair and W an lc center of (X, B) . By [KM98, Theorem 2.44] and [Kol16, Definition 3], W meets the largest open subset $X^{\text{snc}} \subset X$ over which (X, B) is log smooth; hence $W \cap X^{\text{snc}}$ is an irreducible component of an intersection of components of $[B]$. Moreover, W is normal, and there is a $\mathbb{Q}$ -divisor $B_W \geq 0$ on W , the *different* of B on W , such that $K_W + B_W \sim_{\mathbb{Q}} (K_X + B)|_W$ and (W, B_W) is dlt [Fuj11, Lemma 10.2]. The different is defined inductively from the different of a divisor [Kol16, Definition 4]. We call (W, B_W) the *adjunction pair* of W .

Definition 2.4 (Complements, cf. [Bir19, 2.18]). Let $(X/Z \ni z, B)$ be a pair and n a positive integer. We say that $(X/Z \ni z, B^+)$ is an $\mathbb{R}$ -complement of $(X/Z \ni z, B)$ if $B^+ \geq B$ and, over a neighborhood of z , (X, B^+) is lc and $K_X + B^+ \sim_{\mathbb{R}} 0$. If moreover B^+ is a $\mathbb{Q}$ -divisor and $K_X + B^+ \sim_{\mathbb{Q}} 0$ over a neighborhood of z , then we say that $(X/Z \ni z, B^+)$ is a $\mathbb{Q}$ -complement of $(X/Z \ni z, B)$. We say that $(X/Z \ni z, B)$ is $\mathbb{R}$ -complementary if it has an $\mathbb{R}$ -complement.

We say that $(X/Z \ni z, B^+)$ is an n -complement of $(X/Z \ni z, B)$ if over a neighborhood of z , (X, B^+) is lc, $n(K_X + B^+) \sim 0$, and

$$nB^+ \geq n[B] + \lfloor (n+1)\{B\} \rfloor.$$

We say that $(X/Z \ni z, B^+)$ is a *monotonic n -complement* of $(X/Z \ni z, B)$ if moreover $B^+ \geq B$.

If $(X/Z \ni z, B^+)$ is an $\mathbb{R}$ -complement (resp. $\mathbb{Q}$ -complement, n -complement, monotonic n -complement) of $(X/Z \ni z, B)$ for every point $z \in Z$, then we say that $(X/Z, B^+)$ is an $\mathbb{R}$ -complement (resp. $\mathbb{Q}$ -complement, n -complement, monotonic n -complement) of $(X/Z, B)$. If Z is a point, then we omit Z and z .

Definition 2.5 (Hyperstandard sets, [Bir19, 2.2]). Let $\mathfrak{R} \subset [0, 1] \cap \mathbb{Q}$ be a finite set such that $0, 1 \in \mathfrak{R}$. We define

$$\Phi(\mathfrak{R}) := \left\{ 1 - \frac{r}{n} \mid r \in \mathfrak{R}, n \text{ a positive integer} \right\}.$$

We say that a set $I \subset [0, 1]$ is a *hyperstandard set* if $I = \Phi(\mathfrak{R})$ for some such $\mathfrak{R}$. The set $\Phi(\mathfrak{R})$ is a DCC set of rational numbers whose only accumulation point is 1.

Definition 2.6 (b-divisors, cf. [Bir19, 2.7]). Let X be a normal quasi-projective variety. A *birational model* of X is a normal variety Y with a projective birational morphism $Y \rightarrow X$. A *b-divisor* $\mathbf{M}$ over X consists of an $\mathbb{R}$ -divisor $\mathbf{M}_Y$ on every birational model Y of X , such that $\pi_* \mathbf{M}_{Y'} = \mathbf{M}_Y$ for every projective birational morphism $\pi : Y' \rightarrow Y$ of birational models of X . We say that $\mathbf{M}$ *descends* to a birational model X' of X if $\mathbf{M}_{X'}$ is $\mathbb{R}$ -Cartier and $\mathbf{M}_Y = \pi^* \mathbf{M}_{X'}$ for every projective birational morphism $\pi : Y \rightarrow X'$ from a birational model Y of X . We say that $\mathbf{M}$ is *b- $\mathbb{R}$ -Cartier* (resp. *b- $\mathbb{Q}$ -Cartier*) if $\mathbf{M}$ descends to some birational model X' (resp. descends to some X' with $\mathbf{M}_{X'}$ a $\mathbb{Q}$ -divisor). For a positive integer p , we say that $p\mathbf{M}$ is *b-Cartier* if $\mathbf{M}$ descends to some X' such that $p\mathbf{M}_{X'}$ is Cartier. If $X \rightarrow U$ is a projective morphism, then we say that $\mathbf{M}$ is *b-nef/ U* if $\mathbf{M}$ descends to some X' such that $\mathbf{M}_{X'}$ is nef/ U . For an $\mathbb{R}$ -Cartier $\mathbb{R}$ -divisor D on X , we denote by $\overline{D}$ the b-divisor over X with $\overline{D}_Y := \pi^* D$ for every birational model $\pi : Y \rightarrow X$.

Definition 2.7 (Generalized pairs, cf. [FM20, Definitions 2.2 and 2.11]). A *generalized pair* $(X, B + \mathbf{M})/U$ consists of a projective morphism $X \rightarrow U$ of normal quasi-projective varieties, a $\mathbb{Q}$ -divisor $B \geq 0$ on X , and a b- $\mathbb{Q}$ -Cartier b-divisor $\mathbf{M}$ over X which is b-nef/ U , such that $K_X + B + \mathbf{M}_X$ is $\mathbb{Q}$ -Cartier. In [FM20], $\mathbf{M}_{X'}$ is required to be nef rather than nef/ U ; the two conditions agree when U is affine. Let $\pi : Y \rightarrow X$ be a birational model of X to which $\mathbf{M}$ descends, and define B_Y by

$$(2.1) \quad K_Y + B_Y + \mathbf{M}_Y = \pi^*(K_X + B + \mathbf{M}_X).$$

For a prime divisor E on Y , the *generalized log discrepancy* of E with respect to $(X, B + \mathbf{M})$ is $a(E, X, B + \mathbf{M}) := 1 - \text{mult}_E B_Y$; it does not depend on Y . We say that $(X, B + \mathbf{M})/U$ is *generalized lc* (resp. *generalized klt*) if $a(E, X, B + \mathbf{M}) \geq 0$ (resp. > 0) for every prime divisor E over X . We omit U when it is a point or is clear from the context. If $\mathbf{M} = 0$, then $(X, B + \mathbf{M})/U$ is the $\mathbb{Q}$ -pair (X, B) .

Let $f : X \rightarrow Z$ be a contraction and $z \in Z$ a point. A *generalized pair* $(X/Z \ni z, B + \mathbf{M})$ consists of f, z , a $\mathbb{Q}$ -divisor $B \geq 0$ on X , and a b- $\mathbb{Q}$ -Cartier b-divisor $\mathbf{M}$ over X , such that over a neighborhood of z , $\mathbf{M}$ is b-nef/ Z and $K_X + B + \mathbf{M}_X$ is $\mathbb{Q}$ -Cartier. We say that $(X/Z \ni z, B + \mathbf{M})$ is *generalized lc* (resp. *generalized klt*) if $(X, B + \mathbf{M})/Z$ is generalized lc (resp. generalized klt) over a neighborhood of z . Let n be a positive integer. We say that $(X/Z \ni z, B^+ + \mathbf{M})$ is a *monotonic generalized n -complement* of $(X/Z \ni z, B + \mathbf{M})$ if $B^+ \geq B$ and, over a neighborhood of z ,

- (1) $(X, B^+ + \mathbf{M})/Z$ is generalized lc,
- (2) $n(K_X + B^+ + \mathbf{M}_X) \sim 0$, and
- (3) $nB^+ \geq n\lfloor B \rfloor + \lfloor (n+1)\{B\} \rfloor$.

Here the b-divisor $\mathbf{M}$ is part of the data and is unchanged. In [FM20], such a B^+ is called a strong $(0, n)$ -complement of $(X, B + \mathbf{M})$ over z . If $\mathbf{M} = 0$, then a monotonic generalized n -complement of $(X/Z \ni z, B + \mathbf{M})$ is a monotonic n -complement of $(X/Z \ni z, B)$.

Definition 2.8 (Crepan maps, cf. [FG14a, Definition 2.11], [Kol16, Definition 2 and Paragraph 6]). Let (X, B) and (X', B') be two $\mathbb{Q}$ -pairs. A birational map $\phi : X \dashrightarrow X'$ is *crepan* with respect to (X, B) and (X', B') , and we write $\phi : (X, B) \dashrightarrow (X', B')$, if $p^*(K_X + B) = q^*(K_{X'} + B')$ for a common resolution $p : W \rightarrow X, q : W \rightarrow X'$ of ϕ . Here a *common resolution* of ϕ consists of a normal variety W and projective birational morphisms $p : W \rightarrow X$ and $q : W \rightarrow X'$ such that $q = \phi \circ p$. This does not depend on the choice of the common resolution. Such maps are called *B-birational maps* in [FG14a]. If $X \rightarrow V$ and $X' \rightarrow V$ are morphisms and ϕ commutes with them, then we say that ϕ is *over V* . The crepan birational self-maps of (X, B) over V form a group, which we denote by $\text{Bir}_V^c(X, B)$.

A *crepan log structure* $f : (X, B) \rightarrow T$ consists of an lc $\mathbb{Q}$ -pair (X, B) and a contraction $f : X \rightarrow T$ such that $K_X + B \sim_{\mathbb{Q}, T} 0$. An *lc center* of f is the image $f(W)$ of an lc center W of (X, B) ; f has finitely many lc centers. We say that f is *dlt* (resp. *$\mathbb{Q}$ -factorial dlt*) if (X, B) is dlt (resp. X is $\mathbb{Q}$ -factorial and (X, B) is dlt).

We will use the following results of [OAI26a, OAI26b, OAI26c, OAI26d]. We refer to the versions of these preprints indicated in the bibliography.

Following [Bir11, Definition 2.1] and [OAI26a, Section 2], a *log minimal model* of a projective pair (X, B) is a projective pair (X', B') with a birational map $\phi : X \dashrightarrow X'$ such that $B' = \phi_* B + E$ and $K_{X'} + B'$ is nef, where E is the reduced ϕ^{-1} -exceptional divisor. Moreover, we require that $a(D, X, B) < a(D, X', B')$ for every prime divisor D on X which is exceptional over X' . In [Bir11, Definition 2.1], (X', B') is in addition required to be $\mathbb{Q}$ -factorial dlt. It is a *good minimal model* of (X, B) if moreover $K_{X'} + B'$ is semi-ample, that is, $K_{X'} + B' \sim_{\mathbb{R}} g^* A$ for some contraction $g : X' \rightarrow V$ and some ample $\mathbb{R}$ -divisor A on V .

Theorem 2.9 ([OAI26a, Theorem 11.1 and Lemma 2.2]). *Let (X, B) be a projective lc pair such that $K_X + B$ is pseudo-effective. Then (X, B) has a good minimal model (X', B') . Moreover, let $p : W \rightarrow X$ and $q : W \rightarrow X'$ be a common resolution of the induced birational map $X \dashrightarrow X'$. Then there exists a q -exceptional $\mathbb{R}$ -divisor $E \geq 0$ on W such that*

$$(2.2) \quad p^*(K_X + B) = q^*(K_{X'} + B') + E.$$

In the proof of [OAI26a, Proposition 2.5], the good minimal model is constructed as a $\mathbb{Q}$ -factorial dlt log minimal model; see also the proof of [OAI26a, Corollary 12.1].

Theorem 2.10 ([OAI26a, Corollary 14.1]). *Let $\pi : X \rightarrow Y$ be a projective surjective morphism of normal complex analytic varieties. Let $\Delta \geq 0$ be a $\mathbb{Q}$ -divisor on X such that $D := K_X + \Delta$ is $\mathbb{Q}$ -Cartier and (X, Δ) is lc, and assume that D is π -nef over all of Y . Then for every compact subset $W \subset Y$, there exist an analytic open neighborhood $U \supset W$ and a positive integer m such that $mD|_{X_U}$ is Cartier, where $X_U := \pi^{-1}(U)$, and the line bundle $\mathcal{L} := \mathcal{O}_{X_U}(mD|_{X_U})$ is generated relative to $\pi_U : X_U \rightarrow U$. That is, the evaluation map $\pi_U^*(\pi_U)_*\mathcal{L} \rightarrow \mathcal{L}$ is surjective.*

The analytic conventions here are those of [OAI26a, Section 14]. A complex analytic variety is a reduced and irreducible Hausdorff second-countable complex analytic space. A projective morphism is a proper morphism with a relatively ample line bundle. The $\mathbb{Q}$ -Cartier condition is local on X . The divisor D is π -nef over Y if D has non-negative degree on every projective curve contracted by π .

Theorem 2.11 ([OAI26b, Theorem 1.2]). *Let d be a non-negative integer and $I \subset [0, 1] \cap \mathbb{Q}$ a DCC set. Then there exists a positive integer $a(d, I)$ depending only on d and I satisfying the following. Assume that (X, B) is a projective lc $\mathbb{Q}$ -pair of dimension d such that $B \in I$ and $K_X + B \sim_{\mathbb{Q}} 0$. Then*

$$(2.3) \quad a(d, I)(K_X + B) \sim 0.$$

In (2.3), $a(d, I)(K_X + B) \sim 0$ means that $a(d, I)(K_X + B)$ is a principal divisor; in particular $a(d, I)B$ is an integral divisor. A projective pair of dimension 0 is a point with $B = 0$, so we may take $a(0, I) = 1$.

Definition 2.12 (Weights, [OAI26c, Definition 4.1]). Let C be a smooth projective curve, $c \in C$ a closed point, $K := \mathbb{C}(C)$, and $z \in K$ a *rational uniformizer* at c , that is, $\text{ord}_c(z) = 1$. Let F/K be a finitely generated regular field extension of transcendence degree s , that is, the function field of a geometrically integral variety of dimension s over K , and let m be a positive integer. A *rational relative m -canonical form* of F/K is a nonzero element φ of $(\bigwedge^s \Omega_{F/K})^{\otimes m}$. Using the identification of $\bigwedge^{s+1} \Omega_{F/\mathbb{C}}$ with $\bigwedge^s \Omega_{F/K} \otimes_K \Omega_{K/\mathbb{C}}$ given by a fixed wedge order, we associate with φ the m -canonical form

$$\Omega := \varphi \wedge (dz/z)^{\otimes m}$$

of $F/\mathbb{C}$. The *weight* of φ at c is

$$(2.4) \quad w_z(\varphi) := \inf_E \frac{\text{ord}_E(\Omega) + m}{m \text{ord}_E(z)},$$

where E runs over the normalized divisorial valuations of $F/\mathbb{C}$ whose restriction to K is a positive multiple of ord_c . Here $\text{ord}_E(\Omega)$ is the multiplicity of $\text{Div}(\Omega)$ along E on any normal variety with function field F on which E is a prime divisor. The weight depends only on F , φ , and c , and not on the choice of z [OAI26c, Lemma 4.3(i)].

Let V be a normal geometrically integral projective variety over K with function field F . We write $\text{Div}_V(\varphi)$ for the divisor of φ on V , defined as in Notation 2.2 with $\mathbb{C}$ replaced by K and with K_V the divisor of a rational relative top form of V over K . Let $B \geq 0$ be a $\mathbb{Q}$ -divisor on V such that $K_V + B$ is $\mathbb{Q}$ -Cartier. We say that (V, B) is *geometrically lc* (resp. *geometrically klt*) if $(V_{\bar{K}}, B_{\bar{K}})$ is lc (resp. klt), where $\bar{K}$ is an algebraic closure of K .

Theorem 2.13 ([OAI26c, Theorem 4.2 and Proposition 4.5]). *Let s be a non-negative integer and m a positive integer. Then there exists a positive integer $q(s, m)$ depending only on s and m satisfying the following. Let C , c , K , z , F/K , φ , and Ω be as in Definition 2.12. Let V be a normal geometrically integral projective variety of dimension s over K with function field F , and (V, B) a geometrically lc pair such that*

$$\text{Div}_V(\varphi) + mB = 0.$$

Then:

- (1) $q(s, m)w_z(\varphi) \in \mathbb{Z}$.
- (2) *Assume that $s > 0$. Then there exist a contraction $f : N \rightarrow C$ and a $\mathbb{Q}$ -divisor $H \geq 0$ on N which is horizontal/ C satisfying the following. Let $T := (f^*c)_{\text{red}}$. Then N is $\mathbb{Q}$ -factorial, the generic fiber N_η of f is geometrically integral and birational to V over K , $(N, T + H)$ is dlt, $K_N + T + H \sim_{\mathbb{Q}, C} 0$, and*

$$\frac{1}{m} \text{Div}_N(\Omega) + T + H = w_z(\varphi)f^*c$$

over a neighborhood of c . In particular, $\text{Div}_{N_\eta}(\varphi) + mH_\eta = 0$, where $H_\eta := H|_{N_\eta}$. If H has no component of coefficient 1, then (N_η, H_η) is geometrically klt and K_{N_η} is $\mathbb{Q}$ -Cartier.

In Theorem 2.13(2), $\mathbb{C}(N) = K(N_\eta) = F$, so Ω is an m -canonical form on N . We may take $q(0, m) = m$: if $s = 0$, then $V = \text{Spec } K$, $\varphi \in K^*$, and $w_z(\varphi) = \frac{1}{m} \text{ord}_c(\varphi)$ (see the proof of [OAI26c, Theorem 4.2]).

Theorem 2.14 ([OAI26d, Lemma 2.1 and Theorem 6.1]). *Let n be a non-negative integer and m a positive integer. Then there exists a positive integer $L(n, m)$ depending only on n and m satisfying the following. Assume that (X, Δ) is a projective klt $\mathbb{Q}$ -pair of dimension n such that $m(K_X + \Delta) \sim 0$, and let θ be an m -canonical form on X such that $\text{Div}(\theta) = -m\Delta$. Then for every crepant birational self-map ϕ of (X, Δ) , we have $\phi^*\theta = c\theta$ for some $c \in \mathbb{C}^*$, and $c^{L(n, m)} = 1$.*

Here $m(K_X + \Delta) \sim 0$ means that $m(K_X + \Delta) = \text{div}(\psi)$ for some nonzero $\psi \in \mathbb{C}(X)$; then $\theta := \psi^{-1}\omega^{\otimes m}$ satisfies $\text{Div}(\theta) = -m\Delta$, and any two such forms differ by a nonzero constant [OAI26d, (2.1)]. A pair of dimension 0 is a point whose only birational self-map is the identity, so we may take $L(0, m) = 1$.

Let m be a positive even integer, (Y, Γ) a projective dlt $\mathbb{Q}$ -pair, and θ an m -canonical form on Y such that $\text{Div}(\theta) = -m\Gamma$. Let W be an lc center of (Y, Γ) of codimension k with adjunction pair (W, Γ_W) . By Definition 2.3, near the generic point of W the pair (Y, Γ) is log smooth and W is a component of $D_1 \cap \cdots \cap D_k$, where $D_1, \dots, D_k$ are components of $\lfloor \Gamma \rfloor$. Choose local coordinates $x_1, \dots, x_k, y_1, \dots, y_{\dim W}$ at the generic point of W such that $D_i = (x_i = 0)$ for each i , and write

$$\theta = u \left(\frac{dx_1}{x_1} \wedge \cdots \wedge \frac{dx_k}{x_k} \wedge dy_1 \wedge \cdots \wedge dy_{\dim W} \right)^{\otimes m},$$

where u is a unit at the generic point of W because $\text{Div}(\theta) = -m\Gamma$. The *residue* of θ on W is the m -canonical form

$$\theta_W := u|_W (dy_1 \wedge \cdots \wedge dy_{\dim W})^{\otimes m}$$

on W . Since m is even, θ_W does not depend on the choice of the coordinates, and it coincides with the iterated residue of θ along any flag of lc centers between Y and W [OAI26d, Section 7.1]. We set $\theta_Y := \theta$ and $\Gamma_Y := \Gamma$. By [OAI26d, (7.2)], $\text{Div}(\theta_W) = -m\Gamma_W$; in particular, $m(K_W + \Gamma_W) \sim 0$ and $m\Gamma_W$ is an integral divisor. In the following theorem, a *minimal lc center* of (Y, Γ) means a minimal lc center of (Y, Γ) if (Y, Γ) is not klt, and Y itself if (Y, Γ) is klt.

Theorem 2.15 ([OAI26d, Lemmas 7.1 and 7.2]). *Let m be a positive even integer.*

- (1) *Let (Y, Γ) be a projective dlt $\mathbb{Q}$ -pair and θ an m -canonical form on Y such that $\text{Div}(\theta) = -m\Gamma$. Then for any two minimal lc centers A and A' of (Y, Γ) , there exists a crepant birational map $\phi : (A, \Gamma_A) \dashrightarrow (A', \Gamma_{A'})$ such that $\phi^*\theta_{A'} = \theta_A$.*
- (2) *Let (Y, Γ) and (Y', Γ') be two projective dlt $\mathbb{Q}$ -pairs of the same dimension, and θ and θ' m -canonical forms on Y and Y' such that $\text{Div}(\theta) = -m\Gamma$ and $\text{Div}(\theta') = -m\Gamma'$. Let $\tau : Y \dashrightarrow Y'$ be a birational map such that $\tau^*\theta' = r\theta$ for some $r \in \mathbb{C}^*$. Then there exist minimal lc centers A of (Y, Γ) and A' of (Y', Γ') and a crepant birational map $\psi : (A, \Gamma_A) \dashrightarrow (A', \Gamma'_{A'})$ such that $\psi^*\theta'_{A'} = r\theta_A$, where $(A', \Gamma'_{A'})$ is the adjunction pair of A' and $\theta'_{A'}$ is the residue of θ' on A' .*

3. PRELIMINARY REDUCTIONS

Lemma 3.1. *Let n and k be positive integers.*

- (1) *Let $b \leq c$ be real numbers in $[0, 1]$ such that nc is an integer. Then*

$$(3.1) \quad n\lfloor b \rfloor + \lfloor (n+1)\{b\} \rfloor \leq nc.$$

- (2) *Let $(X/Z \ni z, B)$ be a pair. If $(X/Z \ni z, B^+)$ is a monotonic n -complement of $(X/Z \ni z, B)$, then $(X/Z \ni z, B^+)$ is a monotonic kn -complement of $(X/Z \ni z, B)$.*
- (3) *Let $(X/Z \ni z, B + \mathbf{M})$ be a generalized pair such that $kn\mathbf{M}_X$ is a Weil divisor. If $(X/Z \ni z, B^+ + \mathbf{M})$ is a monotonic generalized n -complement of $(X/Z \ni z, B + \mathbf{M})$, then $(X/Z \ni z, B^+ + \mathbf{M})$ is a monotonic generalized kn -complement of $(X/Z \ni z, B + \mathbf{M})$.*

Proof. (1) If $b = 1$, then $c = 1$ and both sides of (3.1) are equal to n . Assume that $b < 1$. Then the left hand side of (3.1) is $\lfloor (n+1)b \rfloor$. Write $nb = l + s$, where l is an integer and $0 \leq s < 1$. If $s = 0$, then $\lfloor (n+1)b \rfloor = l + \lfloor b \rfloor = nb \leq nc$. If $s > 0$, then $\lfloor (n+1)b \rfloor = l + \lfloor s + b \rfloor \leq l + 1 \leq nc$, as nc is an integer and $nc \geq nb > l$.

(2) and (3). We prove (3). The proof of (2) is the same with $\mathbf{M} = 0$, since $nB^+ = n(K_X + B^+) - nK_X$ is a Weil divisor over a neighborhood of z .

Possibly shrinking Z to a neighborhood of z , we may assume that $(X, B^+ + \mathbf{M})$ is generalized lc, $B^+ \geq B$, and $n(K_X + B^+ + \mathbf{M}_X) \sim 0$. Then $kn(K_X + B^+ + \mathbf{M}_X) \sim 0$, and

$$knB^+ = kn(K_X + B^+ + \mathbf{M}_X) - knK_X - kn\mathbf{M}_X$$

is a Weil divisor. Let D be a prime divisor on X , $b := \text{mult}_D B$, and $c := \text{mult}_D B^+$. Since $(X, B^+ + \mathbf{M})$ is generalized lc, $0 \leq b \leq c \leq 1$, and knc is an integer. By (1), $kn\lfloor b \rfloor + \lfloor (kn+1)\{b\} \rfloor \leq knc$. Thus $knB^+ \geq kn\lfloor B \rfloor + \lfloor (kn+1)\{B\} \rfloor$, and $(X/Z \ni z, B^+ + \mathbf{M})$ is a monotonic generalized kn -complement of $(X/Z \ni z, B + \mathbf{M})$. $\square$

Lemma 3.2 (cf. [Bir19, 6.1]). *Let n be a positive integer, $(X/Z \ni z, B)$ and $(Y/Z \ni z, B_Y)$ two pairs, $\phi : X \dashrightarrow Y$ a birational map over Z , and $p : W \rightarrow X$ and $q : W \rightarrow Y$ a common resolution of ϕ . Assume that $q^*(K_Y + B_Y) \geq p^*(K_X + B)$ over a neighborhood of z . If $(Y/Z \ni z, B_Y^+)$ is a monotonic n -complement of $(Y/Z \ni z, B_Y)$, then $(X/Z \ni z, B^+)$ is a monotonic n -complement of $(X/Z \ni z, B)$, where $B^+ := p_*q^*(K_Y + B_Y^+) - K_X$.*

Proof. Possibly shrinking Z to a neighborhood of z , we may assume that $K_X + B$ and $K_Y + B_Y$ are $\mathbb{R}$ -Cartier, $q^*(K_Y + B_Y) \geq p^*(K_X + B)$, (Y, B_Y^+) is lc, $B_Y^+ \geq B_Y$, and $n(K_Y + B_Y^+) = \text{div}(h)$ for some nonzero rational function $h \in \mathbb{C}(Y) = \mathbb{C}(X)$. Then $nq^*(K_Y + B_Y^+) = \text{div}(h)$ on W , so

$$n(K_X + B^+) = p_* \text{div}(h) = \text{div}(h) \quad \text{and} \quad p^*(K_X + B^+) = \frac{1}{n} \text{div}(h) = q^*(K_Y + B_Y^+).$$

In particular, $n(K_X + B^+) \sim 0$. Since $B_Y^+ - B_Y \geq 0$ is $\mathbb{R}$ -Cartier,

$$B^+ - B = p_*(q^*(K_Y + B_Y^+) - p^*(K_X + B)) = p_*q^*(B_Y^+ - B_Y) + p_*(q^*(K_Y + B_Y) - p^*(K_X + B)) \geq 0.$$

Thus $B^+ \geq B \geq 0$, and $a(E, X, B^+) = a(E, Y, B_Y^+) \geq 0$ for any prime divisor E over X , so (X, B^+) is lc. Finally, $nB^+ = \text{div}(h) - nK_X$ is a Weil divisor and every coefficient of B^+ is at most 1, so $nB^+ \geq n\lfloor B \rfloor + \lfloor (n+1)\{B\} \rfloor$ by Lemma 3.1(1). The lemma follows. $\square$

Lemma 3.3. *Let $(X/Z \ni z, B)$ be an lc pair associated with contraction $\pi : X \rightarrow Z$. Assume that $(X/Z \ni z, B^+)$ is an $\mathbb{R}$ -complement of $(X/Z \ni z, B)$. Then $(X/Z \ni z, B)$ has a $\mathbb{Q}$ -complement $(X/Z \ni z, B')$, such that for any prime divisor E over X with $\pi(\text{center}_X E) \ni z$, we have that E is an lc place of (X, B) if and only if E is an lc place of (X, B') .*

Proof. Possibly shrinking Z to a neighborhood of z , we may assume that $(X/Z, B^+)$ is an $\mathbb{R}$ -complement of $(X/Z, B)$.

Write $B^+ = \sum_{i=1}^m b_i B_i$, $\mathbf{b} := (b_1, \dots, b_m)$, and let V be the rational envelope of $\mathbf{b}$ in $\mathbb{R}^m$. Let $B(\mathbf{v}) := \sum_{i=1}^m v_i B_i$ for any $\mathbf{v} = (v_1, \dots, v_m) \in \mathbb{R}^m$.

By $\mathbb{Q}$ -linearity of log discrepancies, any lc place of (X, B^+) is an lc place of $(X, B(\mathbf{v}))$ for any $\mathbf{v} \in V$. By [HLS26, Theorem 5.6], $(X, B(\mathbf{v}))$ is lc for any $\mathbf{v} \in V$ such that $\|\mathbf{v} - \mathbf{b}\| \ll 1$. Pick $\mathbf{v}_0 \in V \cap \mathbb{Q}^m$ such that $\|\mathbf{v}_0 - \mathbf{b}\| \ll 1$. Since B is a $\mathbb{Q}$ -divisor and $B^+ \geq B$, we have that $B(\mathbf{v}_0) \geq B$. By linearity of log discrepancies again and since $\|\mathbf{v}_0 - \mathbf{b}\| \ll 1$, any lc place of $(X, B(\mathbf{v}_0))$ is an lc place of (X, B^+) . By [HLS26, Lemma 5.3], $K_X + B(\mathbf{v}_0) \sim_{\mathbb{Q}, Z} 0$. Thus we may let $B' := B(\mathbf{v}_0)$. $\square$

Lemma 3.4. *Let $h : P \rightarrow S$ be a projective surjective morphism of normal quasi-projective varieties and (P, Θ) an lc pair such that Θ is a $\mathbb{Q}$ -divisor and $K_P + \Theta$ is nef over S . Then $K_P + \Theta$ is semi-ample over S .*

Proof. Let $E := K_P + \Theta$ and r a positive integer such that rE is Cartier. For any positive integer m divisible by r , let $L_m := \mathcal{O}_P(mE)$ and Q_m the cokernel of the evaluation map $h^*h_*L_m \rightarrow L_m$. Then Q_m is coherent, $h(\text{Supp } Q_m)$ is closed in S , and L_m is base point free over $S \setminus h(\text{Supp } Q_m)$.

Fix a closed point $s \in S$. Let P^{an} , S^{an} , h^{an} , Θ^{an} , and E^{an} be the associated complex analytic objects. Then P^{an} and S^{an} are normal complex analytic varieties [SGA1, XII, Propositions 2.1 and 2.4]. Since h factors through a closed immersion $P \rightarrow \mathbb{P}^N \times S$ over S , h^{an} is projective and surjective. Since a log resolution of (P, Θ) induces a log resolution of $(P^{\text{an}}, \Theta^{\text{an}})$ with the same log discrepancies, $(P^{\text{an}}, \Theta^{\text{an}})$ is lc. Since every compact analytic curve in a fiber of h^{an} is algebraic by Chow's theorem, E^{an} is nef over S^{an} . By Theorem 2.10 applied to the compact subset $\{s\}$ of S^{an} , there exist an open neighborhood U_s of s in S^{an} and a positive integer m_s such that $m_s E^{\text{an}}$ is Cartier and base point free over U_s . Replacing m_s with rm_s , we may assume that $r \mid m_s$. Then $L_{m_s}^{\text{an}}$ is base point free over U_s . By [SGA1, XII, Théorème 4.2], $(h_*L_{m_s})^{\text{an}} = h_*^{\text{an}}L_{m_s}^{\text{an}}$, and the analytification of the evaluation map of L_{m_s} is the evaluation map of $L_{m_s}^{\text{an}}$. Since analytification is exact [SGA1, XII, Théorème 1.1], $Q_{m_s}^{\text{an}}$ is the cokernel of the evaluation map of $L_{m_s}^{\text{an}}$, so $Q_{m_s}^{\text{an}}$ vanishes on $(h^{\text{an}})^{-1}(U_s)$. Let $x \in h^{-1}(s)$ be a closed point. Then $Q_{m_s, x} \otimes_{\mathcal{O}_{P, x}} \mathcal{O}_{P^{\text{an}}, x} = (Q_{m_s}^{\text{an}})_x = 0$, and $Q_{m_s, x} = 0$ as $\mathcal{O}_{P, x} \rightarrow \mathcal{O}_{P^{\text{an}}, x}$ is faithfully flat [SGA1, XII, Théorème 1.1]. Thus the closed subset $\text{Supp } Q_{m_s}$ of P contains no closed point of $h^{-1}(s)$, so $\text{Supp } Q_{m_s} \cap h^{-1}(s) = \emptyset$ and $s \notin h(\text{Supp } Q_{m_s})$.

Since every nonempty closed subset of S contains a closed point and S is quasi-compact, there are closed points $s_1, \dots, s_k \in S$ such that the open subsets $S \setminus h(\text{Supp } Q_{m_{s_i}})$ cover S . Let

$m := \text{lcm}(m_{s_1}, \dots, m_{s_k})$. Since $L_m = L_{m_{s_i}}^{\otimes m/m_{s_i}}$ is base point free over $S \setminus h(\text{Supp } Q_{m_{s_i}})$ for each i , L_m is base point free over S , so E is semi-ample over S . $\square$

Proposition 3.5. *Let $\Phi \subset [0, 1] \cap \mathbb{Q}$ be a finite set and $(X/Z \ni z, B)$ an $\mathbb{R}$ -complementary pair such that $B \in \Phi$. Then, possibly shrinking Z to an affine neighborhood of z , there exist a $\mathbb{Q}$ -factorial dlt pair (Y, A) such that $A \in \Phi \cup \{1\}$, a birational map $\phi : X \dashrightarrow Y$ over Z , contractions $g : Y \rightarrow T$ and $\pi : T \rightarrow Z$, and a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor H on T ample over Z , satisfying the following.*

(1) *We have*

$$(3.2) \quad -(K_Y + A) \sim_{\mathbb{Q}} g^* H.$$

(2) *For any common resolution $p : W \rightarrow X$ and $q : W \rightarrow Y$ of ϕ ,*

$$(3.3) \quad q^*(K_Y + A) \geq p^*(K_X + B).$$

In particular, if $(Y/Z \ni z, A)$ has a monotonic n -complement for some positive integer n , then $(X/Z \ni z, B)$ has a monotonic n -complement.

Proof. Step 1. We construct a $\mathbb{Q}$ -factorial dlt pair (V, A_V) birational to X over Z with $A_V \in \Phi \cup \{1\}$, and a dlt pair (V, Δ_V) with $K_V + \Delta_V \sim_{\mathbb{Q}} -t(K_V + A_V)$ for some rational number $t > 0$.

Since $(X/Z \ni z, B)$ is $\mathbb{R}$ -complementary, possibly shrinking Z to a neighborhood of z , we may assume that (X, B) is lc. By Lemma 3.3, $(X/Z \ni z, B)$ has a $\mathbb{Q}$ -complement $(X/Z \ni z, B')$. Possibly shrinking Z to an affine neighborhood of z , we may assume that Z is affine, (X, B') is lc, and $K_X + B' \sim_{\mathbb{Q}} 0$. Let $v : V \rightarrow X$ be a $\mathbb{Q}$ -factorial dlt modification of (X, B') [FG14a, Theorem 2.8]: v is a projective birational morphism, V is $\mathbb{Q}$ -factorial, (V, B'_V) is dlt, and $K_V + B'_V = v^*(K_X + B')$, where $B'_V := v_*^{-1}B' + E_v$ and E_v is the reduced v -exceptional divisor. Let $B_V := v_*^{-1}B$, and let A_V be the $\mathbb{Q}$ -divisor on V such that, for any prime divisor D on V , $\text{mult}_D A_V = 1$ if $\text{mult}_D B'_V = 1$, and $\text{mult}_D A_V = \text{mult}_D B_V$ otherwise. Then $A_V \in \Phi \cup \{1\}$, and $B_V \leq A_V \leq B'_V$ as $B \leq B'$. Since (V, B'_V) is dlt, (V, A_V) is $\mathbb{Q}$ -factorial dlt. Since (X, B) is lc and every v -exceptional prime divisor has coefficient 1 in A_V ,

$$(3.4) \quad v^*(K_X + B) = K_V + B_V + \sum_E (1 - a(E, X, B))E \leq K_V + A_V,$$

where E runs over the v -exceptional prime divisors. Let $G_V := B'_V - A_V \geq 0$ and $\Delta_V := B'_V + tG_V$ for a rational number $t > 0$. No component of G_V is a component of $\lfloor B'_V \rfloor$, and the log discrepancies of the finitely many exceptional divisors of a fixed log resolution of (V, B'_V) depend continuously on t , so (V, Δ_V) is dlt for any $0 < t \ll 1$. We fix such a t . Since $K_V + B'_V \sim_{\mathbb{Q}} 0$,

$$(3.5) \quad K_V + \Delta_V \sim_{\mathbb{Q}} tG_V \sim_{\mathbb{Q}} -t(K_V + A_V).$$

Step 2. We run a $(K_V + \Delta_V)$ -MMP over Z and construct (Y, A) and ϕ satisfying (3.3).

Let (X_0, B_0) be a projective lc pair such that $K_{X_0} + B_0$ is pseudo-effective. By Theorem 2.9, (X_0, B_0) has a good minimal model (X'_0, B'_0) . Since an ample $\mathbb{R}$ -divisor is $\mathbb{R}$ -linearly equivalent to an effective $\mathbb{R}$ -divisor, $K_{X'_0} + B'_0 \sim_{\mathbb{R}} D'_0$ for some $D'_0 \geq 0$, and $K_{X_0} + B_0 \sim_{\mathbb{R}} D_0$ for some $D_0 \geq 0$ by (2.2). Thus [FG14a, Conjecture 5.1] holds in every dimension, and so does [Bir11, Conjecture 1.3] by [FG14a, Lemma 5.3]. By [Bir11, Theorem 1.4] (cf. [FG14a, 2.20]), there is a $(K_V + \Delta_V)$ -MMP over Z , $\psi : V \dashrightarrow Y$, which consists of finitely many divisorial contractions and flips and ends with a log minimal model or a Mori fiber space of $(V/Z, \Delta_V)$. Let $\Delta_Y := \psi_* \Delta_V$ and $G_Y := \psi_* G_V$. Since ψ is a birational contraction, $K_Y + \Delta_Y \sim_{\mathbb{Q}} tG_Y \geq 0$ is pseudo-effective over Z , so the MMP does not end with a Mori fiber space. Thus (Y, Δ_Y) is a log minimal model of $(V/Z, \Delta_V)$. In particular, Y is $\mathbb{Q}$ -factorial, (Y, Δ_Y) is dlt, and $K_Y + \Delta_Y$ is nef over Z (see [Bir11, Definition 2.1]). Let $A := \psi_* A_V$ and $B'_Y := \psi_* B'_V$. Then $A \in \Phi \cup \{1\}$, $\Delta_Y = B'_Y + tG_Y$, and $0 \leq A \leq \Delta_Y$, so (Y, A) is $\mathbb{Q}$ -factorial dlt. Let r be a positive integer and $h \in \mathbb{C}(V)$ a

nonzero rational function such that $r(K_V + B'_V) = \text{div}(h)$. Since ψ is a birational contraction, $r(K_Y + B'_Y) = \psi_* \text{div}(h) = \text{div}(h)$, and (3.5) gives

$$(3.6) \quad K_Y + \Delta_Y \sim_{\mathbb{Q}} -t(K_Y + A).$$

Let $\phi := \psi \circ v^{-1} : X \dashrightarrow Y$, and let $p : W \rightarrow X$ and $q : W \rightarrow Y$ be a common resolution of ϕ . If (3.3) holds on a higher model $W' \rightarrow W$, then it holds on W by pushing forward along $W' \rightarrow W$. Thus, possibly replacing W with a higher model, we may assume that $p = v \circ p_V$ for some morphism $p_V : W \rightarrow V$. Since ψ is a $(K_V + \Delta_V)$ -MMP, the negativity lemma [KM98, Lemma 3.39] gives

$$(3.7) \quad p_V^*(K_V + \Delta_V) - q^*(K_Y + \Delta_Y) \geq 0.$$

Moreover,

$$(3.8) \quad p_V^*(K_V + B'_V) = \frac{1}{r} \text{div}(h) = q^*(K_Y + B'_Y).$$

Since $K_V + A_V = (1 + \frac{1}{t})(K_V + B'_V) - \frac{1}{t}(K_V + \Delta_V)$ and $K_Y + A = (1 + \frac{1}{t})(K_Y + B'_Y) - \frac{1}{t}(K_Y + \Delta_Y)$, (3.7) and (3.8) give

$$q^*(K_Y + A) - p_V^*(K_V + A_V) = \frac{1}{t}(p_V^*(K_V + \Delta_V) - q^*(K_Y + \Delta_Y)) \geq 0.$$

Together with (3.4), $q^*(K_Y + A) \geq p_V^*(K_V + A_V) \geq p_V^*v^*(K_X + B) = p^*(K_X + B)$. This is (3.3).

Step 3. We construct g , π , and H .

By Lemma 3.4, $K_Y + \Delta_Y$ is semi-ample over Z . By (3.6), $-(K_Y + A) \sim_{\mathbb{Q}} \frac{1}{t}(K_Y + \Delta_Y)$ is semi-ample over Z . Let m be a positive integer such that $L := -m(K_Y + A)$ is Cartier and base point free over Z . Since Z is affine, $\mathcal{O}_Y(L)$ is generated by finitely many global sections, which define a morphism $\rho : Y \rightarrow \mathbb{P}^N \times Z$ over Z such that $\mathcal{O}_Y(L) \cong \rho^*\mathcal{O}_{\mathbb{P}^N \times Z}(1)$. Let $Y \xrightarrow{g} T \xrightarrow{\tau} \mathbb{P}^N \times Z$ be the Stein factorization of ρ [Har77, Corollary III.11.5]. Then g is a contraction, T is a normal variety projective over Z , and τ is finite. Let H_T be a Cartier divisor on T such that $\mathcal{O}_T(H_T) \cong \tau^*\mathcal{O}_{\mathbb{P}^N \times Z}(1)$. Then H_T is ample over Z as τ is finite, and $L \sim g^*H_T$. Let $H := \frac{1}{m}H_T$ and $\pi : T \rightarrow Z$ the induced morphism. Then (3.2) holds. Since $X \rightarrow Z$ is a contraction and Y is birational to X over Z , $\pi \circ g : Y \rightarrow Z$ is a contraction, hence so is π . Finally, the last assertion of the proposition follows from (3.3) and Lemma 3.2. $\square$

4. CANONICAL BUNDLE FORMULAS

Lemma 4.1. *Let $f : X \rightarrow T$ be a contraction between normal quasi-projective varieties and $v \in \mathbb{C}(X)$ a nonzero rational function such that $\text{div}(v)$ is vertical over T . Then $v = f^*a$ for some nonzero $a \in \mathbb{C}(T)$. In particular, $\text{div}(v) = f^*\text{div}(a)$.*

Proof. Let F be the generic fiber of f and $K := \mathbb{C}(T)$. Then F is a normal projective variety over K and $H^0(F, \mathcal{O}_F) = K$ as $f_*\mathcal{O}_X = \mathcal{O}_T$. Since $\text{div}(v)$ is vertical over T , $v|_F$ has neither zeros nor poles on F . Thus $v|_F \in H^0(F, \mathcal{O}_F) = K$, and we may let $a := v|_F$ (cf. [CHL24, Lemma 2.5]). $\square$

Definition 4.2. Let $f : (X, B) \rightarrow T$ be a crepant log structure (Definition 2.8) such that B is a $\mathbb{Q}$ -divisor and f is a contraction between normal quasi-projective varieties. Let P be a prime divisor on T . Then P is Cartier over a neighborhood of the generic point of P , and we define

$$t_P := \text{lct}(X, B; f^*P)$$

over the generic point of P (Definition 2.3). Since (X, B) is lc, $t_P \geq 0$. Since f is surjective, there is a prime divisor E on X such that $f(E) = P$, and $\text{mult}_E(B + t_P f^*P) \leq 1$ gives $t_P \leq 1$.

Moreover, $t_P = 1$ for all but finitely many P . Indeed, let $p : U \rightarrow X$ be a log resolution of (X, B) and $K_U + B_U := p^*(K_X + B)$. By generic smoothness [Har77, Corollary III.10.7], there is a dense open subset $T^0 \subset T$ such that U and all irreducible components of intersections of components of $\text{Supp } B_U \cup \text{Exc}(p)$ are smooth over T^0 . Let P be a prime divisor on T such

that $P \cap T^0 \neq \emptyset$. Then over a neighborhood of the generic point of P , $(f \circ p)^*P$ is a reduced divisor and $\text{Supp } B_U \cup \text{Exc}(p) \cup \text{Supp}(f \circ p)^*P$ is a simple normal crossing divisor. Since the coefficients of $B_U + (f \circ p)^*P$ are at most 1, $(X, B + f^*P)$ is lc over a neighborhood of the generic point of P by [KM98, Corollary 2.31], so $t_P \geq 1$. The *discriminant part* of f is the $\mathbb{Q}$ -divisor $B_T := \sum_P (1 - t_P)P$, where P runs over all prime divisors on T . It depends only on f and (X, B) .

Let m be a positive integer, $\psi \in \mathbb{C}(X)$ a nonzero rational function, and L a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor on T . Assume that $m(K_X + B) + \text{div}(\psi) = f^*(mL)$, that is,

$$(4.1) \quad K_X + B + \frac{1}{m} \text{div}(\psi) = f^*L.$$

The *moduli part* of (4.1) is the $\mathbb{Q}$ -divisor

$$\mathbf{M}_T := L - K_T - B_T.$$

Let $\tau : T' \rightarrow T$ be a projective birational morphism from a normal variety, X' the normalization of the main component of $X \times_T T'$, and $\mu : X' \rightarrow X$ and $f' : X' \rightarrow T'$ the induced morphisms. Since $\mathbb{C}(T)$ is algebraically closed in $\mathbb{C}(X)$, f' is a contraction. We define the $\mathbb{Q}$ -divisor $B_{X'}$ by $K_{X'} + B_{X'} = \mu^*(K_X + B)$. For a $\mathbb{Q}$ -divisor D on a normal variety Z such that $K_Z + D$ is $\mathbb{Q}$ -Cartier, we say that (Z, D) is *sub-lc* if $a(E, Z, D) \geq 0$ for any prime divisor E over Z , where $a(E, Z, D)$ is defined as in Definition 2.3. Thus $(X', B_{X'})$ is sub-lc. For any prime divisor P' on T' , let $t_{P'}$ be the largest rational number t such that $(X', B_{X'} + t f'^*P')$ is sub-lc over the generic point of P' . Here sub-lc over the generic point of P' means sub-lc over a neighborhood of the generic point of P' (Notation 2.2). Since log discrepancies are preserved under crepant pullback, $t_{P'}$ may be computed on any normal variety X'' with a projective birational morphism $\mu'' : X'' \rightarrow X'$, using $K_{X''} + B_{X''} := \mu''^*(K_{X'} + B_{X'})$ and $f' \circ \mu''$ in place of $B_{X'}$ and f' (cf. [FG14b, 3.3]). If P' is not τ -exceptional, then τ is an isomorphism over a neighborhood of the generic point of $\tau(P')$ as T is normal, so $t_{P'} = t_{\tau(P')}$. Let $B_{T'} := \sum_{P'} (1 - t_{P'})P'$ and

$$\mathbf{M}_{T'} := \tau^*L - K_{T'} - B_{T'},$$

where $K_{T'}$ is the canonical divisor compatible with K_T (Notation 2.2). Then $B_{T'}$ is a $\mathbb{Q}$ -divisor, $\tau_*B_{T'} = B_T$, and $\tau_*\mathbf{M}_{T'} = \mathbf{M}_T$. The *discriminant b-divisor* $\mathbf{B}$ of f and the *moduli b-divisor* $\mathbf{M}$ of (4.1) are the b-divisors over T with $\mathbf{B}_{T'} := B_{T'}$ and $\mathbf{M}_{T'}$ as above for every T' (Definition 2.6). We have $\tau^*L = K_{T'} + B_{T'} + \mathbf{M}_{T'}$ for every T' .

This is the setting of [FG14b, Definition 3.2 and 3.3–3.4] and of [OAI26c, (7.1)–(7.3)]. In the terminology of [FG14b, Definition 3.2], f is an lc-trivial fibration: the condition $\text{rank } f_*\mathcal{O}_X([\mathbf{A}^*(X, B)]) = 1$ there, where $\mathbf{A}^*(X, B)$ is the discrepancy b-divisor of (X, B) with the components of coefficient -1 removed [FG14b, 2.3], holds because $\mathcal{O}_X([\mathbf{A}^*(X, B)]) \simeq \mathcal{O}_X$ as (X, B) is lc [FG14b, 2.3] and $f_*\mathcal{O}_X = \mathcal{O}_T$. If ψ' and L' also satisfy (4.1) with the same m , then $\text{div}(\psi'/\psi) = m f^*(L' - L)$ is vertical over T , so $\psi' = \psi \cdot f^*a$ for some nonzero $a \in \mathbb{C}(T)$ by Lemma 4.1, $L' = L + \frac{1}{m} \text{div}(a)$, and $\mathbf{M}'_{T'} = \mathbf{M}_{T'} + \frac{1}{m} \text{div}_{T'}(a)$ for every T' .

Theorem 4.3 ([FG14b, Theorem 3.6]). *Let $f : (X, B) \rightarrow T$, m , ψ , and L be as in Definition 4.2, $\pi : T \rightarrow S$ a proper morphism to a variety, and $\mathbf{K}$ the b-divisor over T with $\mathbf{K}_{T'} = K_{T'}$ for every T' (Notation 2.2). Then:*

- (1) $\mathbf{K} + \mathbf{B}$ is b- $\mathbb{Q}$ -Cartier. *That is, there exists a projective birational morphism $T' \rightarrow T$ from a normal variety such that $K_{T'} + B_{T'}$ is $\mathbb{Q}$ -Cartier and $K_W + B_W = \rho^*(K_{T'} + B_{T'})$ for any projective birational morphism $\rho : W \rightarrow T'$ from a normal variety. In particular, $\mathbf{M}$ descends to T' .*
- (2) $\mathbf{M}$ is b-nef over S . *That is, there exists a projective birational morphism $T'' \rightarrow T$ from a normal variety such that $\mathbf{M}$ descends to T'' and $\mathbf{M}_{T''}$ is nef over S .*

Lemma 4.4. *Let d be a positive integer and $\Phi \subset [0, 1] \cap \mathbb{Q}$ a finite set. Then there exists a finite set $\mathfrak{R} \subset [0, 1] \cap \mathbb{Q}$ with $0, 1 \in \mathfrak{R}$ depending only on d and Φ satisfying the following.*

- (1) $\Phi \cup \{1\} \subset \Phi(\mathfrak{R})$. Moreover, if (X, B) is a dlt pair of dimension $\leq d$ such that $B \in \Phi \cup \{1\}$ and W is an lc center of (X, B) , then $B_W \in \Phi(\mathfrak{R})$, where (W, B_W) is the adjunction pair of W (Definition 2.3).

- (2) For any positive integer N such that $N\mathfrak{R} \subset \mathbb{Z}$ and any $b \in \Phi(\mathfrak{R})$,

$$(4.2) \quad N\lfloor b \rfloor + \lfloor (N+1)\{b\} \rfloor = \lceil Nb \rceil.$$

- (3) $\Phi(\mathfrak{R})$ is a DCC set whose only accumulation point is 1.

Proof. Let (X, B) be a dlt pair and W an lc center of (X, B) of codimension k . By Definition 2.3 and [Kol16, Definition 4], there exist lc centers $X = W_0 \supset W_1 \supset \cdots \supset W_k = W$ of (X, B) such that W_i is a prime divisor on W_{i-1} and a component of $\lfloor B_{W_{i-1}} \rfloor$ for each i , where (W_i, B_{W_i}) is the adjunction pair of W_i . Moreover, $(W_{i-1}, B_{W_{i-1}})$ is dlt [Fuj11, Lemma 10.2], and B_{W_i} is the different of $B_{W_{i-1}}$ on W_i , that is, $K_{W_i} + B_{W_i} = (K_{W_{i-1}} + B_{W_{i-1}})|_{W_i}$ is defined by adjunction.

Let $\mathfrak{R}' \subset [0, 1] \cap \mathbb{Q}$ be a finite set with $0, 1 \in \mathfrak{R}'$. By [Bir19, Lemma 3.3] applied with $p = 1$ and $\mathbf{M} = 0$, there is a finite set $\mathfrak{S}(\mathfrak{R}') \subset [0, 1] \cap \mathbb{Q}$ depending only on $\mathfrak{R}'$ satisfying the following. If (X', B') is an lc pair such that $B' \in \Phi(\mathfrak{R}')$ and S' is the normalization of a component of $\lfloor B' \rfloor$, then $B_{S'} \in \Phi(\mathfrak{S}(\mathfrak{R}'))$, where $K_{S'} + B_{S'} := (K_{X'} + B')|_{S'}$ is defined by adjunction. Possibly enlarging $\mathfrak{S}(\mathfrak{R}')$, we may assume that $\mathfrak{R}' \subset \mathfrak{S}(\mathfrak{R}')$. Let $\mathfrak{R}_0 := \{0, 1\} \cup \{1 - b \mid b \in \Phi\}$, $\mathfrak{R}_i := \mathfrak{S}(\mathfrak{R}_{i-1})$ for $1 \leq i \leq d$, and $\mathfrak{R} := \mathfrak{R}_d$. Then $\Phi \cup \{1\} \subset \Phi(\mathfrak{R}_0) \subset \Phi(\mathfrak{R})$. Let (X, B) and W be as in (1) and $W_0 \supset \cdots \supset W_k = W$ as above. Then $B_{W_0} = B \in \Phi(\mathfrak{R}_0)$, and inductively $B_{W_i} \in \Phi(\mathfrak{R}_i)$ for $1 \leq i \leq k$. Since $k \leq d$, $B_W \in \Phi(\mathfrak{R}_k) \subset \Phi(\mathfrak{R})$. This proves (1).

(2) If $b = 1$, then both sides of (4.2) equal N . Assume that $b < 1$. Then $\lfloor b \rfloor = 0$ and $\{b\} = b$. Write $b = 1 - \frac{r}{j}$ with $r \in \mathfrak{R}$, $r > 0$, and j a positive integer. Then Nr is a positive integer. Write $Nr = qj + \rho$ with integers $q \geq 0$ and $0 \leq \rho \leq j - 1$. Since $0 < \rho + r \leq j$,

$$\left\lceil \frac{(N+1)r}{j} \right\rceil = q + \left\lceil \frac{\rho + r}{j} \right\rceil = q + 1 = \left\lfloor \frac{Nr}{j} \right\rfloor + 1.$$

Thus $\lfloor (N+1)b \rfloor = N + 1 - \lceil \frac{(N+1)r}{j} \rceil = N - \lfloor \frac{Nr}{j} \rfloor = \lceil Nb \rceil$, which is (4.2).

(3) is [Bir19, 2.2]. $\square$

Lemma 4.5. Let d be a positive integer and $I \subset [0, 1] \cap \mathbb{Q}$ a DCC set. Let $a(s, I)$ be as in Theorem 2.11 and $m(d, I) := \text{lcm}_{0 \leq s \leq d} a(s, I)$. Let K be a finitely generated field extension of $\mathbb{C}$ and F a normal geometrically integral projective variety over K of dimension $\leq d$. Let $B_F \geq 0$ be a $\mathbb{Q}$ -divisor on F such that $B_F \in I$, (F, B_F) is geometrically lc, and $K_F + B_F \sim_{\mathbb{Q}} 0$. Then $m(d, I)(K_F + B_F) \sim 0$.

Proof. Let $s := \dim F$, $\overline{K}$ an algebraic closure of K , and $(F_{\overline{K}}, B_{\overline{K}})$ the base change of (F, B_F) to $\overline{K}$. Since K is finitely generated over $\mathbb{C}$, $\overline{K}$ is isomorphic to $\mathbb{C}$ as a field. Since $\overline{K}/K$ is separable, $F_{\overline{K}}$ is normal and $B_{\overline{K}} \in I$. Thus $(F_{\overline{K}}, B_{\overline{K}})$ is a projective lc pair of dimension s with $K_{F_{\overline{K}}} + B_{\overline{K}} \sim_{\mathbb{Q}} 0$, and $a(s, I)(K_{F_{\overline{K}}} + B_{\overline{K}}) \sim 0$ by Theorem 2.11 (cf. [OAI26c, Section 2.2]). In particular, $m(d, I)B_F$ is an integral divisor, and the integral divisor $D := m(d, I)(K_F + B_F)$ becomes principal on $F_{\overline{K}}$. By [OAI26c, Lemma 2.4], $D \sim 0$. $\square$

Lemma 4.6. Let $f : (X, B) \rightarrow T$, m , ψ , and L be as in Definition 4.2, $s := \dim X - \dim T$, $\tau : W \rightarrow T$ a projective birational morphism from a smooth variety, P a prime divisor on W , $\alpha := \text{mult}_P \tau^* L$, and t_P as in Definition 4.2. Then there exist a smooth projective curve C , a closed point $c \in C$, and a rational uniformizer $z \in \mathbb{C}(C)$ at c satisfying the following. There exist a normal geometrically integral projective variety V of dimension s over $\mathbb{C}(C)$, a $\mathbb{Q}$ -divisor $B_V \geq 0$ on V with (V, B_V) geometrically lc, and a rational relative m -canonical form φ of $\mathbb{C}(V)/\mathbb{C}(C)$ (Definition 2.12) such that $\text{Div}_V(\varphi) + mB_V = 0$ and

$$(4.3) \quad w_z(\varphi) = \alpha - 1 + t_P = \text{mult}_P \mathbf{M}_W + \text{mult}_P K_W.$$

In particular, $q(s, m) \text{mult}_P \mathbf{M}_W \in \mathbb{Z}$, where $q(s, m)$ is as in Theorem 2.13.

Proof. When T is projective, this is [OAI26c, Lemma 7.3]. We follow its proof. Let $r := \dim T$ and ω the fixed rational top form with $K_X = \text{Div}(\omega)$ (Notation 2.2). Let $p : U \rightarrow X$ be a log resolution of (X, B) such that the induced birational map $g : U \dashrightarrow W$ is a morphism and

$$\Sigma := \text{Supp } B_U \cup \text{Supp } g^*P \cup \text{Exc}(p) \cup \text{Supp } K_U \cup \text{Supp } \text{div}(\psi)$$

is a simple normal crossing divisor, where $K_U + B_U := p^*(K_X + B)$ and $K_U = \text{Div}(\omega)$. A *stratum* of Σ is an irreducible component of an intersection of components of Σ . Then g is a contraction, as $\mathbb{C}(W)$ is algebraically closed in $\mathbb{C}(U)$ and W is normal. Pulling back (4.1) to U , we have

$$(4.4) \quad K_U + B_U + \frac{1}{m} \text{div}(\psi) = g^* \tau^* L.$$

Let $E_1, \dots, E_n$ be the prime divisors on U such that $g(E_i) = P$, $a_i := \text{mult}_{E_i} g^*P$, and $b_i := \text{mult}_{E_i} B_U$. By Definition 4.2 and [KM98, Corollary 2.31], $t_P = \min_i \frac{1-b_i}{a_i}$.

Step 1. We construct a smooth curve $C^\circ \subset W$ through a general point c of P and a smooth model Y over C° .

Let $T^1 \subset T$ be a dense open subset satisfying the following. The morphism τ is an isomorphism over T^1 , and $T^1 \cap \text{Supp } L = \emptyset$. The morphism f is flat over T^1 , and its fibers over T^1 are geometrically normal and geometrically integral. Every irreducible component of $p(\text{Exc}(p))$ is either disjoint from $f^{-1}(T^1)$ or flat over T^1 . The variety U and all strata of Σ are smooth over T^1 . Such T^1 exists. Indeed, the generic fiber of f is normal and geometrically integral as f is a contraction (cf. [OAI26c, proof of Proposition 7.1]). The set of points of T over which the fibers of a flat proper morphism are geometrically normal and geometrically integral is open (standard). Generic flatness and generic smoothness [Har77, Corollary III.10.7] apply to the finitely many morphisms above (cf. [OAI26c, proof of Lemma 7.3]).

If $r = 1$, let $C^\circ := W$ and $c := P$. Assume that $r \geq 2$. By Hironaka's resolution, there is a smooth projective variety $\overline{W}$ containing W as an open subset. Let A be a very ample divisor on $\overline{W}$, $A_1, \dots, A_{r-1} \in |A|$ general members, and $\overline{P}$ the closure of P in $\overline{W}$. By [Har77, Corollaries III.7.9 and III.10.9], $\overline{C} := A_1 \cap \dots \cap A_{r-1}$ is a smooth connected curve, hence integral. Let $C^\circ := \overline{C} \cap W$. The generic point η of C° lies in any given dense open subset of W , and we require that $\eta \in \tau^{-1}(T^1)$. Moreover, $\overline{C}$ misses the singular locus of $\overline{P}$ and meets $\overline{P}$ transversally, and $\overline{C} \cap \overline{P} \neq \emptyset$ as A is ample. Since $\overline{C}$ also misses any given closed subset of $\overline{P}$ of dimension $\leq r - 2$, we may choose $c \in C^\circ \cap P$ outside any given proper closed subset of P . We choose c outside $g(E)$ for any prime divisor $E \subset \text{Supp } g^*P$ on U with $g(E) \neq P$ and outside the components of $\text{Supp } \tau^*L$ other than P .

Let $Y := g^{-1}(C^\circ)$ and $h := g|_Y : Y \rightarrow C^\circ$. By Bertini's theorem [Kle98, Theorem 4.1] applied successively to the base point free linear system $g^*|A|$ on U and on the strata of Σ , $\Sigma + \sum_{j=1}^{r-1} g^*A_j$ is a simple normal crossing divisor on U . In particular, Y is smooth and $\Sigma|_Y$ is a simple normal crossing divisor on Y . Since the fibers of h are fibers of g , they are connected. Thus Y is connected, hence integral, and h is a contraction. Let $K := \mathbb{C}(C^\circ)$ and $F := \mathbb{C}(Y)$. Then F/K is a regular extension of transcendence degree s .

Step 2. We show that there is a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor L_C on C° with $\text{mult}_c L_C = \alpha$ and a rational top form θ_Y on Y with $K_Y := \text{Div}(\theta_Y)$ such that

$$(4.5) \quad K_Y + B_Y + \frac{1}{m} \text{div}(\psi|_Y) = h^* L_C,$$

where $B_Y := B_U|_Y$. We also show that $t_c = t_P$, where t_c is the largest rational number t such that $(Y, B_Y + t h^* c)$ is sub-lc over a neighborhood of c .

If $r = 1$, let $L_C := \tau^*L$ and $\theta_Y := \omega$. Assume that $r \geq 2$. For each j , let $A'_j \in |A|$ be a general member and $a_j \in \mathbb{C}(\overline{W})$ a rational function with $\text{div}(a_j) = A'_j - A_j$. Then $c \notin A'_j$, $\overline{C} \not\subset A'_j$, and $\Sigma + \sum_j g^*A_j + \sum_j g^*A'_j$ is a simple normal crossing divisor on U . Thus $\text{Div}(\omega \cdot g^*(a_1 \cdots a_{r-1})) = K_U + \sum_j g^*A'_j - \sum_j g^*A_j$ has multiplicity -1 along each component of

g^*A_j . Taking the residues of $\omega \cdot g^*(a_1 \cdots a_{r-1})$ along $g^*A_1, \dots, g^*A_{r-1}$ successively (cf. [Har77, Proposition II.8.20]), we obtain a rational top form θ_Y on Y with

$$\mathrm{Div}(\theta_Y) = \left(K_U + \sum_{j=1}^{r-1} g^*A'_j \right) \Big|_Y.$$

Since $\eta \in \tau^{-1}(T^1)$, $C^\circ \not\subset \mathrm{Supp} \tau^*L$. Restricting (4.4) to Y , we get (4.5) with $L_C := (\tau^*L + \sum_j A'_j)|_{C^\circ}$. Since $c \notin A'_j$ for each j , C° meets P transversally at c , and c does not lie on the other components of $\mathrm{Supp} \tau^*L$, we have $\mathrm{mult}_c L_C = \alpha$.

Over a neighborhood of c we have $h^*c = (g^*P)|_Y$. By the choice of c , the prime divisors on Y contained in $\mathrm{Supp} h^*c$ are the irreducible components of $E_i \cap Y$ for $1 \leq i \leq n$, and each component of $E_i \cap Y$ has multiplicity a_i in h^*c and b_i in B_Y . Since $\Sigma|_Y$ is a simple normal crossing divisor, $t_c = \min_i \frac{1-b_i}{a_i} = t_P$ by [KM98, Corollary 2.31].

Step 3. We construct C , z , V , B_V , and φ , and prove (4.3).

Let C be the smooth projective curve containing C° as an open subset, so that $\mathbb{C}(C) = K$, and let $z \in K$ be a rational uniformizer at c . Let

$$\varphi := (\theta_Y/dz)^{\otimes m} \cdot \psi|_Y, \quad \Omega := \varphi \wedge (dz/z)^{\otimes m} = z^{-m} \psi|_Y \theta_Y^{\otimes m}.$$

Then φ is a rational relative m -canonical form of F/K . Let E be a divisorial valuation of $F/\mathbb{C}$ whose restriction to K is a positive multiple of ord_c . Since h is proper, E is a prime divisor on some projective birational model $\pi : Y' \rightarrow Y$ with $\pi(E) \subset h^{-1}(c)$. Let $a_E := \mathrm{ord}_E(z) = \mathrm{mult}_E \pi^*h^*c > 0$ and $b_E := \mathrm{mult}_E B_{Y'}$, where $K_{Y'} + B_{Y'} := \pi^*(K_Y + B_Y)$. By (4.5) and $\mathrm{mult}_c L_C = \alpha$,

$$\mathrm{mult}_E K_{Y'} = \mathrm{mult}_E \pi^*(K_Y + B_Y) - b_E = \alpha a_E - \frac{1}{m} \mathrm{ord}_E(\psi|_Y) - b_E.$$

Thus $\mathrm{ord}_E(\Omega) = -ma_E + m \mathrm{mult}_E K_{Y'} + \mathrm{ord}_E(\psi|_Y) = m(\alpha - 1)a_E - mb_E$, and

$$\frac{\mathrm{ord}_E(\Omega) + m}{m \mathrm{ord}_E(z)} = \alpha - 1 + \frac{1 - b_E}{a_E} = \alpha - 1 + \frac{a(E, Y, B_Y)}{\mathrm{mult}_E \pi^*h^*c}.$$

Taking the infimum over all such E , we get $w_z(\varphi) = \alpha - 1 + t_c = \alpha - 1 + t_P$ by (2.4) and Step 2.

Since $\eta \in \tau^{-1}(T^1)$, we regard η as a point of T with residue field K . Let $V := X_\eta$ be the fiber of f over η . The generic fiber Y_η of h is the fiber of $f \circ p$ over η ; let $p_\eta : Y_\eta \rightarrow V$ be the morphism induced by p , and $\bar{K}$ an algebraic closure of K . By the choice of T^1 , V is a normal geometrically integral projective variety over K of dimension s and Y_η is smooth over K . Moreover, the base change $\Sigma_{\bar{K}}|_{Y_\eta}$ of $\Sigma|_{Y_\eta}$ to $\bar{K}$ is a simple normal crossing divisor on $Y_{\bar{K}}$. Every irreducible component Z of $p(\mathrm{Exc}(p))$ meeting $f^{-1}(T^1)$ is flat over T^1 , so $\dim Z_\eta = \dim Z - r \leq s - 2$. Since p is an isomorphism over $X \setminus p(\mathrm{Exc}(p))$, p_η is an isomorphism over the complement of a closed subset of V of codimension ≥ 2 . Thus p_η is birational, the function field of V is F , and $E|_{Y_\eta}$ is p_η -exceptional for every p -exceptional prime divisor E on U . Let $B_V := (p_\eta)_*(B_Y|_{Y_\eta})$. Since $B_U - \sum_E (1 - a(E, X, B))E = p_*^{-1}B \geq 0$, where E runs over the p -exceptional prime divisors on U , we have $B_V \geq 0$.

Restricting (4.5) to Y_η , we get $\mathrm{Div}_{Y_\eta}(\varphi) + mB_Y|_{Y_\eta} = 0$, as $(h^*L_C)|_{Y_\eta} = 0$ and $K_Y|_{Y_\eta} = \mathrm{Div}_{Y_\eta}(\theta_Y/dz)$, Y_η being smooth over K . Pushing forward to V , we get $\mathrm{Div}_V(\varphi) + mB_V = 0$, where $K_V := \mathrm{Div}_V(\theta_Y/dz)$. In particular, $K_V + B_V$ is $\mathbb{Q}$ -Cartier and $K_{Y_\eta} + B_Y|_{Y_\eta} = p_\eta^*(K_V + B_V)$, as both sides equal $-\frac{1}{m} \mathrm{div}(\psi|_{Y_\eta})$. Now $V_{\bar{K}}$ is normal, $Y_{\bar{K}} \rightarrow V_{\bar{K}}$ is projective birational, $\Sigma_{\bar{K}}$ is a simple normal crossing divisor, and the coefficients of B_Y are at most 1 as (X, B) is lc. By [KM98, Corollary 2.31], $(Y_{\bar{K}}, B_{Y_{\bar{K}}})$ is sub-lc, so $(V_{\bar{K}}, B_{V_{\bar{K}}})$ is lc, where $B_{Y_{\bar{K}}}$ and $B_{V_{\bar{K}}}$ are the base changes of $B_Y|_{Y_\eta}$ and B_V . Thus (V, B_V) is geometrically lc. Finally, $\mathrm{mult}_P \mathbf{M}_W = \alpha - \mathrm{mult}_P K_W - (1 - t_P)$ by Definition 4.2, which gives (4.3). The last assertion follows from Theorem 2.13(1), as $\mathrm{mult}_P K_W \in \mathbb{Z}$. $\square$

The following theorem is essentially [OAI26c, Proposition 7.1 and Theorem 7.2] when T is projective and I is finite.

Theorem 4.7. *Let d be a positive integer and $I \subset [0, 1] \cap \mathbb{Q}$ a DCC set. Then there exist positive integers $m \mid p$ and a DCC set $\Lambda \subset [0, 1] \cap \mathbb{Q}$ depending only on d and I satisfying the following.*

*Assume that $f : (X, B) \rightarrow T$ is a crepant log structure such that f is a contraction between normal quasi-projective varieties, $0 < \dim T \leq \dim X \leq d$, $B \in I$, and $K_X + B \sim_{\mathbb{Q}} f^*L_0$ for some $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor L_0 on T . Then:*

- (1) *There exist a nonzero $\psi \in \mathbb{C}(X)$ and a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor $L \sim_{\mathbb{Q}} L_0$ on T such that $m(K_X + B) + \operatorname{div}(\psi) = f^*(mL)$.*
- (2) *$B_T \geq 0$ and $B_T \in \Lambda$.*
- (3) *For any ψ and L satisfying (4.1) with this m , $\mathbf{M}$ is b - $\mathbb{Q}$ -Cartier, and $p\mathbf{M}_W$ is Cartier for any smooth variety W projective birational over T .*
- (4) *If moreover $T \rightarrow S$ is a projective morphism to a normal quasi-projective variety, then $\mathbf{M}_W$ is nef over S for any smooth variety W projective birational over T such that $\mathbf{M}$ descends to W . In particular, there exists such a W , and $p\mathbf{M}_W$ is Cartier and nef over S .*

Proof. Let $m := m(d, I)$ be as in Lemma 4.5 and $p := \operatorname{lcm}(m, q(0, m), \dots, q(d-1, m))$, where $q(s, m)$ is as in Theorem 2.13. Let $\mathcal{T}$ be the set of all lc thresholds $\operatorname{lc}(X', B'; M')$, where (X', B') is an lc pair of dimension $\leq d$ such that $B' \in I$ and $M' \geq 0$ is a nonzero Cartier divisor on X' . By [HMX14, Theorem 1.1], $\mathcal{T}$ satisfies the ACC. Let $\Lambda := \{1 - t \mid t \in \mathcal{T} \cap [0, 1]\}$, a DCC subset of $[0, 1] \cap \mathbb{Q}$. We show that m , p , and Λ satisfy the requirements.

(1) Let $K := \mathbb{C}(T)$, F the generic fiber of f , $B_F := B|_F$, and $K_F := K_X|_F$. Then K_F is the divisor of the relative top form $\omega/f^*\omega_T$ of F over K , where ω and ω_T are the fixed rational top forms with $K_X = \operatorname{Div}(\omega)$ and $K_T = \operatorname{Div}(\omega_T)$ (Notation 2.2). Since f is a contraction and (X, B) is lc, F is a normal geometrically integral projective variety over K of dimension $\leq d-1$, (F, B_F) is geometrically lc, $B_F \in I$, and $K_F + B_F \sim_{\mathbb{Q}} 0$ (cf. [OAI26c, proof of Proposition 7.1]). By Lemma 4.5, there exists a nonzero $\psi \in \mathbb{C}(X)$ such that $m(K_F + B_F) + \operatorname{div}(\psi|_F) = 0$. Thus $G := m(K_X + B) + \operatorname{div}(\psi)$ is vertical over T . Choose a positive integer k and a nonzero $u \in \mathbb{C}(X)$ such that $k(K_X + B - f^*L_0) = \operatorname{div}(u)$. Then $\operatorname{div}(u^m\psi^k) = k(G - mf^*L_0)$ is vertical over T . By Lemma 4.1, $u^m\psi^k = f^*a$ for some nonzero $a \in \mathbb{C}(T)$, so $kG = f^*(kmL_0 + \operatorname{div}(a))$. Thus $L := L_0 + \frac{1}{km}\operatorname{div}(a)$ satisfies the requirements.

(2) Let P be a prime divisor on T . By Definition 4.2, $0 \leq t_P \leq 1$, so $B_T \geq 0$. Let $T^1 \subset T$ be an open subset containing the generic point of P such that P is Cartier on T^1 , $X^1 := f^{-1}(T^1)$, and $\pi : \tilde{X} \rightarrow X^1$ a log resolution of $(X^1, \operatorname{Supp} B \cup \operatorname{Supp} f^*P)$. By [KM98, Corollary 2.31], t_P is the minimum of $a(E', X, B)/\operatorname{mult}_{E'} \pi^*f^*P$ over the finitely many prime divisors $E' \subset \operatorname{Supp} \pi^*f^*P$ on $\tilde{X}$ with $f(\pi(E')) = P$. Let $T^0 \subset T^1$ be the complement of the union of $f(\pi(E'))$ for the prime divisors $E' \subset \operatorname{Supp} \pi^*f^*P$ with $f(\pi(E')) \neq P$, and $X^0 := f^{-1}(T^0)$. Then $t_P = \operatorname{lc}(X^0, B|_{X^0}; f^*P|_{X^0})$ by [KM98, Corollary 2.31] again. Thus $t_P \in \mathcal{T}$ and $1 - t_P \in \Lambda$.

(3) By Theorem 4.3(1), $\mathbf{M}$ is b - $\mathbb{Q}$ -Cartier. Let W be a smooth variety projective birational over T and P a prime divisor on W . By Lemma 4.6, $q(s, m)\operatorname{mult}_P \mathbf{M}_W \in \mathbb{Z}$, where $s = \dim X - \dim T \leq d-1$. Thus $p\mathbf{M}_W$ is an integral divisor on W , hence Cartier as W is smooth.

(4) By Theorem 4.3(2), there exists a projective birational morphism $T'' \rightarrow T$ from a normal variety such that $\mathbf{M}$ descends to T'' and $\mathbf{M}_{T''}$ is nef over S . Let W be a smooth variety projective birational over T such that $\mathbf{M}$ descends to W , and W' a smooth variety with projective birational morphisms $\rho : W' \rightarrow W$ and $\rho'' : W' \rightarrow T''$ over T . Then $\rho^*\mathbf{M}_W = \mathbf{M}_{W'} = \rho''^*\mathbf{M}_{T''}$ is nef over S . Since ρ is surjective, $\mathbf{M}_W$ is nef over S by the projection formula. The last assertion follows from (3) by taking W to be a resolution of T'' . $\square$

Proposition 4.8. *Let $\mathfrak{R} \subset [0, 1] \cap \mathbb{Q}$ be a finite set with $0, 1 \in \mathfrak{R}$ and p a positive integer. Let*

$$(4.6) \quad \mathfrak{R}[p] := \mathfrak{R} \cup \left\{ \frac{r}{j} - \frac{i}{p} \mid r \in \mathfrak{R}, i, j \text{ positive integers, } i \leq p, j \leq \frac{pr}{i} \right\}.$$

Then $\mathfrak{R}[p]$ is a finite subset of $[0, 1] \cap \mathbb{Q}$ containing 0 and 1. Let $f : (X, B) \rightarrow T$, m , ψ , and L be as in Definition 4.2. Assume that $m \mid p$, $B \in \Phi(\mathfrak{R})$, and $p\mathbf{M}_T$ is an integral divisor. Then $B_T \in \Phi(\mathfrak{R}[p])$.

In particular, if $I = \Phi(\mathfrak{R})$ in Theorem 4.7 and p is as in Theorem 4.7, then we may take $\Lambda := \Phi(\mathfrak{R}[p])$.

Proof. If $r \in \mathfrak{R}$ and i, j are positive integers with $i \leq p$ and $j \leq \frac{pr}{i}$, then $r > 0$, $j \leq p$, and $0 \leq \frac{r}{j} - \frac{i}{p} \leq 1$. Thus $\mathfrak{R}[p]$ is a finite subset of $[0, 1] \cap \mathbb{Q}$, and $0, 1 \in \mathfrak{R} \subset \mathfrak{R}[p]$.

Let P be a prime divisor on T , $t := t_P$, and $\alpha := \text{mult}_P L$. Let E be a prime divisor on X such that $f(E) = P$, $a := \text{mult}_E f^*P$, and $b := \text{mult}_E B$. Then a is a positive integer, $b \in \Phi(\mathfrak{R})$, and $at + b \leq 1$ as $(X, B + tf^*P)$ is lc over the generic point of P . Since P is the only component of $\text{Supp } L$ containing $f(E)$, $\text{mult}_E f^*(mL) = ma\alpha$. Comparing the multiplicities along E of the two sides of $m(K_X + B) + \text{div}(\psi) = f^*(mL)$, we get $m(a\alpha - b) \in \mathbb{Z}$, hence $p(a\alpha - b) \in \mathbb{Z}$ as $m \mid p$. Since $p\mathbf{M}_T$ is integral and $\text{mult}_P \mathbf{M}_T = \alpha - \text{mult}_P K_T - (1 - t)$, we get $p(\alpha + t) \in \mathbb{Z}$. Thus

$$l := p(at + b) = a \cdot p(\alpha + t) - p(a\alpha - b) \in \mathbb{Z},$$

and $0 \leq l \leq p$. Write $b = 1 - \frac{r}{j}$ with $r \in \mathfrak{R}$ and j a positive integer, and let $i := p - l$. Then

$$t = \frac{1}{a} \left(\frac{r}{j} - \frac{i}{p} \right).$$

If $i = 0$, then $1 - t = 1 - \frac{r}{ja} \in \Phi(\mathfrak{R})$. If $i \geq 1$, then $\frac{r}{j} - \frac{i}{p} = at \geq 0$, so $j \leq \frac{pr}{i}$ and $\frac{r}{j} - \frac{i}{p} \in \mathfrak{R}[p]$. Thus $1 - t \in \Phi(\mathfrak{R}[p])$ in both cases, and $B_T \in \Phi(\mathfrak{R}[p])$ (cf. the proof of [CHL24, Proposition 3.1]).

Finally, assume that $I = \Phi(\mathfrak{R})$ in Theorem 4.7. Let ψ and L be as in Theorem 4.7(1) and $\tau : W \rightarrow T$ a resolution. By Theorem 4.7(3), $p\mathbf{M}_W$ is Cartier, so $p\mathbf{M}_T = \tau_*(p\mathbf{M}_W)$ is an integral divisor. Thus $B_T \in \Phi(\mathfrak{R}[p])$, and $\Phi(\mathfrak{R}[p])$ is a DCC set whose only accumulation point is 1 by [Bir19, 2.2]. $\square$

5. CANONICAL BUNDLE FORMULAS OVER FINITE QUOTIENTS

For a dlt pair (X, Δ) and an lc center Z of (X, Δ) we write (Z, Δ_Z) for the adjunction pair of Z (Definition 2.3), so $\Delta_Z = \text{Diff}_Z^* \Delta$ in the notation of [Kol16, Definition 4].

Definition 5.1 ([Kol16, Definitions 9 and 13]). A *standard $\mathbb{P}^1$ -link* is a $\mathbb{Q}$ -factorial dlt pair $(X, D_1 + D_2 + \Delta)$ whose only lc centers are D_1 and D_2 , together with a proper morphism $\pi : X \rightarrow S$ such that $K_X + D_1 + D_2 + \Delta \sim_{\mathbb{Q}, \pi} 0$ and $\pi|_{D_i} : D_i \rightarrow S$ is an isomorphism for $i = 1, 2$. Moreover, every reduced fiber of π is isomorphic to $\mathbb{P}^1$. Let $g : X \rightarrow S$ be a proper morphism from a dlt pair (X, Δ) such that $K_X + \Delta \sim_{\mathbb{Q}, g} 0$. Two lc centers Z_1, Z_2 of (X, Δ) are *directly $\mathbb{P}^1$ -linked* if there is an lc center $W \supset Z_1 \cup Z_2$ with $g(W) = g(Z_1) = g(Z_2)$ such that (W, Δ_W) is crepant birational to a standard $\mathbb{P}^1$ -link $(X', D_1 + D_2 + \Delta')$ with Z_i mapping to D_i . They are *$\mathbb{P}^1$ -linked* if there are lc centers $Z_1 = Z'_1, \dots, Z'_n = Z_2$ such that Z'_i and Z'_{i+1} are directly $\mathbb{P}^1$ -linked for every i .

Let m be a positive even integer such that $m\Delta$ is an integral divisor and the sheaf $\omega_X^{[m]}(m\Delta) := \mathcal{O}_X(m(K_X + \Delta))$ is locally free, and let Z be an lc center of (X, Δ) . Iterating the Poincaré residue maps for divisors gives the *Poincaré residue map*

$$R_{X \rightarrow Z}^m : \omega_X^{[m]}(m\Delta)|_Z \rightarrow \omega_Z^{[m]}(m\Delta_Z) := \mathcal{O}_Z(m(K_Z + \Delta_Z)),$$

which is an isomorphism. Let θ be an m -canonical form on X with $\text{Div}(\theta) = -m\Delta$. Then θ generates $\omega_X^{[m]}(m\Delta)$, and $R_{X \rightarrow Z}^m(\theta|_Z)$ is the residue θ_Z of θ on Z defined by the local formula preceding Theorem 2.15, which makes sense for any dlt pair. In particular, θ_Z generates $\omega_Z^{[m]}(m\Delta_Z)$ and $\text{Div}(\theta_Z) = -m\Delta_Z$ (cf. [OAI26d, (7.2)]).

Theorem 5.2 ([Kol16, Theorem 10 and Proposition 14]). *Let $f : (X, \Delta) \rightarrow S$ be a proper morphism from a dlt pair to a variety such that $K_X + \Delta \sim_{\mathbb{Q},f} 0$, and let $s \in S$ be a point such that $f^{-1}(s)$ is connected.*

- (1) *Let Z be minimal among the lc centers of (X, Δ) such that $s \in f(Z)$, and let W be an lc center of (X, Δ) such that $s \in f(W)$. Then there is an lc center $Z_W \subset W$ such that Z and Z_W are $\mathbb{P}^1$ -linked. In particular, all minimal lc centers Z_i of (X, Δ) with $s \in f(Z_i)$ are $\mathbb{P}^1$ -linked to each other.*
- (2) *Assume moreover that f is a crepant log structure, that m is a positive even integer such that $\omega_X^{[m]}(m\Delta) \simeq f^*\mathcal{L}$ for a line bundle $\mathcal{L}$ on S , and that Z_1, Z_2 are minimal lc centers of (X, Δ) with $f(Z_1) = f(Z_2)$. Then there is a birational map $\phi : Z_2 \dashrightarrow Z_1$ such that*

$$R_{X \rightarrow Z_2}^m(\theta|_{Z_2}) = \phi^* R_{X \rightarrow Z_1}^m(\theta|_{Z_1})$$

as rational sections of $\omega_{Z_2}^{[m]}(m\Delta_{Z_2})$, for every section θ of $f^\mathcal{L} \simeq \omega_X^{[m]}(m\Delta)$ over an open subset of X meeting Z_1 and Z_2 .*

The map ϕ in (2) is a composition of the birational maps induced by $\mathbb{P}^1$ -links; these links are obtained by an MMP over S , so ϕ is over S , and it is crepant birational for the pairs (Z_2, Δ_{Z_2}) and (Z_1, Δ_{Z_1}) [Kol16, proof of Theorem 10 and Proposition 14].

Let $\rho : P \rightarrow V$ be a finite surjective morphism of normal varieties. For a prime divisor F on V we put $\rho^*F := \sum_Q e_Q Q$, where Q runs over the prime divisors of P with $\rho(Q) = F$ and e_Q is the ramification index of ρ along Q . This is the pullback of F from the regular locus of V , whose complement has preimage of codimension at least two in P ; it extends the pullback of $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisors, and ρ^* is injective on $\mathbb{Q}$ -divisors. If ρ is the quotient by a finite group G acting faithfully on P , then G permutes the prime divisors over F transitively, as the fibers of ρ are G -orbits, and $\rho_*\rho^*F = \deg(\rho)F = |G|F$.

Lemma 5.3. *Let G be a finite group acting faithfully on a normal quasi-projective variety P with quotient $\rho : P \rightarrow V$, and let D be a G -invariant $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor on P . Then $D_V := \frac{1}{|G|}\rho_*D$ is a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor on V , and it is the unique $\mathbb{Q}$ -divisor on V with $\rho^*D_V = D$.*

Proof. Let F be a prime divisor on V and $Q_1, \dots, Q_g$ the prime divisors of P over F . Since G permutes the Q_i transitively, they have the same ramification index e and the same degree $f := [\mathbb{C}(Q_i) : \mathbb{C}(F)]$, and $efg = |G|$. Since D is G -invariant, $\text{mult}_{Q_i} D = \text{mult}_{Q_1} D$ for every i . Thus $\text{mult}_{Q_1} \rho^*\rho_*D = e \cdot gf \text{mult}_{Q_1} D = |G| \text{mult}_{Q_1} D$, so $\rho^*D_V = D$. Uniqueness follows from the injectivity of ρ^* .

Let n be a positive integer such that nD is Cartier and $v \in V$ a closed point. Since $\rho^{-1}(v)$ is finite and P is quasi-projective, there is an affine open subset $A \subset P$ containing $\rho^{-1}(v)$; the sheaf $\mathcal{O}_A(nD)$ is generated by global sections, and a general global section does not vanish at any point of $\rho^{-1}(v)$. Hence $nD = \text{div}(\varphi)$ on an open subset of P containing $\rho^{-1}(v)$ for some $\varphi \in \mathbb{C}(P)^*$. As ρ is closed, after shrinking V to a neighborhood of v we may assume that $nD = \text{div}(\varphi)$ on P . Put $N(\varphi) := \prod_{\sigma \in G} \sigma^*\varphi \in \mathbb{C}(P)^G = \mathbb{C}(V)$. Then

$$\rho^* \text{div}_V(N(\varphi)) = \text{div}_P(N(\varphi)) = \sum_{\sigma \in G} \sigma^* \text{div}_P(\varphi) = n \sum_{\sigma \in G} \sigma^* D = n|G|D = \rho^*(n|G|D_V),$$

so $n|G|D_V = \text{div}_V(N(\varphi))$ near v . Thus D_V is $\mathbb{Q}$ -Cartier. $\square$

Lemma 5.4. *Let $d \geq 2$ and b be positive integers, and put*

$$(5.1) \quad q_0 := \text{lcm}_{0 \leq j \leq d-1} q(j, b), \quad a := 2bq_0, \quad H_0 := \text{lcm}_{0 \leq j \leq d-1} L(j, a), \quad p_* := aH_0,$$

where $q(j, b)$ and $L(j, a)$ are the integers of Theorems 2.13 and 2.14. Let C be a smooth curve, $c \in C$ a closed point, $K := \mathbb{C}(C)$, and σ an automorphism of C of finite order such that $\sigma(c) = c$ and σ^ acts on $\mathfrak{m}_c/\mathfrak{m}_c^2$ by multiplication by a primitive e -th root of unity ζ . Let (F, B_F) be a normal geometrically integral projective pair over K of dimension $r \leq d-1$ which is geometrically klt, and let β be a rational relative b -canonical form on F with $\text{Div}(\beta) + bB_F = 0$.*

Let γ be an automorphism of the field $\mathbb{C}(F)$ such that $\gamma|_K = \sigma^*$ and $\gamma(\beta) = \beta$, where γ acts on $(\bigwedge^r \Omega_{\mathbb{C}(F)/K})^{\otimes b}$ through its action on $\Omega_{\mathbb{C}(F)/\mathbb{C}}$. Then

$$e \mid p_* w_z(\beta),$$

where $z \in K$ is a uniformizer at c and $w_z(\beta)$ is the weight of Definition 2.12, computed on the smooth projective curve containing C .

Proof. Let $\bar{C}$ be the smooth projective curve containing C . Then σ extends to an automorphism of $\bar{C}$ of the same order, and $w_z(\beta)$ depends only on K , c and β (Definition 2.12). Let n be the order of σ . Then $\zeta^n = 1$ and $(\sigma^*)^i z \equiv \zeta^i z \pmod{\mathfrak{m}_c^2}$, so

$$u := \frac{1}{n} \sum_{i=0}^{n-1} \zeta^{-i} (\sigma^*)^i z \in \mathcal{O}_{C,c}$$

is a uniformizer at c with $\sigma^* u = \zeta u$. Put $w := w_u(\beta) = w_z(\beta)$ and $\Omega := \beta \wedge (du/u)^{\otimes b}$. By Theorem 2.13(1), $q(r, b)w \in \mathbb{Z}$, hence $aw \in \mathbb{Z}$.

Suppose $r = 0$. Then $F = \text{Spec } K$, $B_F = 0$, $\beta \in K^*$, and $w = \text{ord}_c(\beta)/b$ by (2.4). Write $\beta = u^k v$ with $k = \text{ord}_c(\beta)$ and $v \in \mathcal{O}_{C,c}^*$. Since $\beta = \gamma(\beta) = \sigma^* \beta = \zeta^k u^k \sigma^* v$, we get $v(c) = \zeta^k v(c)$, so $e \mid k$ and $e \mid p_* w = 2q_0 H_0 k$.

Suppose $r > 0$.

Step 1. We construct a dlt pair (N_0, Γ_0) , projective over a neighborhood of c , and a rational a -canonical form θ with $\text{Div}(\theta) = -a\Gamma_0$ and $\gamma^* \theta = \zeta^{-aw} \theta$.

By Theorem 2.13(2), there are a contraction $f : N \rightarrow \bar{C}$ whose generic fiber is birational to F and a $\mathbb{Q}$ -divisor $H \geq 0$ on N horizontal over $\bar{C}$ satisfying the following. Put $T := (f^* c)_{\text{red}}$. Then $(N, T + H)$ is dlt, $K_N + T + H \sim_{\mathbb{Q}, \bar{C}} 0$, $\text{Div}(\beta) + bH_\eta = 0$ on the generic fiber N_η , and

$$\frac{1}{b} \text{Div}_N(\Omega) + T + H = w f^* c$$

on $f^{-1}(C_0)$ for some open neighborhood C_0 of c in C . Since $\text{Div}(\beta) = -bB_F = -bH_\eta$, the pairs (F, B_F) and (N_η, H_η) are crepant birational (cf. [OAI26d, Lemma 2.1]), so (N_η, H_η) is geometrically klt. As H is horizontal, all coefficients of H are less than one. Shrinking C_0 , we may assume that C_0 is σ -stable and that u has neither zeros nor poles on $C_0 \setminus \{c\}$. Put $N_0 := f^{-1}(C_0)$, $\Gamma_0 := (T + H)|_{N_0}$, and

$$\theta := u^{-aw} \Omega^{\otimes a/b}.$$

Then θ is a rational a -canonical form on N_0 , and $\text{Div}(\theta) = -aw f^* c + a(w f^* c - T - H) = -a\Gamma_0$ on N_0 . The pair (N_0, Γ_0) is dlt with $[\Gamma_0] = T$ and $K_{N_0} + \Gamma_0 \sim_{\mathbb{Q}, C_0} 0$.

The automorphism γ of $\mathbb{C}(F) = \mathbb{C}(N)$ defines a birational self-map of N_0 lying over $\sigma|_{C_0}$, which we also denote by γ . Since $\gamma(\beta) = \beta$ and $\sigma^*(du/u) = du/u$, we have $\gamma^* \Omega = \Omega$ and

$$(5.2) \quad \gamma^* \theta = \lambda \theta, \quad \lambda := \zeta^{-aw} \in \mathbb{C}^*.$$

Step 2. We find a projective klt pair (A, Γ_A) of dimension at most r with $a(K_A + \Gamma_A) \sim 0$ and a crepant birational self-map g of (A, Γ_A) with $g^* \theta_A = \lambda \theta_A$, where $\text{Div}(\theta_A) = -a\Gamma_A$.

Every lc center of (N_0, Γ_0) is contained in $[\Gamma_0] = T$, hence is projective. Let $k \geq 1$ be the largest codimension of an lc center of (N_0, Γ_0) and A an lc center of codimension k . Then A is a minimal lc center, and (A, Γ_A) is a projective klt pair, since its lc centers correspond to the lc centers of (N_0, Γ_0) properly contained in A (cf. [Kol16, Definition 18]). Its residue θ_A satisfies $\text{Div}(\theta_A) = -a\Gamma_A$ (Definition 5.1); in particular $a(K_A + \Gamma_A) \sim 0$. At the generic point of A the pair (N_0, Γ_0) is log smooth, and A is a component of the intersection of exactly k components of T .

Since $f \circ \gamma = \sigma \circ f$, the graph of γ lies in $N_0 \times_{C_0} N_0$, where the second factor is viewed over C_0 through $\sigma^{-1} \circ f$. Since f is projective, the closure of the graph is projective over N_0 by either projection. Hence there are a smooth variety W and projective birational morphisms

$p, q : W \rightarrow N_0$ with $q = \gamma \circ p$. We choose W as follows. Let $W_1 \rightarrow N_0$ be a log resolution of (N_0, Γ_0) which is an isomorphism over the log smooth locus of (N_0, Γ_0) (Szabó's resolution lemma; standard). Let $W \rightarrow W_1$ be a composition of blow-ups of smooth centers having simple normal crossings with the accumulated boundary (Hironaka; standard) such that $q := \gamma \circ p$ is a morphism and $\text{Supp } \Gamma_W \cup \text{Exc}(p) \cup \text{Exc}(q)$ is a simple normal crossing divisor, where

$$K_W + \Gamma_W := p^*(K_{N_0} + \Gamma_0).$$

The $\mathbb{Q}$ -divisor Γ_W need not be effective; we call (W, Γ_W) a *sub-pair*, and it is log smooth. By (5.2), $\text{Div}_W(q^*\theta) = \text{Div}_W(p^*\theta) = -a\Gamma_W$, so γ is crepant and $K_W + \Gamma_W = q^*(K_{N_0} + \Gamma_0)$ as well, the canonical divisors being chosen compatibly.

We claim that there is an irreducible component Z of an intersection of k coefficient-one components of Γ_W with $p(Z) = A$. The strict transform of A on W_1 is such a component. A blow-up whose center does not contain the generic point of the current Z does not change Z near its generic point. Suppose that the center contains the generic point of Z . Since the center has simple normal crossings with the boundary and any further equation would cut Z properly, it is locally the intersection of h of the k coefficient-one components through Z . If $h = 1$, the blow-up is an isomorphism near Z . If $h \geq 2$, the exceptional divisor has log discrepancy $h - h = 0$, hence coefficient one, and in each chart it meets the strict transforms of the other $k - 1$ components in a codimension- k stratum dominating Z . This proves the claim by induction (cf. [OAI26d, proof of Lemma 7.2]).

Blowing up Z (or taking Z itself if $k = 1$) gives an lc place of (W, Γ_W) , hence of (N_0, Γ_0) with respect to both p and q . Thus $A' := q(Z)$ is an lc center of (N_0, Γ_0) with $\dim A' \leq \dim Z = \dim A$. By the maximality of k , A' is a minimal lc center of codimension k , with projective klt adjunction pair $(A', \Gamma_{A'})$, and $q|_Z : Z \rightarrow A'$ and $p|_Z : Z \rightarrow A$ are generically finite and dominant.

We show that $q|_Z$ is birational and that

$$(5.3) \quad (q^*\theta)_Z = (q|_Z)^*\theta_{A'},$$

where $(q^*\theta)_Z$ is the residue of $q^*\theta$ along Z , defined by the same local formula as in Theorem 2.15 at the generic point of the log smooth sub-pair (W, Γ_W) . At the generic points of A' and Z choose regular parameters $x_1, \dots, x_k$ and $z_1, \dots, z_k$ of the coefficient-one components through A' and Z , completed by coordinates $y_1, \dots, y_{n-k}$ and $w_1, \dots, w_{n-k}$ along A' and Z , where $n := \dim N_0$. No other component of $\text{Supp } \Gamma_0$, respectively of $\text{Supp } \Gamma_W \cup \text{Exc}(q)$, contains A' , respectively Z , by the simple normal crossing property. Hence $\theta = g(dx_1/x_1 \wedge \dots \wedge dx_k/x_k \wedge dy_1 \wedge \dots \wedge dy_{n-k})^{\otimes a}$ with g a unit at the generic point of A' , $\theta_{A'} = g|_{A'}(dy_1 \wedge \dots \wedge dy_{n-k})^{\otimes a}$, and $q^*x_i = u_i \prod_j z_j^{e_{ij}}$ with u_i units at the generic point of Z and e_{ij} non-negative integers. Let M be the matrix expressing $q^*(dx_i/x_i)$ and q^*dy_i in the basis $dz_j/z_j, dw_i$. Modulo the ideal of Z , M is block triangular with diagonal blocks $E := (e_{ij})$ and the Jacobian matrix J of $q|_Z$. Since $\text{Div}_W(q^*\theta) = -a\Gamma_W$, the coefficient $(g \circ q)(\det M)^a$ of $q^*\theta$ in the basis $(dz_1/z_1 \wedge \dots \wedge dz_k/z_k \wedge dw_1 \wedge \dots \wedge dw_{n-k})^{\otimes a}$ is a unit at the generic point of Z , so $\det E \neq 0$, $\det J \neq 0$, and

$$(q^*\theta)_Z = (\det E)^a (q|_Z)^*\theta_{A'}.$$

Choose positive real numbers $\alpha_1, \dots, \alpha_k$ linearly independent over $\mathbb{Q}$ and let v be the monomial valuation of $\mathbb{C}(W) = \mathbb{C}(N_0)$ at the generic point of Z with $v(z_j) = \alpha_j$. Let R be a regular local ring and v' a real valuation centered on R whose values on a regular system of parameters of R are positive and $\mathbb{Q}$ -linearly independent. Then the value group of v' is generated by these values, and the residue field of v' is that of R [OAI26d, (7.6)]. Applied to $\mathcal{O}_{W, \eta_Z}$, this gives value group $\sum_j \mathbb{Z}\alpha_j$ and residue field $\mathbb{C}(Z)$. The values $v(x_i) = \sum_j e_{ij}\alpha_j$ are positive and $\mathbb{Q}$ -linearly independent because $\det E \neq 0$, so the same statement applied to $\mathcal{O}_{N_0, \eta_{A'}}$ gives value group $\sum_i \mathbb{Z}v(x_i)$ and residue field $\mathbb{C}(A')$. Comparing, E is unimodular and $\mathbb{C}(Z) = \mathbb{C}(A')$. Thus $q|_Z$ is birational and $(\det E)^a = 1$ as a is even, which proves (5.3). The same argument shows that $p|_Z$ is birational and $(p^*\theta)_Z = (p|_Z)^*\theta_A$.

Put $\psi := q|_Z \circ (p|_Z)^{-1} : A \dashrightarrow A'$. By (5.2) and (5.3), $\psi^*\theta_{A'} = \lambda\theta_A$. Now $f_0 : (N_0, \Gamma_0) \rightarrow C_0$ is a dlt crepant log structure, $\omega_{N_0}^{[a]}(a\Gamma_0)$ is trivialized by θ , and A, A' are minimal lc centers with $f(A) = f(A') = \{c\}$. Theorem 5.2(2) gives a birational map $\phi : A \dashrightarrow A'$ with $\phi^*\theta_{A'} = \theta_A$. Then $g := \phi^{-1} \circ \psi$ is a birational self-map of A with $g^*\theta_A = \lambda\theta_A$, and g is crepant for (A, Γ_A) by [OAI26d, Lemma 2.1]. Since $\dim A \leq r \leq d-1$, Theorem 2.14 gives $\lambda^{L(\dim A, a)} = 1$, and $L(\dim A, a) \mid H_0$. Thus $\zeta^{-awH_0} = 1$, so $e \mid awH_0 = p_*w$. This finishes the proof. $\square$

Theorem 5.5. *Let $d \geq 2$ and m be positive integers, and put*

$$(5.4) \quad \begin{aligned} E_0 &:= \text{lcm}_{0 \leq j \leq d-1} L(j, m), & b &:= mE_0, \\ q_0 &:= \text{lcm}_{0 \leq j \leq d-1} q(j, b), & a &:= 2bq_0, \\ H_0 &:= \text{lcm}_{0 \leq j \leq d-1} L(j, a), & p_v &:= aH_0, \end{aligned}$$

where $q(j, b)$ and $L(j, \cdot)$ are the integers of Theorems 2.13 and 2.14. Let $h : (X, B) \rightarrow P$ be a crepant log structure such that h is a contraction of normal quasi-projective varieties, $0 < \dim P \leq \dim X \leq d$, B is a $\mathbb{Q}$ -divisor, and the generic fiber (X_η, B_η) of h is geometrically klt over $\mathbb{C}(P)$. Let $\psi \in \mathbb{C}(X)$ be a nonzero rational function and D a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor on P such that

$$(5.5) \quad K_X + B + \frac{1}{m} \text{div}(\psi) = h^*D,$$

and let B_P and $\mathbf{M}$ be the discriminant part and the moduli part of (5.5) (Definition 4.2). Let G be a finite group acting faithfully on P with quotient $\rho : P \rightarrow V$. Assume that every $\sigma \in G$ lifts to a crepant birational self-map of (X, B) over σ , that is, there is a crepant birational self-map $\tilde{\sigma}$ of (X, B) with $h \circ \tilde{\sigma} = \sigma \circ h$. Let $V \rightarrow S$ be a projective morphism to a normal quasi-projective variety. Let ω_V be a rational top form on V , $\omega_P := \rho^*\omega_V$, and ω_X a rational top form on X , with canonical divisors K_V , K_P and K_X . Then the following hold.

(1) *There exists $\chi \in \mathbb{C}(P)^*$ such that the rational b -canonical form $\theta_0 := (\chi \circ h)\psi^{E_0}\omega_X^{\otimes b}$ satisfies*

$$(5.6) \quad \text{Div}(\theta_0) + bB = h^*(bD + \text{div}(\chi))$$

and $\tilde{\sigma}^*\theta_0 = \theta_0$ for every crepant birational self-map $\tilde{\sigma}$ of (X, B) lying over an element of G .

(2) *Put $D_\beta := D + \frac{1}{b} \text{div}(\chi)$ and let $\mathbf{M}_\beta$ be the moduli part of (5.6), so that $\mathbf{M}_{\beta, P'} = \mathbf{M}_{P'} + \frac{1}{b} \text{div}_{P'}(\chi)$ for every birational model P' of P . Let U be a normal variety with a projective birational morphism $\tau : U \rightarrow V$, $\rho_U : P_U \rightarrow U$ the normalization of U in $\mathbb{C}(P)$, $\tau_U : P_U \rightarrow P$ the induced morphism, $K_U := \text{Div}(\tau^*\omega_V)$, and $K_{P_U} := \text{Div}(\tau_U^*\omega_P)$. There exist a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor D_V on V , a b - $\mathbb{Q}$ -Cartier b -divisor $\mathbf{M}^V$ over V , and, for every such U , a $\mathbb{Q}$ -divisor B_U on U , such that $B_V \geq 0$ and*

$$(5.7) \quad D_\beta = \rho^*D_V,$$

$$(5.8) \quad \mathbf{M}_{\beta, P_U} = \rho_U^*\mathbf{M}_U^V,$$

$$(5.9) \quad K_{P_U} + B_{P_U} = \rho_U^*(K_U + B_U),$$

$$(5.10) \quad \tau^*D_V = K_U + B_U + \mathbf{M}_U^V$$

for every such U . If $\mathbf{M}^V$ descends to U , then B_U is the $\mathbb{Q}$ -divisor defined by (2.1) for $(V, B_V + \mathbf{M}^V)$. For $U = V$, (5.9) and (5.10) read $K_P + B_P = \rho^*(K_V + B_V)$ and $D_V = K_V + B_V + \mathbf{M}_V^V$.

(3) *$(V, B_V + \mathbf{M}^V)/S$ is a generalized lc pair. For every U as in (2), every prime divisor F on U , and every prime divisor Q on P_U over F , with ramification index e ,*

$$(5.11) \quad 1 - \text{mult}_Q B_{P_U} = e(1 - \text{mult}_F B_U).$$

- (4) For every smooth variety U projective and birational over V , $p_v \mathbf{M}_U^V$ is a Cartier divisor, and $\mathbf{M}_U^V$ is nef over S if $\mathbf{M}^V$ descends to U .

Proof. Put $K := \mathbb{C}(P)$, $K_0 := \mathbb{C}(V) = K^G$, and $r := \dim X - \dim P$. Let $\omega_{X/P}$ be the rational relative top form with $\omega_X = \omega_{X/P} \wedge h^* \omega_P$, and put $\alpha := \psi \omega_{X/P}^{\otimes m}$, a rational relative m -canonical form on X_η . By (5.5), $\text{Div}_{X_\eta}(\alpha) = -mB_\eta$.

Step 1. We construct χ and prove (1).

Let $\mathcal{A}$ be the set of crepant birational self-maps of (X, B) lying over elements of G . Then $\mathcal{A}$ is a group, and sending $\tilde{\sigma}$ to σ defines a homomorphism $\mathcal{A} \rightarrow G$, which is surjective by assumption and whose kernel is $\mathcal{A}_0 := \text{Bir}_P^c(X, B)$.

Let $\tilde{\sigma} \in \mathcal{A}$ lie over σ and let $p, q : W \rightarrow X$ be a common resolution of $\tilde{\sigma}$, so that $q = \tilde{\sigma} \circ p$. Since $\tilde{\sigma}$ is crepant, $p^*(K_X + B) = q^*(K_X + B)$ on W , the canonical divisors being chosen compatibly with a fixed K_W (Definition 2.8); write $K_W + \Gamma_W$ for this divisor. For every rational n -canonical form θ' on X , unwinding the definitions gives

$$(5.12) \quad \text{Div}_W(p^*\theta') = p^*(\text{Div}_X(\theta') + nB) - n\Gamma_W, \quad \text{Div}_W(q^*\theta') = q^*(\text{Div}_X(\theta') + nB) - n\Gamma_W.$$

Suppose that $\text{Div}_X(\theta') + nB = h^*(nD')$ for a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor D' on P . Since $h \circ q = \sigma \circ h \circ p$, (5.12) gives $\text{div}_W(q^*\theta'/p^*\theta') = (h \circ p)^*(n\sigma^*D' - nD')$. Thus the rational function $q^*\theta'/p^*\theta'$ has zero divisor on the generic fiber of $h \circ p$, which is a normal projective geometrically integral variety over K , so it is the pullback of some $c \in K^*$. Hence

$$(5.13) \quad \tilde{\sigma}^*\theta' = (c \circ h)\theta' \quad \text{and} \quad \sigma^*D' = D' + \frac{1}{n} \text{div}(c).$$

The rational relative m -canonical forms α' on X_η with $\text{Div}_{X_\eta}(\alpha') = -mB_\eta$, together with 0, form a one-dimensional K -vector space $\mathcal{L}$. Indeed, the ratio α'/α is a rational function with zero divisor on the normal projective variety X_η , hence lies in $H^0(X_\eta, \mathcal{O}_{X_\eta})^* = K^*$ (cf. Lemma 4.1). Applying (5.13) to $\theta := \psi \omega_X^{\otimes m} = \alpha \wedge (h^*\omega_P)^{\otimes m}$, with $n = m$ and $D' = D$, and using $\sigma^*\omega_P = \omega_P$, we get $\tilde{\sigma}^*\alpha = c\alpha$ with $c \in K^*$. Thus $\mathcal{A}$ acts on $\mathcal{L}$ and on $\mathcal{L}^{\otimes E_0}$ by pullback, and $\tilde{\sigma}^*(x\alpha') = \sigma^*(x)\tilde{\sigma}^*\alpha'$ for $x \in K$.

We claim that $\mathcal{A}_0$ acts trivially on $\mathcal{L}^{\otimes E_0}$. Let $g \in \mathcal{A}_0$ and $g^*\alpha = c\alpha$ with $c \in K^*$; it suffices to show $c^{L(r,m)} = 1$. If $r = 0$, then $\mathcal{L} = K$ and $c = 1$. Let $r > 0$. Let $p, q : W \rightarrow X$ be a common resolution of g over P such that W is smooth and $\Sigma := \text{Supp } \Gamma_W \cup \text{Exc}(p) \cup \text{Exc}(q)$ is a simple normal crossing divisor, where $K_W + \Gamma_W := p^*(K_X + B) = q^*(K_X + B)$. Let $P^0 \subset P$ be a nonempty open subset satisfying the following. The variety P^0 is smooth, ω_P generates ω_{P^0} , $c \in \mathcal{O}(P^0)^*$, and $P^0 \cap \text{Supp } D = \emptyset$. The morphism h is flat over P^0 with geometrically normal and geometrically integral fibers. The variety W and all strata of Σ are smooth over P^0 , and Σ and B are horizontal over P^0 . Such P^0 exists by generic flatness, generic smoothness, and the openness of geometric normality and geometric integrality in flat proper families, since the generic fiber of h is normal and geometrically integral (cf. [OAI26c, proofs of Proposition 7.1 and Lemma 7.3] and the proof of Lemma 4.6). Let $t \in P^0$ be a point, closed or not, and let $X_t, W_t, B_t, \Gamma_{W,t}$ and Σ_t be the fibers over $\text{Spec } \mathbb{C}(t)$ and the restrictions to them. Then X_t is a normal geometrically integral projective variety of dimension r over $\mathbb{C}(t)$, W_t is smooth and geometrically integral, Σ_t is a simple normal crossing divisor, and $p_t, q_t : W_t \rightarrow X_t$ are birational. The identity $K_W + \Gamma_W = p^*(K_X + B) = q^*(K_X + B)$ restricts to $K_{W_t} + \Gamma_{W,t} = p_t^*(K_{X_t} + B_t) = q_t^*(K_{X_t} + B_t)$ (cf. [OAI26c, proof of Proposition 7.1]). Since (X_η, B_η) is geometrically klt and Σ is horizontal over P^0 , the coefficients of $\Gamma_{W,t}$ are less than one, so (X_t, B_t) is geometrically klt. Let θ_t be the restriction to X_t of $\psi \omega_{X/P}^{\otimes m}$. Since $\text{Div}(\theta) = -mB$ over P^0 by (5.5) and ω_P generates ω_{P^0} , $\text{Div}_{X_t}(\theta_t) = -mB_t$. Restricting $q^*\theta = (c \circ h \circ p)p^*\theta$ to W_t shows that $g_t := q_t \circ p_t^{-1}$ satisfies $g_t^*\theta_t = c(t)\theta_t$. Let t be a closed point. By [OAI26d, Lemma 2.1], g_t is a crepant birational self-map of the projective klt pair (X_t, B_t) , so Theorem 2.14 gives $c(t)^{L(r,m)} = 1$. As this holds for every closed point $t \in P^0$, $c^{L(r,m)} = 1$. Since $L(r,m) \mid E_0$, the claim follows.

Consequently the action of $\mathcal{A}$ on $\mathcal{L}^{\otimes E_0}$ factors through G : for $\sigma \in G$ there is $c_\sigma \in K^*$ with $\tilde{\sigma}^*v = c_\sigma v$ for every lift $\tilde{\sigma}$ of σ , where $v := \alpha^{\otimes E_0}$. As $(\tilde{\sigma}\tilde{\tau})^* = \tilde{\tau}^*\tilde{\sigma}^*$, we have $c_{\sigma\tau} = \tau^*(c_\sigma)c_\tau$. The automorphisms τ^* ($\tau \in G$) of K over K_0 are linearly independent over K , so there is $x \in K$ with

$$\chi := \sum_{\tau \in G} c_\tau \tau^*(x) \neq 0,$$

and $\sigma^*(\chi)c_\sigma = \sum_{\tau} \sigma^*(c_\tau)c_\sigma(\tau\sigma)^*(x) = \sum_{\tau} c_{\tau\sigma}(\tau\sigma)^*(x) = \chi$ for every $\sigma \in G$. Put $\beta := \chi v = \chi \alpha^{\otimes E_0}$. Then $\tilde{\sigma}^*\beta = \sigma^*(\chi)c_\sigma v = \beta$ for every $\tilde{\sigma} \in \mathcal{A}$, and $\theta_0 = \beta \wedge (h^*\omega_P)^{\otimes b}$. Hence $\tilde{\sigma}^*\theta_0 = \theta_0$ for every $\tilde{\sigma} \in \mathcal{A}$, and

$$\begin{aligned} \text{Div}(\theta_0) &= h^*\text{div}(\chi) + E_0\text{div}(\psi) + bK_X \\ &= h^*\text{div}(\chi) + E_0(h^*(mD) - m(K_X + B)) + bK_X = h^*(bD + \text{div}(\chi)) - bB. \end{aligned}$$

This proves (1).

Step 2. We prove (2) and (3), and the nefness in (4).

Applying (5.13) to θ_0 , with $n = b$, $D' = D_\beta$ and $c = 1$, gives $\sigma^*D_\beta = D_\beta$ for every $\sigma \in G$. Lemma 5.3 gives D_V with (5.7).

Let U , τ , ρ_U , τ_U , K_U and K_{P_U} be as in (2). Then G acts faithfully on P_U over U , and $U = P_U/G$ since ρ_U is finite and U is normal. The morphism τ_U is projective and birational. Let t_Q be the threshold at a prime divisor Q of P_U of the sub-lc crepant pullback over P_U (Definition 4.2), so $\text{mult}_Q B_{P_U} = 1 - t_Q$. Every $\tilde{\sigma} \in \mathcal{A}$ is a crepant birational self-map over the automorphism σ_U of P_U induced by σ . We claim that $t_{\sigma_U(Q)} = t_Q$. Let $p, q : W \rightarrow X$ be a common resolution of $\tilde{\sigma}$ such that $h \circ p$ lifts to a morphism $h_W : W \rightarrow P_U$, and $K_W + \Gamma_W := p^*(K_X + B) = q^*(K_X + B)$. Since $h \circ q = \sigma \circ h \circ p$, the morphism $h \circ q$ lifts to $\sigma_U \circ h_W$, and $(\sigma_U \circ h_W)^*\sigma_U(Q) = h_W^*Q$. By Definition 4.2, t_Q and $t_{\sigma_U(Q)}$ are both computed on the sub-pair (W, Γ_W) with these structure morphisms, which proves the claim. Hence B_{P_U} , K_{P_U} and $\mathbf{M}_{\beta, P_U} = \tau_U^*D_\beta - K_{P_U} - B_{P_U}$ are G -invariant. For a prime divisor F on U choose a prime divisor Q of P_U over F , with ramification index e_Q , and put

$$\text{mult}_F B_U := 1 - \frac{1 - \text{mult}_Q B_{P_U}}{e_Q};$$

this does not depend on Q since G permutes the prime divisors over F transitively. At the generic point of Q , write $v = u^{e_Q}w$ with u and v local equations of Q and F and w a unit. Then $\rho_U^*dv = u^{e_Q-1}(e_Q w du + u dw)$, so $\text{mult}_Q K_{P_U} = e_Q \text{mult}_F K_U + e_Q - 1$ (the Hurwitz formula). Hence

$$K_{P_U} + B_{P_U} = \rho_U^*(K_U + B_U).$$

Put $\mathbf{M}_U^V := \tau^*D_V - K_U - B_U$. Then $\rho_U^*\mathbf{M}_U^V = \tau_U^*D_\beta - K_{P_U} - B_{P_U} = \mathbf{M}_{\beta, P_U}$. Let $U' \rightarrow U$ be a birational morphism of such models. The thresholds t_Q and the ramification indices of the prime divisors not contracted by $P_{U'} \rightarrow P_U$ are the same on $P_{U'}$ and P_U . Thus $B_{U'}$ pushes forward to B_U , and the divisors $\mathbf{M}_{U'}^V$ form a b-divisor $\mathbf{M}^V$ over V . This gives (5.8), (5.9), (5.10) and (5.11) for every U . By Definition 4.2, $0 \leq t_Q \leq 1$ for every prime divisor Q on P , so $B_P \in [0, 1]$ and $B_V \in [0, 1]$ by (5.11). In particular $B_V \geq 0$.

We show that $\mathbf{M}^V$ is b- $\mathbb{Q}$ -Cartier. By Theorem 4.3(2), applied to (5.6) and the projective morphism $P \rightarrow S$, and after resolving the model obtained there, $\mathbf{M}_\beta$ descends to a smooth variety P' projective and birational over P with $\mathbf{M}_{\beta, P'}$ nef over S . By [Har77, Theorem II.7.17], P' is the blow-up of P along a coherent ideal sheaf $\mathcal{I}$. Let P'' be the normalization of the blow-up of P along the G -invariant ideal sheaf $\prod_{\sigma \in G} \sigma^*\mathcal{I}$. Then G acts on P'' , $P'' \rightarrow P$ is projective and birational, and it factors through P' by [Har77, Proposition II.7.14], because $\mathcal{I}\mathcal{O}_{P''}$ divides the invertible ideal $\prod_{\sigma} \sigma^*\mathcal{I} \cdot \mathcal{O}_{P''}$ on the integral scheme P'' and is therefore invertible. Since P'' is quasi-projective, every G -orbit in P'' lies in an affine open subset, so the quotient P''/G exists [SGA1, Exposé V, Proposition 1.8]; it is a normal variety, proper and birational over V . By Chow's lemma [Har77, Exercise II.4.10] and normalization, there is a

normal variety U_0 projective and birational over P''/G , hence over V , and P_{U_0} dominates P'' and P' . Let U be normal, projective and birational over U_0 . Then $\mathbf{M}_{\beta, P_U}$ is the pullback of $\mathbf{M}_{\beta, P'}$, hence $\mathbb{Q}$ -Cartier and G -invariant, so by Lemma 5.3 and the injectivity of ρ_U^* , $\mathbf{M}_U^V$ is $\mathbb{Q}$ -Cartier. For a birational morphism $\pi : U' \rightarrow U$ of such models, with $\pi_P : P_{U'} \rightarrow P_U$, we get $\rho_{U'}^* \mathbf{M}_{U'}^V = \mathbf{M}_{\beta, P_{U'}} = \pi_P^* \mathbf{M}_{\beta, P_U} = \rho_U^* \pi^* \mathbf{M}_U^V$, so $\mathbf{M}_{U'}^V = \pi^* \mathbf{M}_U^V$. Thus $\mathbf{M}^V$ descends to U_0 .

Let U be smooth, projective and birational over V , such that $\mathbf{M}^V$ descends to U , and let U_1 be a smooth variety projective and birational over both U and U_0 . Then $\mathbf{M}_{U_1}^V$ is the pullback of $\mathbf{M}_U^V$, and $\rho_{U_1}^* \mathbf{M}_{U_1}^V = \mathbf{M}_{\beta, P_{U_1}}$ is the pullback of $\mathbf{M}_{\beta, P'}$, hence nef over S . For a curve $\Sigma \subset U_1$ contracted over S , choose a curve $\Sigma' \subset P_{U_1}$ mapping finitely onto Σ ; then $\deg(\Sigma' \rightarrow \Sigma)(\mathbf{M}_{U_1}^V \cdot \Sigma) = \mathbf{M}_{\beta, P_{U_1}} \cdot \Sigma \geq 0$. Thus $\mathbf{M}_{U_1}^V$ is nef over S , and so is $\mathbf{M}_U^V$. In particular $(V, B_V + \mathbf{M}_U^V)/S$ is a generalized pair, with $K_V + B_V + \mathbf{M}_U^V = D_V$ $\mathbb{Q}$ -Cartier. On every U to which $\mathbf{M}^V$ descends, $K_U + B_U + \mathbf{M}_U^V = \tau^* D_V$ by construction, so B_U is the divisor defined by (2.1), and $\text{mult}_F B_U \leq 1$ because $\text{mult}_Q B_{P_U} = 1 - t_Q \leq 1$. Hence $(V, B_V + \mathbf{M}^V)/S$ is generalized lc.

Step 3. We show that $p_v \mathbf{M}_U^V$ is Cartier for every smooth variety U projective and birational over V .

Let F be a prime divisor on U and Q a prime divisor of P_U over F , with ramification index e . By (5.8), $\text{mult}_Q \mathbf{M}_{\beta, P_U} = e \text{mult}_F \mathbf{M}_U^V$. Since U is smooth, it suffices to show

$$(5.14) \quad p_v \text{mult}_F \mathbf{M}_U^V = \frac{p_v \text{mult}_Q \mathbf{M}_{\beta, P_U}}{e} \in \mathbb{Z}.$$

Let $I \subset G$ be the inertia group of Q , that is, the group of elements of G fixing Q pointwise. It is cyclic of order e , generated by an element σ acting on the conormal space of Q at its generic point by a primitive e -th root of unity ζ (standard). Put $k := \dim P$. Let $\sigma' \in I \setminus \{1\}$. The fixed locus of σ' is smooth at a general point of Q , and its tangent space there is the fixed subspace of the tangent space of P_U (standard). Since σ' acts on the conormal space of Q by a nontrivial power of ζ , the fixed locus of σ' coincides with Q near a general point of Q . Let $P_U^1 \subset P_U$ be the complement of the I -stable closed subsets $\text{Sing}(P_U)$, $\text{Sing}(Q)$, and the closure of $\text{Fix}(\sigma') \setminus Q$ for $\sigma' \in I \setminus \{1\}$; none of them contains Q . Let $A \subset P_U^1$ be an affine open subset meeting Q , and $P^\circ := \bigcap_{\sigma' \in I} \sigma'(A)$, which is affine as P_U is separated. Then P° is a smooth I -stable affine open subset meeting Q , and every element of $I \setminus \{1\}$ has fixed locus $Q^\circ := Q \cap P^\circ$. Let $\pi : P^\circ \rightarrow Y := P^\circ/I$ be the quotient, so $Y = \text{Spec } \mathcal{O}(P^\circ)^I$. At a point of Q° , averaging local coordinates over I gives a regular system of parameters $u, t_2, \dots, t_k$ of P° with u a local equation of Q , $\sigma^* u = \zeta u$ and $\sigma^* t_i = t_i$. Hence Y is smooth with local parameters $u^e, t_2, \dots, t_k$, and π is a cyclic cover of degree e which is étale outside Q° and totally ramified along Q° .

Let $\mu : \tilde{X} \rightarrow X$ be a projective birational morphism from a smooth variety such that $g : \tilde{X} \rightarrow P_U$, $\tau_U \circ g = h \circ \mu$, is a morphism, and such that, writing $K_{\tilde{X}} + B^c := \mu^*(K_X + B)$ and $\tilde{\theta}_0 := \mu^* \theta_0$, the divisor $\text{Supp } B^c \cup \text{Supp } g^* Q \cup \text{Supp } \text{Div}(\tilde{\theta}_0) \cup \text{Exc}(\mu)$ has simple normal crossings. As in (5.12),

$$\text{Div}(\tilde{\theta}_0) + bB^c = g^*(bD_{P_U}), \quad D_{P_U} := \tau_U^* D_\beta.$$

Let $\tilde{\sigma} \in \mathcal{A}$ be a lift of σ and $p, q : \tilde{W} \rightarrow \tilde{X}$ a common resolution of $\tilde{\sigma}$ with $\tilde{W}$ smooth, so that $g \circ q = \sigma_U \circ g \circ p$. The morphism g has connected fibers, because K is algebraically closed in $\mathbb{C}(X)$ and P_U is normal; the same holds for $g \circ p$.

Let $\bar{Y}$ be a projective closure of Y . If $k \geq 2$, let $L_1, \dots, L_{k-1}$ be general members of a very ample linear system on $\bar{Y}$, and let C_Y be a connected component of $L_1 \cap \dots \cap L_{k-1} \cap Y$ meeting $\pi(Q^\circ)$; if $k = 1$, put $C_Y := Y$. We apply Bertini's theorem [Kle98, Theorem (4.1)] to the linear systems $|L_j|$ pulled back to the smooth varieties $Y, P^\circ, \tilde{X}^\circ := g^{-1}(P^\circ)$ and $\tilde{W}^\circ := (g \circ p)^{-1}(P^\circ)$, and to the strata of $\tilde{X}^\circ$ with respect to the marked divisors above. These pullbacks are base point free, and the following hold. The curve C_Y is smooth and meets the divisor $\pi(Q^\circ)$ transversally at finitely many points. These points, and the generic point of C_Y , avoid any proper closed subset of $\pi(Q^\circ)$, respectively of Y , chosen in advance, since $L_1 \cap \dots \cap L_{k-1}$ meets a closed subset

of $\bar{Y}$ of dimension at most $k-1$ in finitely many points. The curve $C := \pi^{-1}(C_Y)$ is smooth and I -stable, and it is irreducible because it is smooth and has a single point over any point of $C_Y \cap \pi(Q^\circ)$. The varieties $\tilde{X}_C := g^{-1}(C)$ and $\tilde{W}_C := (g \circ p)^{-1}(C)$ are smooth and irreducible, the restrictions to $\tilde{X}_C$ of the marked divisors have simple normal crossings, and the restrictions $p_C, q_C : \tilde{W}_C \rightarrow \tilde{X}_C$ are projective birational, being surjective with connected fibers between varieties of the same dimension. Put $g_C := g|_{\tilde{X}_C} : \tilde{X}_C \rightarrow C$.

Fix a point $c \in C \cap Q$. Then $\sigma|_C$ has order e , fixes c , and acts on $\mathfrak{m}_c/\mathfrak{m}_c^2$ by ζ , since C is transverse to Q at c . Let η_C be the generic point of C . Since η_C is a general point of P° , we may assume that τ_U is an isomorphism near η_C , that χ is regular and invertible at $\eta := \tau_U(\eta_C)$, and that $\eta \in P^0$, where P^0 is the open subset of Step 1. Thus the generic fiber of g_C is birational to the fiber $F_C := X_\eta$ of h over η . By the choice of P^0 , (F_C, B_C) , $B_C := B|_{F_C}$, is a normal geometrically integral projective geometrically klt pair of dimension $r \leq d-1$ over $\mathbb{C}(\eta) = \mathbb{C}(C)$. Put

$$\tilde{\beta} := \chi \psi^{E_0} \omega_{\tilde{X}/P}^{\otimes b}, \quad \text{so that} \quad \tilde{\theta}_0 = \tilde{\beta} \wedge (g^* \omega_{P_U})^{\otimes b}.$$

The relative form $\tilde{\beta}$ restricts to a rational relative b -canonical form β_C on F_C with $\text{Div}(\beta_C) + bB_C = 0$.

Since $\tilde{\sigma}^* \theta_0 = \theta_0$ and $\sigma_U^* \omega_{P_U} = \omega_{P_U}$, we have $q^* \tilde{\beta} = p^* \tilde{\beta}$ as relative forms on $\tilde{W}$ over P_U . Restricting to $\tilde{W}_C$ gives $q_C^* \beta_C = p_C^* \beta_C$. Hence the birational self-map $q_C \circ p_C^{-1}$ of $\tilde{X}_C$ lies over $\sigma|_C$ and defines an automorphism γ of $\mathbb{C}(F_C) = \mathbb{C}(\tilde{X}_C)$ with $\gamma|_{\mathbb{C}(C)} = \sigma^*$ and $\gamma(\beta_C) = \beta_C$.

We compute the weight of β_C at c . Let $u \in \mathcal{O}_{P_U, c}$ be a local equation of Q and $\ell_1, \dots, \ell_{k-1}$ local equations at c of $\pi^* L_1, \dots, \pi^* L_{k-1}$. Then $\ell_1, \dots, \ell_{k-1}, u$ are local coordinates at c , $u|_C$ is a uniformizer at c , and $\omega_{P_U} = \varpi d\ell_1 \wedge \dots \wedge d\ell_{k-1} \wedge du$ near c with $\text{div}(\varpi) = K_{P_U}$ near c . Since c is a general point of Q , no component of $\text{Supp } K_{P_U} \cup \text{Supp } D_{P_U}$ other than Q passes through c , and every component of $g^* Q$ meeting $g^{-1}(c)$ dominates Q . Put $\alpha_Q := \text{mult}_Q D_{P_U}$. Consider the rational b -canonical form

$$\Theta := \frac{\tilde{\theta}_0}{g^*(\varpi \ell_1 \dots \ell_{k-1})^b} = \tilde{\beta} \wedge g^* \left(\frac{d\ell_1}{\ell_1} \wedge \dots \wedge \frac{d\ell_{k-1}}{\ell_{k-1}} \wedge du \right)^{\otimes b}$$

on the preimage under g of a neighborhood of c . Its iterated residue along the smooth divisors $g^* \pi^* L_j$, $1 \leq j \leq k-1$, is $\Theta_C := \beta_C \wedge (g_C^* du)^{\otimes b}$. Near $g_C^{-1}(c)$, adjunction gives

$$\begin{aligned} \text{Div}_{\tilde{X}_C}(\Theta_C) &= \left(\text{Div}(\Theta) + b \sum_j g^* \pi^* L_j \right) \Big|_{\tilde{X}_C} = -bB_C^c + b g_C^* ((D_{P_U} - K_{P_U})|_C) \\ &= -bB_C^c + b(\alpha_Q - \text{mult}_Q K_{P_U}) g_C^* c, \end{aligned}$$

where $B_C^c := B^c|_{\tilde{X}_C}$. Hence $\Omega_C := \beta_C \wedge (du/u)^{\otimes b} = u^{-b} \Theta_C$ satisfies

$$\frac{1}{b} \text{Div}_{\tilde{X}_C}(\Omega_C) + B_C^c = (\alpha_Q - \text{mult}_Q K_{P_U} - 1) g_C^* c$$

near $g_C^{-1}(c)$. Every divisorial valuation E of $\mathbb{C}(F_C)$ over $\mathbb{C}$ whose restriction to $\mathbb{C}(C)$ is a positive multiple of ord_c has a center in $g_C^{-1}(c)$, as g_C is proper. Pulling back the last identity to a model on which E is a divisor, we get

$$\frac{\frac{1}{b} \text{ord}_E(\Omega_C) + 1}{\text{ord}_E(u)} = \alpha_Q - \text{mult}_Q K_{P_U} - 1 + \frac{1 - b_E}{\text{ord}_E(u)},$$

where b_E is the coefficient of E in the crepant pullback of B_C^c . The infimum over E of the last quotient is the lc threshold of the sub-pair $(\tilde{X}_C, B_C^c)$ with respect to $g_C^* c$ over c . On the log smooth sub-pair $(\tilde{X}_C, B_C^c)$ this threshold is the minimum of $(1 - b_i)/a_i$ over the components of $g^* Q$ meeting $g^{-1}(c)$, where b_i is the coefficient in B^c and a_i the multiplicity in $g^* Q$. Indeed, these

components restrict to smooth divisors of $\tilde{X}_C$ with the same coefficients and multiplicities. This minimum is t_Q (cf. [OAI26c, proof of Lemma 7.3] and Lemma 4.6). By (2.4) and Definition 4.2,

$$(5.15) \quad w_u(\beta_C) = \alpha_Q - \text{mult}_Q K_{P_U} - 1 + t_Q = \text{mult}_Q \mathbf{M}_{\beta, P_U}.$$

The data $C, c, \sigma|_C, \zeta, (F_C, B_C), \beta_C$ and γ satisfy the hypotheses of Lemma 5.4 for d and b , and the integer p_* of (5.1) is p_v . Hence $e \mid p_v w_u(\beta_C) = p_v \text{mult}_Q \mathbf{M}_{\beta, P_U}$ by (5.15). This proves (5.14) and finishes the proof. $\square$

6. SECTIONS ON THE UNION OF LC CENTERS OVER A MINIMAL LC CENTER

Definition 6.1 (Sources and springs, [Kol16, Definition 18]). Let $q : (Y, A) \rightarrow T$ be a dlt crepant log structure (Definition 2.8) and V an lc center of q . A *source* of V is an lc center W of (Y, A) such that $q(W) = V$ and W is minimal with respect to inclusion among the lc centers of (Y, A) with image V . We always consider a source together with its adjunction pair (W, Θ_W) (Definition 2.3). Let $W \xrightarrow{h} P \xrightarrow{\rho} V$ be the Stein factorization of $q|_W$ [Har77, Corollary III.11.5], so that P is normal, h is a contraction, and ρ is finite. We call P the *spring* of V . By [Kol16, Theorem 10 and Definition 18], any two sources of V are crepant birational over V , and their springs are isomorphic over V .

Let (W, Θ_W) , h , P , and ρ be as in Definition 6.1. Since h is a contraction, $\mathbb{C}(P)$ is the algebraic closure of $\mathbb{C}(V)$ in $\mathbb{C}(W)$, and P is the normalization of V in $\mathbb{C}(P)$. Thus every crepant birational self-map γ of (W, Θ_W) over V induces an automorphism $\bar{\gamma}$ of P over V with $h \circ \gamma = \bar{\gamma} \circ h$. We regard $\text{Aut}(\mathbb{C}(P)/\mathbb{C}(V))$ as the group of automorphisms of P over V , through the normalization. Sending γ to $\bar{\gamma}$ defines a group homomorphism

$$\text{Bir}_V^c(W, \Theta_W) \rightarrow \text{Aut}(\mathbb{C}(P)/\mathbb{C}(V)).$$

Theorem 6.2 ([Kol16, Proposition 19 and Definition 18]). *Let $q : (Y, A) \rightarrow T$ be a $\mathbb{Q}$ -factorial dlt crepant log structure, V an lc center of q , (W, Θ_W) a source of V , and $W \xrightarrow{h} P \xrightarrow{\rho} V$ its Stein factorization. Then:*

- (1) $\mathbb{C}(P)/\mathbb{C}(V)$ is a finite Galois extension.
- (2) The homomorphism $\text{Bir}_V^c(W, \Theta_W) \rightarrow \text{Gal}(\mathbb{C}(P)/\mathbb{C}(V))$ is surjective.

We recall the notion of quasi-log canonical pairs [Fuj14, 2.6, Definitions 3.1 and 3.2]. A *globally embedded simple normal crossing pair* (R, Δ_R) consists of a simple normal crossing divisor R on a smooth variety M and an $\mathbb{R}$ -divisor B on M such that $\text{Supp}(B + R)$ is a simple normal crossing divisor and B and R have no common irreducible components. Here $\Delta_R := B|_R$. Let $\nu : R^\nu \rightarrow R$ be the normalization and $K_{R^\nu} + \Xi := \nu^*(K_R + \Delta_R)$. A *stratum* of (R, Δ_R) is an irreducible component of R or the ν -image of an lc center of the sub-pair (R^ν, Ξ) , that is, of the center of a prime divisor F over R^ν with $a(F, R^\nu, \Xi) = 0$. A *quasi-log canonical pair* $[T, \omega]$ consists of a variety T , an $\mathbb{R}$ -Cartier $\mathbb{R}$ -divisor ω on T , and a proper morphism $f : (R, \Delta_R) \rightarrow T$ from a globally embedded simple normal crossing pair with the following properties. We have $\Delta_R \leq 1$, $f^*\omega \sim_{\mathbb{R}} K_R + \Delta_R$, and $\mathcal{O}_T \simeq f_*\mathcal{O}_R(\lceil -\Delta_R^{\leq 1} \rceil)$, where $\Delta_R^{\leq 1}$ is the sum of the components of Δ_R with coefficient < 1 . The *qlc centers* of $[T, \omega]$ are the images of the strata of (R, Δ_R) .

Theorem 6.3 ([Fuj14, Theorem 3.4(ii)]). *Let $[T, \omega]$ be a quasi-log canonical pair and $x \in T$ a point. Then the set of qlc centers of $[T, \omega]$ passing through x has a unique minimal element W_x , and W_x is normal at x .*

Lemma 6.4. *Let $q : (Y, A) \rightarrow T$ be a crepant log structure and ω a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor on T such that $K_Y + A \sim_{\mathbb{Q}} q^*\omega$. Then $[T, \omega]$ is a quasi-log canonical pair whose qlc centers are T and the lc centers of q .*

Proof. Let $\mu : \tilde{Y} \rightarrow Y$ be a log resolution of (Y, A) and $K_{\tilde{Y}} + \Delta := \mu^*(K_Y + A)$. Since (Y, A) is lc and $A \geq 0$, we have $\Delta \leq 1$, and every component of Δ with negative coefficient is μ -exceptional. Thus $\lceil -\Delta^{\leq 1} \rceil$ is an effective μ -exceptional divisor, so $\mu_*\mathcal{O}_{\tilde{Y}}(\lceil -\Delta^{\leq 1} \rceil) = \mathcal{O}_Y$ as

Y is normal, and $(q\mu)_*\mathcal{O}_{\tilde{Y}}(\lceil -\Delta^{\leq 1} \rceil) = q_*\mathcal{O}_Y = \mathcal{O}_T$. The pair $(\tilde{Y}, \Delta)$ is a globally embedded simple normal crossing pair: take $M := \tilde{Y} \times \mathbb{A}^1$, $R := \tilde{Y} \times \{0\}$, and $B := \Delta \times \mathbb{A}^1$. Since $K_{\tilde{Y}} + \Delta \sim_{\mathbb{Q}} (q\mu)^*\omega$, the proper morphism $q\mu : (\tilde{Y}, \Delta) \rightarrow T$ makes $[T, \omega]$ a quasi-log canonical pair. As $\tilde{Y}$ is smooth, the strata of the globally embedded simple normal crossing pair $(\tilde{Y}, \Delta)$ are $\tilde{Y}$ and the lc centers of $(\tilde{Y}, \Delta)$. Since $a(F, \tilde{Y}, \Delta) = a(F, Y, A)$ for every prime divisor F over Y and $\mu(\text{center}_{\tilde{Y}} F) = \text{center}_Y F$, the images under μ of the lc centers of $(\tilde{Y}, \Delta)$ are the lc centers of (Y, A) . Hence the qlc centers of $[T, \omega]$ are T and the images under q of the lc centers of (Y, A) . $\square$

Theorem 6.5 ([Fuj17, Theorem 6.4]). *Let (X, Δ) be an lc pair and $\pi : X \rightarrow V$ a proper morphism to a variety. Let L be a Cartier divisor on X such that $H := L - (K_X + \Delta)$ is a π -semi-ample $\mathbb{R}$ -divisor on X . Let $D \geq 0$ be an $\mathbb{R}$ -divisor on X such that $D \sim_{\mathbb{R}, \pi} tH$ for some positive real number t , that is, $D \sim_{\mathbb{R}} tH + \pi^*B$ for some $\mathbb{R}$ -Cartier $\mathbb{R}$ -divisor B on V . Let Z be the union of the lc centers of (X, Δ) contained in $\text{Supp } D$. Then the restriction map*

$$\pi_*\mathcal{O}_X(L) \rightarrow \pi_*\mathcal{O}_Z(L)$$

is surjective.

Here an $\mathbb{R}$ -Cartier $\mathbb{R}$ -divisor D on X is π -semi-ample if $D \sim_{\mathbb{R}} \sum_i a_i D_i$ for finitely many π -semi-ample Cartier divisors D_i and positive real numbers a_i [Fuj11, Definition 4.11].

Set-up 6.6. Let $d \geq 2$ be an integer and $\Phi \subset [0, 1] \cap \mathbb{Q}$ a finite set. Let $\mathfrak{R}$ be the finite set given by Lemma 4.4 for d and Φ , and $\Gamma := \Phi(\mathfrak{R})$. Let $Y \xrightarrow{q} T \xrightarrow{\pi} U$ be contractions of normal quasi-projective varieties such that U is affine and $\dim Y \leq d$. Let (Y, A) be a $\mathbb{Q}$ -factorial dlt pair such that $A \in \Phi$ and

$$(6.1) \quad -(K_Y + A) \sim_{\mathbb{Q}} q^*H$$

for some $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor H on T which is ample over U . In particular $q : (Y, A) \rightarrow T$ is a $\mathbb{Q}$ -factorial dlt crepant log structure. Let $V \subsetneq T$ be an lc center of q which is minimal with respect to inclusion among all lc centers of q . Let Z_V be the reduced union of all lc centers of (Y, A) with image V . Since V is minimal, an lc center of (Y, A) is contained in Z_V if and only if its image is V . Fix a source (W, Θ_W) of V and its Stein factorization $W \xrightarrow{h} P \xrightarrow{\rho} V$. For an lc center C of (Y, A) with $q(C) = V$, let $C \xrightarrow{f_C} P_C \xrightarrow{\rho_C} V$ be the Stein factorization of $q|_C$ and (C, Θ_C) the adjunction pair of C ; by Lemma 4.4(1), $\Theta_C \in \Gamma$. A source of V contained in C dominates P_C , and we have $K_C + \Theta_C \sim_{\mathbb{Q}} -f_C^*\rho_C^*(H|_V)$, where $H|_V$ denotes the restriction of a $\mathbb{Q}$ -Cartier representative of H whose support does not contain V .

Let $\mu : \tilde{Y} \rightarrow Y$ be a log resolution of (Y, A) which is an isomorphism over the largest open subset Y^{snc} of Y over which (Y, A) is log smooth (standard). Write

$$(6.2) \quad K_{\tilde{Y}} + P_1 + E = \mu^*(K_Y + A), \quad P_1 := \mu_*^{-1}[A],$$

where the canonical divisors are chosen compatibly (Notation 2.2). Since (Y, A) is dlt, every μ -exceptional prime divisor has positive log discrepancy ([KM98, Theorem 2.44], see also [Kol16, Definition 3]), so every coefficient of E is less than 1. The coefficients of the μ -exceptional components of E may be negative, and P_1 and $\text{Supp } E$ have no common components. A stratum of P_1 is an irreducible component of an intersection of components of P_1 . If $\tilde{C}$ is a stratum of codimension k and $D_1, \dots, D_k$ are the components of P_1 containing $\tilde{C}$, then no other component of $P_1 + \text{Supp } E$ contains $\tilde{C}$. Hence the exceptional divisor of the blow-up of $\tilde{Y}$ along $\tilde{C}$ has log discrepancy $k - k = 0$ with respect to $(\tilde{Y}, P_1 + E)$. Thus every stratum of P_1 is an lc center of $(\tilde{Y}, P_1 + E)$, and $\mu(\tilde{C})$ is an lc center of (Y, A) , because $(\tilde{Y}, P_1 + E)$ and (Y, A) have the same lc places. Conversely, let C be an lc center of (Y, A) . By Definition 2.3, C meets Y^{snc} , and $C \cap Y^{\text{snc}}$ is an irreducible component of an intersection of components of $[A]$. Hence the closure $\tilde{C}$ of $\mu^{-1}(C \cap Y^{\text{snc}})$ is a stratum of P_1 , and $\mu_C := \mu|_{\tilde{C}} : \tilde{C} \rightarrow C$ is birational. Since μ is an

isomorphism over the generic point of C , $\tilde{C}$ is the only stratum of P_1 whose image is C . Thus μ induces a bijection between the strata of P_1 and the lc centers of (Y, A) , and if $C' \subset C$ are lc centers of (Y, A) , then $\tilde{C}' \subset \tilde{C}$. Let $\tilde{Z}_V$ be the reduced union of all strata of P_1 mapping onto V , so that $\mu(\tilde{Z}_V) = Z_V$, and $\tilde{W}$ the stratum with $\mu(\tilde{W}) = W$. Every irreducible component of the intersection of two strata of P_1 is a stratum of P_1 , since P_1 is a simple normal crossing divisor. If the two strata are contained in $\tilde{Z}_V$, then so is this component, by the minimality of V .

For a positive integer N we define the Cartier divisor

$$(6.3) \quad L_N := -N(K_{\tilde{Y}} + P_1) - \lfloor (N+1)E \rfloor$$

on $\tilde{Y}$. A rational function φ is a section of $\mathcal{O}_{\tilde{Y}}(L_N)$ over an open subset if and only if the rational $(-N)$ -canonical form $\beta := \varphi \omega^{\otimes (-N)}$ satisfies $\text{Div}(\beta) \geq NP_1 + \lfloor (N+1)E \rfloor$ there, where ω is the fixed rational top form with $K_{\tilde{Y}} = \text{Div}(\omega)$. We identify $\mathcal{O}_{\tilde{Y}}(L_N)$ with this sheaf of $(-N)$ -canonical forms.

Residues along strata. We keep the notation of Set-up 6.6, and we also regard $\tilde{Y}$ as a stratum of P_1 , the intersection of the empty set of components, with $\mu(\tilde{Y}) = Y$. Let $\tilde{C}$ be a stratum of P_1 , $D_1, \dots, D_k$ the components of P_1 containing $\tilde{C}$, $C := \mu(\tilde{C})$, and

$$S_C := (P_1 - D_1 - \dots - D_k)|_{\tilde{C}}, \quad E_C := E|_{\tilde{C}}.$$

Then S_C is a reduced divisor and $S_C + \text{Supp } E_C$ is a simple normal crossing divisor on $\tilde{C}$. The coefficients of E_C are those of E , and $\lfloor (N+1)E \rfloor|_{\tilde{C}} = \lfloor (N+1)E_C \rfloor$. The strata of P_1 contained in $\tilde{C}$ are $\tilde{C}$ and the strata of S_C . If $\tilde{C}' \subset \tilde{C}$ is a stratum of P_1 and $D'_1, \dots, D'_l$ are the components of S_C containing $\tilde{C}'$, then $(S_C - D'_1 - \dots - D'_l)|_{\tilde{C}'} = S_{C'}$ and $E_C|_{\tilde{C}'} = E_{C'}$.

Iterating the adjunction isomorphism [Har77, Proposition II.8.20] along $D'_1, \dots, D'_l$ gives an isomorphism $\omega_{\tilde{C}}(D'_1 + \dots + D'_l)|_{\tilde{C}'} \simeq \omega_{\tilde{C}'}$, which is well defined up to sign, and its m -th tensor power is well defined for every even integer m (cf. [Kol16, Definition 13]). Let m be an even integer and θ a rational m -canonical form on $\tilde{C}$ such that no component of $\text{Div}(\theta) + m(D'_1 + \dots + D'_l)$ contains $\tilde{C}'$. Then θ is a rational section of $\omega_{\tilde{C}}^{\otimes m}(m(D'_1 + \dots + D'_l))$ which is regular and nonzero at the generic point of $\tilde{C}'$, and we denote by $\theta_{\tilde{C}'}$ the rational m -canonical form on $\tilde{C}'$ corresponding to $\theta|_{\tilde{C}'}$. We call $\theta_{\tilde{C}'}$ the *residue* of θ on $\tilde{C}'$. By construction,

$$(6.4) \quad \text{Div}(\theta_{\tilde{C}'}) = (\text{Div}(\theta) + m(D'_1 + \dots + D'_l))|_{\tilde{C}'},$$

The residue is multiplicative: $(\theta \otimes \theta')_{\tilde{C}'} = \theta_{\tilde{C}'} \otimes \theta'_{\tilde{C}'}$ for forms of even degrees, and $(g\theta)_{\tilde{C}'} = g|_{\tilde{C}'} \theta_{\tilde{C}'}$ for a rational function g on $\tilde{C}$ which is regular and nonzero at the generic point of $\tilde{C}'$. It is transitive along flags of strata, it commutes with étale base change, and in local coordinates at the generic point of $\tilde{C}'$ it is given by the formula recalled before Theorem 2.15. In particular, let C' be an lc center of the dlt pair (C, Θ_C) , and let θ be a rational m -canonical form on C , viewed on $\tilde{C}$, such that the residue $\theta_{\tilde{C}'}$ is defined. Near the generic point of C' , the pair (C, Θ_C) is log smooth and $K_C + \Theta_C$ is Cartier, so θ is a rational section of $\omega_C^{[m]}(m\Theta_C)$ there which is regular and nonzero at the generic point of C' . Then $\theta_{\tilde{C}'}$ is the image of $\theta|_{C'}$ under the Poincaré residue map $R_{C \rightarrow C'}^m$ of Definition 5.1. Indeed, both are determined by their values at the generic point of C' , where μ_C is an isomorphism and both are given by the formula recalled before Theorem 2.15.

Applying this to local generators of $\mathcal{O}_{\tilde{Y}}(L_N)$, whose divisors are $NP_1 + \lfloor (N+1)E \rfloor$ locally, we obtain for every even positive integer N a canonical isomorphism

$$(6.5) \quad \mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{C}} \simeq \mathcal{O}_{\tilde{C}}(-NK_{\tilde{C}} - NS_C - \lfloor (N+1)E_C \rfloor),$$

where the right hand side is the sheaf of rational $(-N)$ -canonical forms β on $\tilde{C}$ with $\text{Div}(\beta) \geq NS_C + \lfloor (N+1)E_C \rfloor$. For a stratum $\tilde{C}' \subset \tilde{C}$, the restriction map $\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{C}} \rightarrow \mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{C}'}$ sends

a section β to $\beta_{\tilde{C}'}$, under (6.5), where $\beta_{\tilde{C}'} := 0$ if β vanishes along $\tilde{C}'$, that is, if some component of $\text{Div}(\beta) - N(D'_1 + \cdots + D'_l)$ contains $\tilde{C}'$. We also write $\beta|_{\tilde{C}'}$ for $\beta_{\tilde{C}'}$.

Lemma 6.7. *Notation as in Set-up 6.6. Let C be an lc center of (Y, A) or $C := Y$, let $\tilde{C}$ be the stratum of P_1 with $\mu(\tilde{C}) = C$, and $\mu_C := \mu|_{\tilde{C}}$. Then:*

(1) *We have*

$$(6.6) \quad K_{\tilde{C}} + S_C + E_C = \mu_C^*(K_C + \Theta_C),$$

where $K_{\tilde{C}}$ and K_C are the divisors of one rational top form on C and $\Theta_Y := A$. In particular, $\Theta_C = (\mu_C)_*(S_C + E_C)$, and for every integer m and every rational m -canonical form θ on C ,

$$(6.7) \quad \text{Div}_{\tilde{C}}(\theta) = \mu_C^*(\text{Div}_C(\theta) + m\Theta_C) - m(S_C + E_C).$$

(2) *Let N be a positive even integer such that $N\Re \subset \mathbb{Z}$. Then*

$$(6.8) \quad (\mu_C)_*\mathcal{O}_{\tilde{C}}(-NK_{\tilde{C}} - NS_C - \lfloor (N+1)E_C \rfloor) = \mathcal{O}_C(-NK_C - \lceil N\Theta_C \rceil)$$

as subsheaves of the constant sheaf of rational $(-N)$ -canonical forms. That is, for a rational $(-N)$ -canonical form β on C and an open subset $C^\circ \subset C$, we have $\text{Div}_{\tilde{C}}(\beta) \geq NS_C + \lfloor (N+1)E_C \rfloor$ on $\mu_C^{-1}(C^\circ)$ if and only if $\text{Div}_C(\beta) \geq \lceil N\Theta_C \rceil$ on C° . In particular, $(\mu_C)_*(\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{C}}) \simeq \mathcal{O}_C(-NK_C - \lceil N\Theta_C \rceil)$ by (6.5).

Proof. (1) If $C = Y$, then (6.6) is (6.2). Assume that C is an lc center of (Y, A) , and let $D_1, \dots, D_k$ be the components of P_1 containing $\tilde{C}$. Let m be a positive even integer such that $m(K_Y + A)$ is Cartier. Let σ be an m -canonical form on Y such that $G := \text{Div}_Y(\sigma) + mA$ is a Cartier divisor whose support does not contain C . For example, we may take $\sigma := \psi\omega_Y^{\otimes m}$ for a rational function ψ with $\text{div}(\psi) = -m(K_Y + A)$ near the generic point of C . Then σ is a rational section of $\omega_Y^{[m]}(mA)$ with divisor G , so by [Kol16, Definition 13] the Poincaré residue $\theta_C := R_{Y \rightarrow C}^m(\sigma|_C)$ is a rational section of $\omega_C^{[m]}(m\Theta_C)$ with divisor $G|_C$, that is,

$$\text{Div}_C(\theta_C) + m\Theta_C = G|_C.$$

By (6.2), $\text{Div}_{\tilde{Y}}(\sigma) = \mu^*(\text{Div}_Y(\sigma) - mK_Y) + mK_{\tilde{Y}} = \mu^*G - m(P_1 + E)$. Hence

$$\text{Div}_{\tilde{Y}}(\sigma) + m(D_1 + \cdots + D_k) = \mu^*G - m(P_1 - D_1 - \cdots - D_k) - mE,$$

and no component of this divisor contains $\tilde{C}$. Thus the residue $\sigma_{\tilde{C}}$ is defined, and by (6.4) its divisor is $\mu_C^*(G|_C) - m(S_C + E_C)$. As $\sigma_{\tilde{C}} = \theta_C$, we get

$$\text{Div}_{\tilde{C}}(\theta_C) = \mu_C^*(\text{Div}_C(\theta_C) + m\Theta_C) - m(S_C + E_C).$$

On the other hand, writing $\theta_C = \varphi\omega_C^{\otimes m}$, we have $\text{Div}_{\tilde{C}}(\theta_C) = \mu_C^*\text{div}_C(\varphi) + mK_{\tilde{C}} = \mu_C^*(\text{Div}_C(\theta_C) - mK_C) + mK_{\tilde{C}}$. Comparing the two expressions and dividing by m gives (6.6). Pushing forward (6.6) gives $\Theta_C = (\mu_C)_*(S_C + E_C)$, and the same computation with an arbitrary θ gives (6.7).

(2) Let Q be a prime divisor on C and F its strict transform on $\tilde{C}$. Since S_C and $\text{Supp } E_C$ have no common components, F is a component of S_C if and only if $\text{mult}_Q \Theta_C = 1$, and F is a component of E_C with coefficient e if and only if $\text{mult}_Q \Theta_C = e \in (0, 1)$. Since $\Theta_C \in \Gamma = \Phi(\Re)$ and $N\Re \subset \mathbb{Z}$, Lemma 4.4(2) gives

$$(6.9) \quad (\mu_C)_*(NS_C + \lfloor (N+1)E_C \rfloor) = N\lceil \Theta_C \rceil + \lfloor (N+1)\{\Theta_C\} \rfloor = \lceil N\Theta_C \rceil.$$

Let β be a rational $(-N)$ -canonical form on C . We have $\text{Div}_C(\beta) = (\mu_C)_*\text{Div}_{\tilde{C}}(\beta)$. If $\text{Div}_{\tilde{C}}(\beta) \geq NS_C + \lfloor (N+1)E_C \rfloor$ on $\mu_C^{-1}(C^\circ)$, then $\text{Div}_C(\beta) \geq \lceil N\Theta_C \rceil$ on C° by (6.9).

Conversely, assume that $\text{Div}_C(\beta) \geq \lceil N\Theta_C \rceil$ on C° . Write $\beta = \varphi\omega_C^{\otimes(-N)}$. Then $N(K_C + \Theta_C)$ and $\text{Div}_C(\beta) - NK_C = \text{div}(\varphi)$ are $\mathbb{Q}$ -Cartier, so $\text{Div}_C(\beta) - N\Theta_C$ is a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor, and it is effective on C° . By (6.7),

$$\text{Div}_{\tilde{C}}(\beta) - N(S_C + E_C) = \mu_C^*(\text{Div}_C(\beta) - N\Theta_C) \geq 0$$

on $\mu_C^{-1}(C^\circ)$. Let F be a prime divisor on $\mu_C^{-1}(C^\circ)$ and e its coefficient in $S_C + E_C$. Then $\text{mult}_F \text{Div}_{\tilde{C}}(\beta) \geq Ne$, hence $\text{mult}_F \text{Div}_{\tilde{C}}(\beta) \geq \lceil Ne \rceil$. If F is a component of S_C , then $\lceil Ne \rceil = N$ is the coefficient of F in $NS_C + \lfloor (N+1)E_C \rfloor$. Otherwise $e < 1$, and

$$\lfloor (N+1)e \rfloor \leq \lceil Ne \rceil,$$

because $(N+1)e < Ne + 1 \leq \lceil Ne \rceil + 1$. Thus $\text{Div}_{\tilde{C}}(\beta) \geq NS_C + \lfloor (N+1)E_C \rfloor$ on $\mu_C^{-1}(C^\circ)$. This proves (6.8), and the last assertion follows from (6.5). $\square$

Lemma 6.8. *Let $f : (X, \Delta) \rightarrow S$ be a crepant log structure such that f is a contraction between normal quasi-projective varieties. Let a be a positive integer and θ a rational a -canonical form on X such that*

$$\text{Div}(\theta) + a\Delta = f^*(aD)$$

for some $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor D on S . Let N be a positive multiple of a and β a nonzero rational $(-N)$ -canonical form on X . Then $\text{Div}(\beta) \geq N\Delta$ if and only if

$$\beta = (f^*u)\theta^{\otimes(-N/a)}$$

for some nonzero $u \in \mathbb{C}(S)$ such that $E := \frac{1}{N} \text{div}_S(u) - D \geq 0$. In this case u is unique, E is $\mathbb{Q}$ -Cartier, and

$$(6.10) \quad \frac{1}{N} \text{Div}(\beta) - \Delta = f^*E.$$

*Moreover, for an open subset $S^\circ \subset S$, we have $\text{Div}(\beta) \geq N\Delta$ on $f^{-1}(S^\circ)$ if and only if $\beta = (f^*u)\theta^{\otimes(-N/a)}$ with $E \geq 0$ on S° .*

Proof. Put $v := \beta \otimes \theta^{\otimes(N/a)}$, a nonzero rational function on X . Then

$$\text{div}(v) = \text{Div}(\beta) + \frac{N}{a} \text{Div}(\theta) = (\text{Div}(\beta) - N\Delta) + f^*(ND).$$

Assume that $\text{Div}(\beta) \geq N\Delta$ on $f^{-1}(S^\circ)$. Since $f^*(ND)$ is vertical over S , the horizontal part of $\text{div}(v)$ is effective, so v has no poles on the generic fiber F of f . As f is a contraction, F is a normal projective variety over $\mathbb{C}(S)$ with $H^0(F, \mathcal{O}_F) = \mathbb{C}(S)$, so $u := v|_F$ is a nonzero element of $\mathbb{C}(S)$ and $v = f^*u$ (cf. the proof of Lemma 4.1). Thus $\text{Div}(\beta) - N\Delta = f^*(\text{div}(u) - ND) = f^*(NE)$, which is (6.10), and E is $\mathbb{Q}$ -Cartier. Since f is surjective, for every prime divisor Q on S° some component of f^*Q dominates Q , and the coefficient of this component in $f^*(NE)$ has the sign of $\text{mult}_Q E$. Hence $E \geq 0$ on S° . The uniqueness of u is clear.

Conversely, if $\beta = (f^*u)\theta^{\otimes(-N/a)}$ with $E \geq 0$ on S° , then $\text{Div}(\beta) = f^* \text{div}(u) - \frac{N}{a} \text{Div}(\theta) = N\Delta + f^*(NE) \geq N\Delta$ on $f^{-1}(S^\circ)$. $\square$

Since $\text{Div}(\beta)$ is an integral divisor, $\text{Div}(\beta) \geq N\Delta$ if and only if $\text{Div}(\beta) \geq \lceil N\Delta \rceil$. Thus Lemma 6.8 describes the sections of $\mathcal{O}_X(-NK_X - \lceil N\Delta \rceil)$ over $f^{-1}(S^\circ)$: they are the forms $(f^*u)\theta^{\otimes(-N/a)}$ with $u \in \mathbb{C}(S)$ and $\text{div}(u) \geq ND$ on S° , together with 0.

Lemma 6.9. *Let $f : (X, \Delta) \rightarrow S$ be a dlt crepant log structure such that f is a contraction between normal quasi-projective varieties. Let a be a positive even integer and θ a rational a -canonical form on X such that $\text{Div}(\theta) + a\Delta = f^*(aD)$ for some $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor D on S . Let Z_1 and Z_2 be minimal lc centers of (X, Δ) such that $f(Z_1) = f(Z_2) = S$, with adjunction pairs (Z_i, Δ_{Z_i}) . Then:*

- (1) *For $i = 1, 2$, the Poincaré residue $\theta_{Z_i} := R_{X \rightarrow Z_i}^a(\theta|_{Z_i})$ of Definition 5.1, taken over the open subset of S where aD is Cartier, is a nonzero rational a -canonical form on Z_i , and*

$$\text{Div}(\theta_{Z_i}) + a\Delta_{Z_i} = (f|_{Z_i})^*(aD).$$

- (2) There exists a crepant birational map $\phi : (Z_2, \Delta_{Z_2}) \dashrightarrow (Z_1, \Delta_{Z_1})$ over S such that $\phi^*\theta_{Z_1} = \theta_{Z_2}$.
- (3) Let $Z_1 \xrightarrow{h_1} P_1 \xrightarrow{\pi_1} S$ be the Stein factorization of $f|_{Z_1}$ and τ an automorphism of P_1 over S . Then there exists a crepant birational self-map γ of (Z_1, Δ_{Z_1}) over S such that $h_1 \circ \gamma = \tau \circ h_1$ and $\gamma^*\theta_{Z_1} = \theta_{Z_1}$.

Proof. Let $S^\circ \subset S$ be a dense open subset over which aD is Cartier, and put $X^\circ := f^{-1}(S^\circ)$. On X° , the divisor $a(K_X + \Delta) = f^*(aD) - \text{Div}(\theta)$ is Cartier, so $a\Delta$ is integral, the sheaf $\omega_X^{[a]}(a\Delta)$ of rational a -canonical forms θ' with $\text{Div}(\theta') + a\Delta \geq 0$ is locally free, and θ is a rational section of it with divisor $f^*(aD)$. Since Z_i dominates S , its generic point lies over S° and no component of $f^*(aD)$ contains Z_i . Thus $\theta|_{Z_i}$ is a nonzero rational section of $\omega_X^{[a]}(a\Delta)|_{Z_i}$ over S° , and $\theta_{Z_i} := R_{X \rightarrow Z_i}^a(\theta|_{Z_i})$ is a nonzero rational a -canonical form on Z_i .

(1) Let k be a positive integer such that kaD is Cartier. Then $ka(K_X + \Delta) = f^*(kaD) - k\text{Div}(\theta)$ is Cartier on X . As above, $\omega_X^{[ka]}(ka\Delta)$ is locally free, $\theta^{\otimes k}$ is a rational section of it with divisor $f^*(kaD)$, and $\theta^{\otimes k}|_{Z_i}$ is a nonzero rational section of $\omega_X^{[ka]}(ka\Delta)|_{Z_i}$ with divisor $(f|_{Z_i})^*(kaD)$. Since $R_{X \rightarrow Z_i}^{ka}$ is an isomorphism onto $\omega_{Z_i}^{[ka]}(ka\Delta_{Z_i})$, the rational ka -canonical form $\eta := R_{X \rightarrow Z_i}^{ka}(\theta^{\otimes k}|_{Z_i})$ on Z_i satisfies $\text{Div}(\eta) + ka\Delta_{Z_i} = (f|_{Z_i})^*(kaD)$. At the generic point of Z_i , the residue maps are given by the formula recalled before Theorem 2.15, which is multiplicative, so $\eta = \theta_{Z_i}^{\otimes k}$. Dividing by k gives (1).

Let $i, j \in \{1, 2\}$, and let $\phi : Z_j \dashrightarrow Z_i$ be a birational map over S such that $\phi^*\theta_{Z_i} = \theta_{Z_j}$. We claim that ϕ is crepant with respect to (Z_j, Δ_{Z_j}) and (Z_i, Δ_{Z_i}) . Let $p : Z' \rightarrow Z_j$ and $r : Z' \rightarrow Z_i$ be a common resolution of ϕ , so that $r = \phi \circ p$ and $f|_{Z_j} \circ p = f|_{Z_i} \circ r$, and let $K_{Z'} + \Gamma_p := p^*(K_{Z_j} + \Delta_{Z_j})$ and $K_{Z'} + \Gamma_r := r^*(K_{Z_i} + \Delta_{Z_i})$. By the computation of (5.12) and (1),

$$\text{Div}_{Z'}(p^*\theta_{Z_j}) = p^*(f|_{Z_j})^*(aD) - a\Gamma_p \quad \text{and} \quad \text{Div}_{Z'}(r^*\theta_{Z_i}) = r^*(f|_{Z_i})^*(aD) - a\Gamma_r.$$

Since $p^*\theta_{Z_j} = p^*\phi^*\theta_{Z_i} = r^*\theta_{Z_i}$ and $p^*(f|_{Z_j})^*(aD) = r^*(f|_{Z_i})^*(aD)$, we get $\Gamma_p = \Gamma_r$. This proves the claim.

(2) The fiber of f over the generic point of S is connected [Har77, Corollary III.11.3], and $Z_1 \cap X^\circ$ and $Z_2 \cap X^\circ$ are minimal lc centers of $(X^\circ, \Delta|_{X^\circ})$ with image S° . The form θ is a section of $\omega_X^{[a]}(a\Delta) \simeq f^*\mathcal{O}_{S^\circ}(aD)$ over the open subset $X^\circ \setminus \text{Supp } f^*(aD)$, which meets Z_1 and Z_2 . Theorem 5.2(2), applied to $f : X^\circ \rightarrow S^\circ$ and $\mathcal{L} := \mathcal{O}_{S^\circ}(aD)$, gives a birational map $\phi : Z_2 \dashrightarrow Z_1$ with $\theta_{Z_2} = \phi^*\theta_{Z_1}$. The map ϕ is a composition of the birational maps induced by $\mathbb{P}^1$ -links, and these are over S° , because the $\mathbb{P}^1$ -links in the proof of [Kol16, Theorem 10] are obtained by an MMP over the base [Kol16, proof of Theorem 10 and Proposition 14]. Thus ϕ is over S , and it is crepant by the claim. This proves (2).

(3) By [Kol16, Proposition 19], $\mathbb{C}(P_1)/\mathbb{C}(S)$ is a Galois extension. Its Galois group is the group $\mathcal{H}$ of automorphisms of P_1 over S , as P_1 is the normalization of S in $\mathbb{C}(P_1)$. Shrinking S° , we may assume that π_1 is étale over S° , and we put $P_1^\circ := \pi_1^{-1}(S^\circ)$. Then $P_1^\circ \rightarrow S^\circ$ is a finite étale Galois cover with group $\mathcal{H}$, so $P_1^\circ \times_{S^\circ} P_1^\circ$ is the disjoint union of the graphs of the elements of $\mathcal{H}$. Put $X_1 := X \times_S P_1^\circ$, with projections $\tau_X : X_1 \rightarrow X$ and $f_1 : X_1 \rightarrow P_1^\circ$, and $\Delta_1 := \tau_X^*\Delta$. Since τ_X is étale and the generic fiber of f is geometrically integral, X_1 is a normal variety, f_1 is a contraction, (X_1, Δ_1) is dlt, and $K_{X_1} + \Delta_1 \sim_{\mathbb{Q}, P_1^\circ} 0$. The lc centers of (X_1, Δ_1) are the irreducible components of the preimages of the lc centers of (X, Δ) . The form $\theta_1 := \tau_X^*\theta$ satisfies $\text{Div}(\theta_1) + a\Delta_1 = f_1^*(a\pi_1^*D)$. The preimage of Z_1 in X_1 is $Z_1 \times_S P_1^\circ = Z_1 \times_{P_1} (P_1 \times_S P_1^\circ)$, the disjoint union of the subvarieties

$$Z_\sigma := \{(z, \sigma(h_1(z))) \mid z \in h_1^{-1}(P_1^\circ)\}, \quad \sigma \in \mathcal{H},$$

each of which is mapped isomorphically onto $h_1^{-1}(P_1^\circ)$ by τ_X . We write $j_\sigma : Z_\sigma \rightarrow Z_1$ for this isomorphism; it identifies $(Z_\sigma, \Delta_{Z_\sigma})$ with (Z_1, Δ_{Z_1}) , and $f_1|_{Z_\sigma} = \sigma \circ h_1 \circ j_\sigma$. Since Z_1 is a minimal lc center of (X, Δ) , the pair (Z_1, Δ_{Z_1}) is klt by the correspondence of lc centers in [Kol16,

Definition 18], so the Z_σ are minimal lc centers of (X_1, Δ_1) , with image P_1° . By (2) applied to f_1 , θ_1 , Z_{id} , and Z_τ , there is a crepant birational map $\phi : Z_\tau \dashrightarrow Z_{\text{id}}$ over P_1° with $\phi^*(\theta_1)_{Z_{\text{id}}} = (\theta_1)_{Z_\tau}$. Since Poincaré residues commute with étale base change, $(\theta_1)_{Z_\sigma} = j_\sigma^* \theta_{Z_1}$ for every σ . Put $\gamma := j_{\text{id}} \circ \phi \circ j_\tau^{-1}$, a birational self-map of Z_1 . Then $\gamma^* \theta_{Z_1} = (j_\tau^{-1})^* \phi^* j_{\text{id}}^* \theta_{Z_1} = (j_\tau^{-1})^* (\theta_1)_{Z_\tau} = \theta_{Z_1}$. Moreover, $f_1|_{Z_{\text{id}}} \circ \phi = f_1|_{Z_\tau}$ gives $h_1 \circ j_{\text{id}} \circ \phi = \tau \circ h_1 \circ j_\tau$, that is, $h_1 \circ \gamma = \tau \circ h_1$. As τ is over S , γ is over S , and γ is crepant by the claim applied with $i = j = 1$. This proves (3). $\square$

Lemma 6.10. *Let $h : (W, \Theta_W) \rightarrow P$ be a crepant log structure such that h is a contraction between normal quasi-projective varieties, and let g be a birational self-map of W with $h \circ g = h$. Assume that g is crepant over a dense open subset $P^\circ \subset P$, that is, the restriction of g to $h^{-1}(P^\circ)$ is a crepant birational self-map of $(h^{-1}(P^\circ), \Theta_W|_{h^{-1}(P^\circ)})$. Then $g \in \text{Bir}_P^c(W, \Theta_W)$.*

Proof. Let $p, q : W' \rightarrow W$ be a common resolution of g , so that $q = g \circ p$ and $h \circ q = h \circ p$, and put $K_{W'} + \Gamma_p := p^*(K_W + \Theta_W)$ and $K_{W'} + \Gamma_q := q^*(K_W + \Theta_W)$. We show that $\Gamma_p = \Gamma_q$. Since $K_W + \Theta_W \sim_{\mathbb{Q}} h^* L$ for some $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor L on P , we have $\Gamma_p - \Gamma_q \sim_{\mathbb{Q}} (hp)^* L - (hq)^* L = 0$. Since g is crepant over P° , $\Gamma_p - \Gamma_q$ is supported in $(hp)^{-1}(P \setminus P^\circ)$. Thus a positive multiple of $\Gamma_p - \Gamma_q$ is the divisor of a rational function on W' , and this divisor is vertical over P . As hp is a contraction, Lemma 4.1 gives a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor F on P with $\Gamma_p - \Gamma_q = (hp)^* F$, that is,

$$p^*(K_W + \Theta_W) = q^*(K_W + \Theta_W + h^* F).$$

Let Q be a prime divisor on P and $t_Q := \text{lct}(W, \Theta_W; h^* Q)$ over the generic point of Q (Definition 4.2), so that $0 \leq t_Q \leq 1$ and $(W, \Theta_W + t_Q h^* Q)$ is lc over the generic point of Q . Over a neighborhood of the generic point of Q , we have $h^* F = \text{mult}_Q F \cdot h^* Q$, so

$$p^*(K_W + \Theta_W + t_Q h^* Q) = q^*(K_W + \Theta_W + (t_Q + \text{mult}_Q F) h^* Q)$$

there. Suppose that $\text{mult}_Q F > 0$. Since p and q are birational morphisms over P , the last identity shows that the pair $(W, \Theta_W + (t_Q + \text{mult}_Q F) h^* Q)$ has the same log discrepancies over the generic point of Q as $(W, \Theta_W + t_Q h^* Q)$, so it is lc over the generic point of Q . This contradicts the definition of t_Q . Hence $\text{mult}_Q F \leq 0$. Applying the same argument to g^{-1} , with the common resolution q, p and with $-F$ in place of F , we get $\text{mult}_Q F \geq 0$. Thus $F = 0$ and $\Gamma_p = \Gamma_q$. $\square$

Lemma 6.11. *Let X be a smooth variety, D a simple normal crossing divisor on X , and $Z \subset X$ the reduced union of finitely many strata of D , that is, of irreducible components of intersections of components of D . Let $Z_1, \dots, Z_r$ be the irreducible components of Z and $\mathcal{L}$ an invertible sheaf on X . Then for all $i \neq j$ the scheme-theoretic intersection $Z_i \cap Z_j$ is reduced, and the sequence*

$$0 \rightarrow \mathcal{L}|_Z \rightarrow \bigoplus_{i=1}^r \mathcal{L}|_{Z_i} \rightarrow \bigoplus_{i < j} \mathcal{L}|_{Z_i \cap Z_j}$$

is exact, where the last map sends $(s_i)_i$ to $(s_i|_{Z_i \cap Z_j} - s_j|_{Z_i \cap Z_j})_{i < j}$. In particular, sections s_i of $\mathcal{L}|_{Z_i}$ over the open subsets $Z_i \cap X^\circ$, for an open subset $X^\circ \subset X$, such that s_i and s_j agree on $Z_i \cap Z_j \cap X^\circ$ for all i, j glue to a unique section of $\mathcal{L}|_Z$ over $Z \cap X^\circ$.

Proof. Tensoring with $\mathcal{L}^{-1}$, we may assume that $\mathcal{L} = \mathcal{O}_X$. The statements are local, and it suffices to check the exactness of a sequence of coherent sheaves at the closed points. Let $x \in Z$ be a closed point. Since the components of D through x are smooth and meet transversally, there is a regular system of parameters $x_1, \dots, x_n$ of $\mathcal{O}_{X,x}$ such that these components are $(x_1 = 0), \dots, (x_p = 0)$. Every Z_i containing x is a stratum of D , and the intersection of the components of D containing Z_i is smooth, hence irreducible near x . Thus the ideal of Z_i in $\mathcal{O}_{X,x}$ is $I_i := (x_j \mid j \in J_i)$ for a subset $J_i \subset \{1, \dots, p\}$. The ideal of $Z_i \cap Z_j$ is $I_i + I_j = (x_k \mid k \in J_i \cup J_j)$, which is prime, so $Z_i \cap Z_j$ is reduced near x . The ideal of Z is $\bigcap_i I_i$. Since the completion $\hat{\mathcal{O}}_{X,x} = \mathbb{C}[[x_1, \dots, x_n]]$ is faithfully flat over $\mathcal{O}_{X,x}$, it suffices to show that the sequence

$$0 \rightarrow \hat{\mathcal{O}}_{X,x} / \bigcap_i I_i \rightarrow \bigoplus_i \hat{\mathcal{O}}_{X,x} / I_i \rightarrow \bigoplus_{i < j} \hat{\mathcal{O}}_{X,x} / (I_i + I_j)$$

is exact, where the ideals are extended to the completion; here i and j run over the indices with $x \in Z_i$ and $x \in Z_j$. Exactness at the first term is clear. Let $(f_i)_i$ be in the kernel of the last map. The ring $\widehat{\mathcal{O}}_{X,x}/I_i$ is the ring of power series in the variables x_k with $k \notin J_i$, and we write f_i as such a power series. For a monomial m in $x_1, \dots, x_n$, let $S(m)$ be the set of indices i such that no variable of m lies in $\{x_k \mid k \in J_i\}$. If $i, j \in S(m)$, then m is a nonzero monomial in $\widehat{\mathcal{O}}_{X,x}/(I_i + I_j)$, so the coefficients of m in f_i and in f_j agree. Let c_m be this common coefficient if $S(m) \neq \emptyset$, and $c_m := 0$ otherwise. Then $f := \sum_m c_m m$ satisfies $f \equiv f_i \pmod{I_i}$ for every i . This proves the exactness at the middle term, and the last assertion follows by applying the exact sequence on X° . $\square$

Theorem 6.12. *Notation as in Set-up 6.6. Let $\mathcal{A} := \text{Bir}_V^c(W, \Theta_W)$ and $G := \text{Gal}(\mathbb{C}(P)/\mathbb{C}(V))$.*

- (1) *V is normal, (W, Θ_W) is klt, and $\rho : P \rightarrow V$ is the quotient of P by G . Every element of G is induced by an element of $\mathcal{A}$.*
- (2) *Let $b_{\mathfrak{R}}$ be a positive integer such that $b_{\mathfrak{R}}\mathfrak{R} \subset \mathbb{Z}$ and $b_{\mathfrak{R}}\Phi \subset \mathbb{Z}$, let m be the integer given by Theorem 4.7 for d and Γ , and let $m_0 := \text{lcm}(m, a(0, \Gamma), \dots, a(d-1, \Gamma))$, where $a(s, \Gamma)$ is as in Theorem 2.11. Put*

$$(6.11) \quad a_0 := 2\text{lcm}(b_{\mathfrak{R}}, m_0), \quad Q_0 := \text{lcm}_{0 \leq s \leq d-1} L(s, a_0), \quad N_0 := a_0 Q_0,$$

where $L(s, a_0)$ is as in Theorem 2.14. Then N_0 is a positive even integer depending only on d and Φ , and $N_0\Phi \subset \mathbb{Z}$, $N_0\mathfrak{R} \subset \mathbb{Z}$. For every positive multiple N of N_0 the following hold.

- (3) *Let C be an lc center of (Y, A) with $q(C) = V$ and $\tilde{C} \subset \tilde{Z}_V$ its stratum. Then $(\mu|_{\tilde{C}})^*(\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{C}}) = \mathcal{O}_C(-NK_C - \lceil N\Theta_C \rceil)$ by (6.5) and (6.8). In particular, for every open subset $U' \subset V$, the sections of $\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{W}}$ over $(q\mu)^{-1}(U') \cap \tilde{W}$ are the rational $(-N)$ -canonical forms β on W such that*

$$(6.12) \quad \text{Div}(\beta) \geq \lceil N\Theta_W \rceil \quad \text{on } (\rho h)^{-1}(U'),$$

together with 0.

- (4) *Let $U' \subset V$ be an open subset, β a form satisfying (6.12), and $\gamma \in \mathcal{A}$. Then $\gamma^*\beta$ satisfies (6.12). The resulting action of $\mathcal{A}$ on the sheaf $(\rho h)_*\mathcal{O}_W(-NK_W - \lceil N\Theta_W \rceil)$ on V is trivial on the kernel of $\mathcal{A} \rightarrow G$, so it factors through G . We denote by $((\rho h)_*\mathcal{O}_W(-NK_W - \lceil N\Theta_W \rceil))^G$ the subsheaf of $\mathcal{A}$ -invariant sections.*
- (5) *Restriction from $\tilde{Z}_V$ to $\tilde{W}$, followed by the identification in (3), induces an isomorphism of sheaves on V*

$$(6.13) \quad (q\mu|_{\tilde{Z}_V})^*(\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{Z}_V}) \simeq ((\rho h)_*\mathcal{O}_W(-NK_W - \lceil N\Theta_W \rceil))^G.$$

In particular, the restriction map $H^0(\tilde{Z}_V, \mathcal{O}_{\tilde{Y}}(L_N)) \rightarrow H^0(\tilde{W}, \mathcal{O}_{\tilde{Y}}(L_N))$ is injective, and

$$(6.14) \quad \text{im} [H^0(\tilde{Z}_V, \mathcal{O}_{\tilde{Y}}(L_N)) \rightarrow H^0(\tilde{W}, \mathcal{O}_{\tilde{Y}}(L_N))] = H^0(W, \mathcal{O}_W(-NK_W - \lceil N\Theta_W \rceil))^G.$$

- (6) *The restriction map $H^0(\tilde{Y}, \mathcal{O}_{\tilde{Y}}(L_N)) \rightarrow H^0(\tilde{Z}_V, \mathcal{O}_{\tilde{Y}}(L_N))$ is surjective.*
- (7) *Let $U' \subset V$ be an open subset, $\beta \neq 0$ an $\mathcal{A}$ -invariant form satisfying (6.12), and s the section of $\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{Z}_V}$ over $(q\mu)^{-1}(U')$ with $s|_{\tilde{W}} = \beta$ given by (5). Let C be an lc center of (Y, A) with $q(C) = V$. Then there exist an a_0 -canonical form θ_C^0 on C and a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor D_C on P_C with*

$$\text{Div}(\theta_C^0) + a_0\Theta_C = f_C^*(a_0D_C),$$

and a unique nonzero $a_C \in \mathbb{C}(P_C)$ with the following properties. Put

$$(6.15) \quad \theta_C := \frac{(\theta_C^0)^{\otimes (N/a_0)}}{f_C^{*}a_C}, \quad E'_C := \frac{1}{N} \text{div}_{P_C}(a_C) - D_C, \quad \Theta_C^+ := \Theta_C + f_C^*E'_C.$$

Then $E'_C \geq 0$ on $\rho_C^{-1}(U')$, the N -canonical form θ_C satisfies $\text{Div}(\theta_C) = -N\Theta_C^+$, and $s|_{\tilde{C}}$ corresponds under (3) to θ_C^{-1} . Moreover, let W_C be a source of V contained in C , with Stein factorization $W_C \xrightarrow{h_C} P_{W_C} \rightarrow V$. Then $s|_{\tilde{W}_C}$ is the crepant transport of β , that is, $s|_{\tilde{W}_C} = (\phi^{-1})^*\beta$ for every crepant birational map $\phi : (W, \Theta_W) \dashrightarrow (W_C, \Theta_{W_C})$ over V , and

$$(6.16) \quad \frac{1}{N} \text{Div}(s|_{\tilde{W}_C}) - \Theta_{W_C} = (f_C|_{W_C})^* E'_C.$$

In (7), the section $s|_{\tilde{W}_C}$ does not depend on ϕ by (4), since two such maps differ by an element of $\mathcal{A}$, and such maps exist by Theorem 5.2(1) and [Kol16, Definition 18], because W and W_C are sources of V . By (6.10), $\Theta_C^+ - \Theta_C = f_C^* E'_C$ is the difference between the boundary $\frac{1}{N} \text{Div}(\theta_C^{-1})$ defined by the section and Θ_C , and $N(K_C + \Theta_C^+) \sim 0$ over $\rho_C^{-1}(U')$.

Proof. Step 1. We prove (1).

By (6.1), $K_Y + A \sim_{\mathbb{Q}} q^*(-H)$, so Lemma 6.4 shows that $[T, -H]$ is a quasi-log canonical pair whose qlc centers are T and the lc centers of q . Let $x \in V$ be a point and W_x the minimal qlc center through x given by Theorem 6.3. Then $W_x \subset V \subsetneq T$, so W_x is an lc center of q contained in V , and $W_x = V$ by the minimality of V . Hence V is normal at every point.

Let W' be any source of V . By [Kol16, Definition 18], the lc centers of $(W', \Theta_{W'})$ are the lc centers of (Y, A) properly contained in W' . Such an lc center C' satisfies $q(C') \subset V$, hence $q(C') = V$ by the minimality of V , which contradicts the minimality of W' . Thus $(W', \Theta_{W'})$ is klt. In particular every source of V is a minimal lc center of (Y, A) .

By Theorem 6.2(1), $\mathbb{C}(P)/\mathbb{C}(V)$ is Galois, so $\mathbb{C}(P)^G = \mathbb{C}(V)$. As G acts on P over V , the morphism $P/G \rightarrow V$ is finite and birational onto the normal variety V , hence an isomorphism. The last assertion of (1) is Theorem 6.2(2).

Step 2. We fix the constants in (2), and for every lc center C of (Y, A) with $q(C) = V$ we construct an a_0 -canonical form θ_C^0 and describe the sections of $\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{C}}$ and their restrictions to the sources of V contained in C .

The integers in (6.11) depend only on d and Φ , since $\mathfrak{R}$ and Γ do, and N_0 is even. As $b_{\mathfrak{R}} \mid N_0$, we have $N_0\Phi \subset \mathbb{Z}$ and $N_0\mathfrak{R} \subset \mathbb{Z}$. Fix a positive multiple N of N_0 .

Let C be an lc center of (Y, A) with $q(C) = V$. By Set-up 6.6, $f_C : (C, \Theta_C) \rightarrow P_C$ is a dlt crepant log structure, f_C is a contraction, $\Theta_C \in \Gamma$, $\dim C \leq d-1$, and $K_C + \Theta_C \sim_{\mathbb{Q}} f_C^* L_C^0$ with $L_C^0 := -\rho_C^*(H|_V)$. If $\dim V > 0$, then Theorem 4.7(1), applied to d , Γ , and f_C , gives a nonzero $\psi \in \mathbb{C}(C)$ and a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor D_C on P_C such that $m(K_C + \Theta_C) + \text{div}(\psi) = f_C^*(mD_C)$; we put $\psi_C := \psi^{a_0/m}$. If $\dim V = 0$, then P_C is a point and C is a projective variety with $K_C + \Theta_C \sim_{\mathbb{Q}} 0$, so $a(\dim C, \Gamma)(K_C + \Theta_C) \sim 0$ by Theorem 2.11; since $a(\dim C, \Gamma) \mid a_0$, there is a nonzero $\psi_C \in \mathbb{C}(C)$ with $a_0(K_C + \Theta_C) + \text{div}(\psi_C) = 0$, and we put $D_C := 0$. In both cases the a_0 -canonical form $\theta_C^0 := \psi_C \omega_C^{\otimes a_0}$ satisfies

$$(6.17) \quad \text{Div}(\theta_C^0) + a_0\Theta_C = f_C^*(a_0D_C).$$

Let $U' \subset V$ be an open subset. We apply Lemma 6.8 to f_C , θ_C^0 , and $a_0 \mid N$. Together with Lemma 6.7(2) and (6.5), it shows that the sections of $\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{C}}$ over $(q\mu)^{-1}(U') \cap \tilde{C}$ are 0 and the forms

$$(6.18) \quad \beta_C = (f_C^* u) (\theta_C^0)^{\otimes (-N/a_0)}, \quad u \in \mathbb{C}(P_C) \setminus \{0\}, \quad \text{div}(u) \geq ND_C \text{ on } \rho_C^{-1}(U').$$

Let W' be a source of V contained in C , let $W' \xrightarrow{h'} P' \xrightarrow{\rho'} V$ be its Stein factorization, and let $\pi' : P' \rightarrow P_C$ be the morphism over V induced by $f_C|_{W'}$, so that $f_C|_{W'} = \pi' \circ h'$; it exists because $P' = \text{Spec}_V(q|_{W'})_* \mathcal{O}_{W'}$, and it is finite and surjective. Let $D'_1, \dots, D'_l$ be the components of S_C containing $\tilde{W}'$. By (6.7) and (6.17),

$$\text{Div}_{\tilde{C}}(\theta_C^0) + a_0(D'_1 + \dots + D'_l) = (f_C \mu_C)^*(a_0D_C) - a_0(S_C - D'_1 - \dots - D'_l) - a_0E_C.$$

No component of the right hand side contains $\widetilde{W}'$: $\widetilde{W}'$ dominates P_C , and $\widetilde{W}'$ is a stratum of the simple normal crossing divisor $S_C + \text{Supp } E_C$. Hence the residue

$$\theta_{W'} := (\theta_C^0)_{\widetilde{W}'}$$

is a well-defined nonzero a_0 -canonical form on W' , and by (6.4), Set-up 6.6, and Lemma 6.7(1) applied to W' ,

$$(6.19) \quad \text{Div}(\theta_{W'}) + a_0 \Theta_{W'} = (f_C|_{W'})^*(a_0 D_C) = h'^*(a_0 \pi'^* D_C).$$

Moreover $\theta_{W'}$ is the Poincaré residue $R_{C \rightarrow W'}^{a_0}(\theta_C^0|_{W'})$, as explained after (6.4). For a section β_C as in (6.18), the rational function $f_C^* u$ is regular and nonzero at the generic point of $\widetilde{W}'$, so the multiplicativity of residues gives

$$(6.20) \quad \beta_C|_{\widetilde{W}'} = ((f_C^* u)|_{W'}) \theta_{W'}^{\otimes(-N/a_0)} = (h'^* \pi'^* u) \theta_{W'}^{\otimes(-N/a_0)} \neq 0.$$

In particular, restriction from $\widetilde{C}$ to $\widetilde{W}'$ is injective on the sections of $\mathcal{O}_{\widetilde{Y}}(L_N)|_{\widetilde{C}}$ over $(q\mu)^{-1}(U') \cap \widetilde{C}$.

Step 3. We prove (3) and (4).

Since N is even and $N\Re \subset \mathbb{Z}$, (3) is Lemma 6.7(2) together with (6.5).

Let (W_1, Θ_1) and (W_2, Θ_2) be sources of V with Stein factorizations $W_i \xrightarrow{h_i} P_i \xrightarrow{\rho_i} V$, let $\phi : (W_1, \Theta_1) \dashrightarrow (W_2, \Theta_2)$ be a crepant birational map over V , and let β be a rational $(-N)$ -canonical form on W_2 with $\text{Div}(\beta) \geq N\Theta_2$ on $(\rho_2 h_2)^{-1}(U')$. We claim that $\text{Div}(\phi^* \beta) \geq N\Theta_1$ on $(\rho_1 h_1)^{-1}(U')$. Let $p : W' \rightarrow W_1$ and $r : W' \rightarrow W_2$ be a common resolution of ϕ , so that $r = \phi \circ p$, and let $K_{W'} + \Gamma' := p^*(K_{W_1} + \Theta_1) = r^*(K_{W_2} + \Theta_2)$. Since ϕ is over V , we have $r^{-1}((\rho_2 h_2)^{-1}(U')) = p^{-1}((\rho_1 h_1)^{-1}(U'))$. By the computation of (5.12),

$$\text{Div}_{W'}(r^* \beta) = r^*(\text{Div}(\beta) - N\Theta_2) + N\Gamma'.$$

The $\mathbb{Q}$ -divisor $\text{Div}(\beta) - N\Theta_2$ is $\mathbb{Q}$ -Cartier, being the difference of $\text{Div}(\beta) - NK_{W_2}$ and $N(K_{W_2} + \Theta_2)$, and it is effective over U' . Hence $r^*(\text{Div}(\beta) - N\Theta_2) \geq 0$ over U' , and pushing forward by p gives $\text{Div}_{W_1}(\phi^* \beta) = p_* r^*(\text{Div}(\beta) - N\Theta_2) + N\Theta_1 \geq N\Theta_1$ over U' . This proves the claim, and the first assertion of (4) is the case $W_1 = W_2 = W$, $\phi = \gamma$.

Let $\mathcal{A}_0$ be the kernel of $\mathcal{A} \rightarrow G$, that is, the set of $\gamma \in \mathcal{A}$ with $h \circ \gamma = h$. We construct a rational N_0 -canonical form θ_0 on W and a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor D_0 on P such that $\text{Div}(\theta_0) + N_0 \Theta_W = h^*(N_0 D_0)$ and $\gamma^* \theta_0 = \theta_0$ for every $\gamma \in \mathcal{A}$. If $\dim V = 0$, then W is projective, (W, Θ_W) is klt, and $\text{Div}(\theta_W^0) = -a_0 \Theta_W$ by (6.17). Theorem 2.14 gives $\gamma^* \theta_W^0 = c_\gamma \theta_W^0$ with $c_\gamma^{L(\dim W, a_0)} = 1$ for every $\gamma \in \mathcal{A}$; as $\dim W \leq d - 1$, $c_\gamma^{Q_0} = 1$, and we put $\theta_0 := (\theta_W^0)^{\otimes Q_0}$ and $D_0 := 0$. If $\dim V > 0$, we apply Theorem 5.5 to $h : (W, \Theta_W) \rightarrow P$, with $m := a_0$, $\psi := \psi_W$, $D := D_W$, the group G acting on P with quotient ρ , and $S := U$. Its hypotheses hold: h is a contraction, $0 < \dim P \leq \dim W \leq d$, the generic fiber of h is geometrically klt because (W, Θ_W) is klt, (6.17) is (5.5), every element of G lifts to an element of $\mathcal{A}$ by (1), and $V \rightarrow U$ is projective. Its constants are $E_0 = \text{lcm}_{0 \leq j \leq d-1} L(j, a_0) = Q_0$ and $b = a_0 Q_0 = N_0$. Theorem 5.5(1) gives a nonzero $\chi \in \mathbb{C}(P)$ such that $\theta_0 := (\chi \circ h) \psi_W^{Q_0} \omega_W^{\otimes N_0}$ satisfies $\text{Div}(\theta_0) + N_0 \Theta_W = h^*(N_0 D_W + \text{div}(\chi))$ and $\gamma^* \theta_0 = \theta_0$ for every $\gamma \in \mathcal{A}$; we put $D_0 := D_W + \frac{1}{N_0} \text{div}(\chi)$.

Now let β satisfy (6.12) and $\gamma \in \mathcal{A}_0$. By Lemma 6.8 applied to h , θ_0 , and $N_0 \mid N$, we have $\beta = (h^* u) \theta_0^{\otimes(-N/N_0)}$ for some nonzero $u \in \mathbb{C}(P)$. Since $h \circ \gamma = h$ and $\gamma^* \theta_0 = \theta_0$,

$$\gamma^* \beta = (h^* u) (\gamma^* \theta_0)^{\otimes(-N/N_0)} = \beta.$$

Thus $\mathcal{A}_0$ acts trivially, which proves (4).

Step 4. We show that the restriction map in (5) is injective.

Let $U' \subset V$ be an open subset and s a section of $\mathcal{O}_{\widetilde{Y}}(L_N)|_{\widetilde{Z}_V}$ over $(q\mu)^{-1}(U')$ such that $s|_{\widetilde{W}} = 0$. Let $\widetilde{C} \subset \widetilde{Z}_V$ be a stratum, $C := \mu(\widetilde{C})$, and W_C a source of V contained in C . By Step

2, $s|_{\tilde{C}} = 0$ if $s|_{\tilde{W}_C} = 0$. Since $\tilde{Z}_V$ is reduced and is the union of its strata, it suffices to show that $s|_{\tilde{W}_C} = 0$ for every source W_C of V .

Let v be the generic point of V . The fiber $q^{-1}(v)$ is connected [Har77, Corollary III.11.3], and W and W_C are minimal lc centers of (Y, A) with $v \in q(W) \cap q(W_C)$ by Step 1. By Theorem 5.2(1), W and W_C are $\mathbb{P}^1$ -linked. By Definition 5.1, there exist lc centers $W = Z_1, Z_2, \dots, Z_n = W_C$ of (Y, A) and lc centers $W_i'' \supset Z_i \cup Z_{i+1}$ for $1 \leq i < n$ such that $q(W_i'') = q(Z_i) = q(Z_{i+1}) = V$ and $(W_i'', \Theta_{W_i''})$ is crepant birational to a standard $\mathbb{P}^1$ -link with Z_i and Z_{i+1} mapping to its two lc centers. Since the two lc centers of a standard $\mathbb{P}^1$ -link are prime divisors, its lc places are the valuations along them, and crepant birational pairs have the same lc places. Thus Z_i and Z_{i+1} are the only lc centers of $(W_i'', \Theta_{W_i''})$, their adjunction pairs are klt by the correspondence of lc centers in [Kol16, Definition 18], and they are sources of V . If $s|_{\tilde{Z}_i} = 0$, then $s|_{\tilde{W}_i''} = 0$ by Step 2 applied to W_i'' and its source Z_i , hence $s|_{\tilde{Z}_{i+1}} = 0$. By induction on i , $s|_{\tilde{W}_C} = 0$.

Step 5. We show that there is a dense open subset $V^0 \subset V$ with the following property. For every $\sigma \in G$ there is a birational self-map γ_σ of W with $h \circ \gamma_\sigma = \sigma \circ h$ whose restriction to $(\rho h)^{-1}(V^0)$ is a crepant birational self-map of $((\rho h)^{-1}(V^0), \Theta_W|_{(\rho h)^{-1}(V^0)})$, such that $\gamma_\sigma^*(s|_{\tilde{W}}) = s|_{\tilde{W}}$ for every open subset $U' \subset V$ and every section s of $\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{Z}_V}$ over $(q\mu)^{-1}(U')$.

If $\dim V = 0$, then $P = V$ is a point, G is trivial, and $V^0 := V$ and $\gamma_{\text{id}} := \text{id}_W$ have the required property. Assume that $\dim V > 0$. Since $\mathbb{C}(P)/\mathbb{C}(V)$ is a finite separable extension, $\mathbb{C}(P) = \mathbb{C}(V)[x]/(p)$ for a monic irreducible polynomial p with coefficients in $\mathbb{C}(V)$. Let $T_1 \subset T$ be an affine open subset meeting V such that the coefficients of p are regular on $V \cap T_1$, and let $\tilde{p} \in \mathcal{O}(T_1)[x]$ be a monic polynomial whose restriction to $V \cap T_1$ is p . Let $T^0 \subset T_1$ be the open subset where the discriminant of $\tilde{p}$ does not vanish; it meets V because p is separable. Put $V^0 := V \cap T^0$ and $P^0 := \rho^{-1}(V^0)$. The scheme $\text{Spec } \mathcal{O}(T^0)[x]/(\tilde{p})$ is finite and étale over T^0 , and its restriction over V^0 is $\text{Spec } \mathcal{O}(V^0)[x]/(p)$, which is finite and étale over V^0 with function field $\mathbb{C}(P)$, hence equal to P^0 . Let T' be the connected component of $\text{Spec } \mathcal{O}(T^0)[x]/(\tilde{p})$ containing P^0 . Then T' is a normal variety, $\tau : T' \rightarrow T^0$ is finite and étale, and $\tau^{-1}(V^0) = P^0$ with $\tau|_{P^0} = \rho|_{P^0}$. In particular $P^0 \rightarrow V^0$ is a finite étale Galois cover with group G , so $P^0 \times_{V^0} P^0$ is the disjoint union of the graphs of the elements of G .

Let $Y^0 := q^{-1}(T^0)$, $Y' := Y^0 \times_{T^0} T'$, with projections $\tau_Y : Y' \rightarrow Y^0$ and $q' : Y' \rightarrow T'$, and $A' := \tau_Y^* A$. Let $\tilde{Y}' := \tilde{Y} \times_{T^0} T'$, with projections $\tau_{\tilde{Y}} : \tilde{Y}' \rightarrow \tilde{Y}$ and $\mu' : \tilde{Y}' \rightarrow Y'$, and let $P'_1, E',$ and L'_N be the pullbacks of $P_1, E,$ and L_N . Since τ_Y and $\tau_{\tilde{Y}}$ are étale and the generic fibers of q and $q\mu$ are geometrically integral, Y' and $\tilde{Y}'$ are normal varieties and q' is a contraction. The pair (Y', A') is dlt of dimension $\leq d$ with $A' \in \Phi$, and μ' is a log resolution of (Y', A') with $K_{\tilde{Y}'} + P'_1 + E' = \mu'^*(K_{Y'} + A')$. The lc centers of (Y', A') are the irreducible components of the preimages of the lc centers of (Y^0, A) , and μ' is an isomorphism over their generic points. The strata of P'_1 are the irreducible components of the preimages of the strata of P_1 . If C is an lc center of (Y, A) with $q(C) = V$, then the irreducible components of $\tau_Y^{-1}(C)$ are isomorphic over P^0 to $C \times_{P_C} P^0$ for the finitely many morphisms $P^0 \rightarrow P_C$ over V^0 , because P_C is étale over V^0 . Hence they are lc centers of (Y', A') with image P^0 and with connected fibers over P^0 . Conversely every lc center of (Y', A') with image P^0 arises in this way. Therefore Set-up 6.6, the residue formalism, Lemma 6.7, and Step 2 apply to (Y', A') , μ', P'_1, E', L'_N , the lc center P^0 of q' , and the lc centers of (Y', A') with image P^0 , with V and the P_C replaced by P^0 . Indeed, their proofs use only that q' is a contraction with $K_{Y'} + A' \sim_{\mathbb{Q}} -q'^* \tau^* H$, that (Y', A') is dlt with $A' \in \Phi$, and that μ' is a log resolution of (Y', A') which is an isomorphism over the generic points of its lc centers. The differentials of A' lie in Γ by Lemma 4.4(1). Let $\tilde{Z}'$ be the union of the strata of P'_1 with image P^0 ; it is the preimage of $\tilde{Z}_V \cap (q\mu)^{-1}(V^0)$.

Put $W^0 := (\rho h)^{-1}(V^0)$. Then $\tau_Y^{-1}(W^0) = W^0 \times_{V^0} P^0$ is the disjoint union of the subvarieties

$$W_\sigma := \{(w, \sigma(h(w))) \mid w \in W^0\}, \quad \sigma \in G,$$

and the first projection $j_\sigma : W_\sigma \rightarrow W^0$ is an isomorphism identifying $(W_\sigma, \Theta_{W_\sigma})$ with (W^0, Θ_{W^0}) , with $q'|_{W_\sigma} = \sigma \circ h \circ j_\sigma$. Since (W, Θ_W) is klt, the W_σ are minimal lc centers of (Y', A') with image P^0 . Let s be a section of $\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{Z}_V}$ over $(q\mu)^{-1}(U')$, $\beta := s|_{\tilde{W}}$, and $s' := \tau_Y^* s$, a section of $\mathcal{O}_{\tilde{Y}'}(L'_N)|_{\tilde{Z}'}$ over $(q'\mu')^{-1}(\tau^{-1}(U' \cap V^0))$. Since the residue isomorphisms (6.5) commute with étale base change, $s'|_{\tilde{W}_\sigma} = j_\sigma^* \beta$ for every $\sigma \in G$, where $\tilde{W}_\sigma$ is the stratum of P_1' with image W_σ .

Fix $\sigma \in G$. The fiber of q' over the generic point of P^0 is connected [Har77, Corollary III.11.3]. By Theorem 5.2(1) and Definition 5.1, as in Step 4, there exist minimal lc centers $W_{\text{id}} = Z'_1, Z'_2, \dots, Z'_n = W_\sigma$ of (Y', A') and lc centers $W''_i \supset Z'_i \cup Z'_{i+1}$ of (Y', A') with $q'(W''_i) = q'(Z'_i) = P^0$, for $1 \leq i < n$, such that Z'_i and Z'_{i+1} are the only lc centers of $(W''_i, \Theta_{W''_i})$. Let $f''_i := q'|_{W''_i} : W''_i \rightarrow P^0$; it is a contraction. By Step 2 applied to (Y', A') and W''_i , there are an a_0 -canonical form θ''_i on W''_i and a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor D''_i on P^0 with $\text{Div}(\theta''_i) + a_0 \Theta_{W''_i} = f''_{i*}(a_0 D''_i)$, and

$$s'|_{\tilde{W}''_i} = (f''_{i*} u_i) (\theta''_i)^{\otimes (-N/a_0)} \quad \text{and} \quad s'|_{\tilde{Z}'_j} = ((f''_{i*} u_i)|_{Z'_j}) (\theta''_i)^{\otimes (-N/a_0)}_{Z'_j} \quad (j = i, i+1)$$

for some $u_i \in \mathbb{C}(P^0)$, by (6.18) and (6.20); here $u_i = 0$ is allowed. By Lemma 6.9(2), applied to f''_i , θ''_i , Z'_i , and Z'_{i+1} , there is a crepant birational map $\phi_i : Z'_{i+1} \dashrightarrow Z'_i$ over P^0 with $\phi_i^*(\theta''_i)_{Z'_i} = (\theta''_i)_{Z'_{i+1}}$. Since ϕ_i is over P^0 , $(f''_{i*} u_i)|_{Z'_{i+1}} = \phi_i^*((f''_{i*} u_i)|_{Z'_i})$, and therefore

$$s'|_{\tilde{Z}'_{i+1}} = \phi_i^*(s'|_{\tilde{Z}'_i}) \quad (1 \leq i < n).$$

Put $\Psi := \phi_{n-1}^{-1} \circ \dots \circ \phi_1^{-1} : W_{\text{id}} \dashrightarrow W_\sigma$, a crepant birational map over P^0 with $\Psi^*(s'|_{\tilde{W}_\sigma}) = s'|_{\tilde{W}_{\text{id}}}$, and $\gamma := j_\sigma \circ \Psi \circ j_{\text{id}}^{-1}$. Then γ is a birational self-map of W over V , and its restriction to W^0 is a crepant birational self-map of (W^0, Θ_{W^0}) . Since $q'|_{W_\sigma} \circ \Psi = q'|_{W_{\text{id}}}$, we get $\sigma \circ h \circ j_\sigma \circ \Psi = h \circ j_{\text{id}}$, that is, $h \circ \gamma = \sigma^{-1} \circ h$. Moreover

$$\gamma^* \beta = (j_{\text{id}}^{-1})^* \Psi^* j_\sigma^* \beta = (j_{\text{id}}^{-1})^* \Psi^*(s'|_{\tilde{W}_\sigma}) = (j_{\text{id}}^{-1})^*(s'|_{\tilde{W}_{\text{id}}}) = \beta.$$

The map γ was constructed from the chain $Z'_1, \dots, Z'_n$ alone and does not depend on s . Replacing σ by σ^{-1} , we obtain γ_σ as required.

Step 6. We prove (5) and (7).

Let $U' \subset V$ be an open subset, s a section of $\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{Z}_V}$ over $(q\mu)^{-1}(U')$, and $\beta := s|_{\tilde{W}}$. Let $\gamma \in \mathcal{A}$ lie over $\sigma \in G$, and let V^0 and γ_σ be as in Step 5. Then $g := \gamma_\sigma^{-1} \circ \gamma$ is a birational self-map of W with $h \circ g = h$, and its restriction to $(\rho h)^{-1}(V^0)$ is a crepant birational self-map of $((\rho h)^{-1}(V^0), \Theta_W|_{(\rho h)^{-1}(V^0)})$. By Lemma 6.10, applied to h and $P^0 := \rho^{-1}(V^0)$, we have $g \in \mathcal{A}_0$. By (4) and Step 5, $\gamma^* \beta = g^* \gamma_\sigma^* \beta = g^* \beta = \beta$. Thus β is $\mathcal{A}$ -invariant. Together with Step 4, it remains to show that every nonzero $\mathcal{A}$ -invariant β satisfying (6.12) is the restriction of a section s over $(q\mu)^{-1}(U')$, and that (7) holds for s .

Let β be such a form. For every lc center C of (Y, A) with $q(C) = V$ we fix a source W_C of V contained in C , with Stein factorization $W_C \xrightarrow{h_C} P_{W_C} \xrightarrow{\rho_{W_C}} V$ and with the finite surjective morphism $\pi_C : P_{W_C} \rightarrow P_C$ of Step 2, so that $f_C|_{W_C} = \pi_C \circ h_C$. By Theorem 5.2(1) and [Kol16, Definition 18], there is a crepant birational map $\phi_C : (W, \Theta_W) \dashrightarrow (W_C, \Theta_{W_C})$ over V . Put $\beta_{W_C} := (\phi_C^{-1})^* \beta$. By the claim in Step 3, $\text{Div}(\beta_{W_C}) \geq \lceil N \Theta_{W_C} \rceil$ on $(\rho_{W_C} h_C)^{-1}(U')$. Since β is $\mathcal{A}$ -invariant and two choices of ϕ_C differ by an element of $\mathcal{A}$, the form β_{W_C} does not depend on ϕ_C , and it is invariant under $\text{Bir}_V^c(W_C, \Theta_{W_C}) = \phi_C \mathcal{A} \phi_C^{-1}$. More generally, for every source W' of V we define $\beta_{W'}$ in the same way, so that $\psi^* \beta_{W_2} = \beta_{W_1}$ for every crepant birational map $\psi : (W_1, \Theta_{W_1}) \dashrightarrow (W_2, \Theta_{W_2})$ over V between sources of V .

By (6.19), the form $\theta_{W_C} = (\theta_C^0)_{\tilde{W}_C}$ satisfies $\text{Div}(\theta_{W_C}) + a_0 \Theta_{W_C} = h_C^*(a_0 \pi_C^* D_C)$. Lemma 6.8, applied to h_C , θ_{W_C} , and $a_0 \mid N$, gives a nonzero $a \in \mathbb{C}(P_{W_C})$ with

$$\beta_{W_C} = (h_C^* a) \theta_{W_C}^{\otimes (-N/a_0)} \quad \text{and} \quad \frac{1}{N} \text{div}(a) - \pi_C^* D_C \geq 0 \quad \text{on } \rho_{W_C}^{-1}(U').$$

The extension $\mathbb{C}(P_{W_C})/\mathbb{C}(V)$ is Galois by Theorem 6.2(1), applied to the source W_C , and $\mathbb{C}(V) \subset \mathbb{C}(P_C) \subset \mathbb{C}(P_{W_C})$. Hence $\mathbb{C}(P_{W_C})/\mathbb{C}(P_C)$ is Galois, its Galois group H_C is the group of automorphisms of P_{W_C} over P_C , and $\mathbb{C}(P_{W_C})^{H_C} = \mathbb{C}(P_C)$. Let $\tau \in H_C$. By Step 2, $\theta_{W_C} = R_{C \rightarrow W_C}^{a_0}(\theta_C^0|_{W_C})$. By Lemma 6.9(3), applied to the dlt crepant log structure $f_C : (C, \Theta_C) \rightarrow P_C$, the form θ_C^0 , the minimal lc center W_C of (C, Θ_C) , and τ , there is a crepant birational self-map γ_τ of (W_C, Θ_{W_C}) over P_C with $h_C \circ \gamma_\tau = \tau \circ h_C$ and $\gamma_\tau^* \theta_{W_C} = \theta_{W_C}$. As $\gamma_\tau \in \text{Bir}_V^c(W_C, \Theta_{W_C})$,

$$\beta_{W_C} = \gamma_\tau^* \beta_{W_C} = (h_C^* \tau^* a) \theta_{W_C}^{\otimes(-N/a_0)},$$

so $\tau^* a = a$ for every $\tau \in H_C$, and $a = \pi_C^* a_C$ for a unique nonzero $a_C \in \mathbb{C}(P_C)$. Put $E'_C := \frac{1}{N} \text{div}(a_C) - D_C$. Then $\pi_C^* E'_C = \frac{1}{N} \text{div}(a) - \pi_C^* D_C \geq 0$ on $\rho_{W_C}^{-1}(U') = \pi_C^{-1}(\rho_C^{-1}(U'))$, and since π_C is finite and surjective, $E'_C \geq 0$ on $\rho_C^{-1}(U')$. By (6.18),

$$\beta_C := (f_C^* a_C) (\theta_C^0)^{\otimes(-N/a_0)}$$

is a section of $\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{C}}$ over $(q\mu)^{-1}(U') \cap \tilde{C}$, which we denote by s_C , and $s_C|_{\tilde{W}_C} = (h_C^* \pi_C^* a_C) \theta_{W_C}^{\otimes(-N/a_0)} = \beta_{W_C}$ by (6.20).

We show that $s_C|_{\tilde{W}'} = \beta_{W'}$ for every source W' of V contained in C . By Step 1, W_C and W' are minimal lc centers of (C, Θ_C) with image P_C . Lemma 6.9(2), applied to f_C and θ_C^0 , gives a crepant birational map $\psi : (W', \Theta_{W'}) \dashrightarrow (W_C, \Theta_{W_C})$ over P_C with $\psi^* \theta_{W_C} = \theta_{W'}$. Since ψ is over P_C , $(f_C^* a_C)|_{W'} = \psi^*((f_C^* a_C)|_{W_C})$, so (6.20) gives

$$s_C|_{\tilde{W}'} = \psi^* \left(((f_C^* a_C)|_{W_C}) \theta_{W_C}^{\otimes(-N/a_0)} \right) = \psi^* \beta_{W_C} = \beta_{W'}.$$

Now let $\tilde{C}_1$ and $\tilde{C}_2$ be irreducible components of $\tilde{Z}_V$ and $\tilde{R}$ an irreducible component of $\tilde{C}_1 \cap \tilde{C}_2$. By Set-up 6.6, $\tilde{R}$ is a stratum of P_1 contained in $\tilde{Z}_V$; let $R := \mu(\tilde{R})$ and W_R a source of V contained in R . By the transitivity of residues and the previous paragraph, $s_{C_1}|_{\tilde{R}}$ and $s_{C_2}|_{\tilde{R}}$ are sections of $\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{R}}$ over $(q\mu)^{-1}(U') \cap \tilde{R}$ whose restrictions to $\tilde{W}_R$ both equal β_{W_R} . By Step 2, $s_{C_1}|_{\tilde{R}} = s_{C_2}|_{\tilde{R}}$. Since $\tilde{C}_1 \cap \tilde{C}_2$ is reduced (Lemma 6.11), s_{C_1} and s_{C_2} agree on $\tilde{C}_1 \cap \tilde{C}_2$. By Lemma 6.11, applied on the open subset $(q\mu)^{-1}(U')$ of $\tilde{Y}$, the sections s_C for the irreducible components $\tilde{C}$ of $\tilde{Z}_V$ glue to a section s of $\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{Z}_V}$ over $(q\mu)^{-1}(U')$. If $\tilde{C} \supset \tilde{W}$ is an irreducible component of $\tilde{Z}_V$, then $s|_{\tilde{W}} = s_C|_{\tilde{W}} = \beta_W = \beta$. This proves (6.13), and (6.14) is the case $U' = V$.

Finally, let C be any lc center of (Y, A) with $q(C) = V$, and let $\tilde{C}_1 \supset \tilde{C}$ be an irreducible component of $\tilde{Z}_V$. Then $s|_{\tilde{W}_C} = s_{C_1}|_{\tilde{W}_C} = \beta_{W_C} = s_C|_{\tilde{W}_C}$, so $s|_{\tilde{C}} = s_C$ by Step 2. Thus $s|_{\tilde{C}}$ corresponds to $\beta_C = \theta_C^{-1}$ with θ_C as in (6.15), and $\text{Div}(\theta_C) = -\text{Div}(\beta_C) = -N\Theta_C - f_C^*(NE'_C) = -N\Theta_C^+$ by (6.10). Let W' be any source of V contained in C , with Stein factorization $W' \xrightarrow{h'} P' \rightarrow V$ and with $\pi' : P' \rightarrow P_C$ as in Step 2. Then $s|_{\tilde{W}'} = s_C|_{\tilde{W}'} = \beta_{W'}$ is the crepant transport of β , and $\beta_{W'} = (h'^* \pi'^* a_C) \theta_{W'}^{\otimes(-N/a_0)}$ by (6.20). By (6.19), Lemma 6.8 applies to h' , $\theta_{W'}$, $\pi'^* D_C$, and $u := \pi'^* a_C$, and (6.10) gives

$$\frac{1}{N} \text{Div}(\beta_{W'}) - \Theta_{W'} = h'^* \left(\frac{1}{N} \text{div}(\pi'^* a_C) - \pi'^* D_C \right) = h'^* \pi'^* E'_C = (f_C|_{W'})^* E'_C,$$

which is (6.16). This proves (7).

Step 7. We prove (6).

Put $\Gamma_N := P_1 + \{(N+1)E\}$. Then $(\tilde{Y}, \Gamma_N)$ is log smooth with coefficients in $[0, 1]$, hence lc [KM98, Corollary 2.31], and its lc centers are the strata of $[\Gamma_N] = P_1$ (Definition 2.3). By (6.2) and (6.3),

$$(6.21) \quad L_N - (K_{\tilde{Y}} + \Gamma_N) = -(N+1)(K_{\tilde{Y}} + P_1 + E) = -(N+1)\mu^*(K_Y + A) \sim_{\mathbb{Q}} (N+1)(q\mu)^* H,$$

which is semi-ample over U . Let k be a positive integer such that kH is Cartier and very ample over U , and let $\mathcal{J} \subset \mathcal{O}_T$ be the ideal sheaf of V . Replacing k by a multiple, we may assume that $\mathcal{J} \otimes \mathcal{O}_T(kH)$ is generated by its global sections, since U is affine and π is projective [Har77, Theorem II.5.17]. Let $V_1, \dots, V_r$ be the lc centers of q not contained in V and η_i the generic point of V_i . Since $\eta_i \notin V$, the global sections of $\mathcal{J} \otimes \mathcal{O}_T(kH)$ vanishing at η_i form a proper linear subspace of $H^0(T, \mathcal{J} \otimes \mathcal{O}_T(kH))$. As $\mathbb{C}$ is infinite, there is a global section of $\mathcal{J} \otimes \mathcal{O}_T(kH) \subset \mathcal{O}_T(kH)$ vanishing at none of the η_i . Let D_T be its zero divisor. Then D_T is an effective Cartier divisor with $D_T \sim kH$, $V \subset \text{Supp } D_T$, and $V_i \not\subset \text{Supp } D_T$ for every i . Put $\tilde{D} := (q\mu)^* D_T \geq 0$. By (6.21), $\tilde{D} \sim_{\mathbb{Q}} \frac{k}{N+1} (L_N - (K_{\tilde{Y}} + \Gamma_N))$. A stratum $\tilde{C}$ of P_1 is contained in $\text{Supp } \tilde{D} = (q\mu)^{-1}(\text{Supp } D_T)$ if and only if the lc center $q\mu(\tilde{C})$ of q is contained in $\text{Supp } D_T$, if and only if $q\mu(\tilde{C}) \subset V$, if and only if $q\mu(\tilde{C}) = V$ by the minimality of V . Hence the union of the lc centers of $(\tilde{Y}, \Gamma_N)$ contained in $\text{Supp } \tilde{D}$ is $\tilde{Z}_V$. Theorem 6.5, applied to $(\tilde{Y}, \Gamma_N)$, the proper morphism $\pi q\mu : \tilde{Y} \rightarrow U$, L_N , and $\tilde{D}$, shows that $(\pi q\mu)_* \mathcal{O}_{\tilde{Y}}(L_N) \rightarrow (\pi q\mu)_* (\mathcal{O}_{\tilde{Y}}(L_N)|_{\tilde{Z}_V})$ is surjective. Since U is affine, taking global sections gives (6). This finishes the proof. $\square$

7. COMPLEMENTS OF GENERALIZED PAIRS OF FANO TYPE

Let $X \rightarrow Z$ be a projective morphism of normal quasi-projective varieties. We say that X is of *Fano type over Z* if there exists a $\mathbb{Q}$ -divisor $\Delta \geq 0$ on X such that (X, Δ) is klt and $-(K_X + \Delta)$ is nef and big over Z (cf. [Bir19, 2.10], [FM20, Definition 2.10]).

Let Y be a smooth variety and D a reduced snc divisor on Y with irreducible components $D_1, \dots, D_m$. A *stratum* of D is an irreducible component of $D_{i_1} \cap \dots \cap D_{i_c}$ for some distinct indices $i_1, \dots, i_c$, where $c \geq 0$. In particular, Y is a stratum of D . Every stratum of D is smooth, and D has finitely many strata.

Lemma 7.1. *Let Y be a smooth quasi-projective variety, D a reduced snc divisor on Y , $\mathcal{L}$ an invertible sheaf on Y , and $H \subset H^0(Y, \mathcal{L})$ a finite-dimensional subspace which generates $\mathcal{L}$. Let $a \in H$ be a general element and L_a the divisor of zeros of a . Then L_a is smooth, L_a contains no stratum of D , and $D + L_a$ is an snc divisor.*

Proof. Let Σ be a stratum of D and H_Σ the image of the restriction map $H \rightarrow H^0(\Sigma, \mathcal{L}|_\Sigma)$. Then H_Σ generates $\mathcal{L}|_\Sigma$. For $a \in H$ with $a|_\Sigma \neq 0$, the scheme $L_a \cap \Sigma$ is the divisor of zeros of $a|_\Sigma$ on Σ . If $\dim H_\Sigma = 1$, then $a|_\Sigma$ vanishes nowhere on Σ , so $L_a \cap \Sigma = \emptyset$. If $\dim H_\Sigma \geq 2$, then H_Σ defines a base point free linear system on the smooth variety Σ , and a general member of this linear system is smooth by [Kle98, Theorem 4.1]. Since the restriction map is linear and surjective onto H_Σ , for a general $a \in H$ we have $a|_\Sigma \neq 0$ and $L_a \cap \Sigma$ is either empty or a smooth divisor on Σ . As D has finitely many strata, we may choose a such that this holds for every stratum Σ of D . Taking $\Sigma = Y$, L_a is smooth. Since $a|_\Sigma \neq 0$ for every stratum Σ , L_a contains no stratum of D .

Let $y \in L_a$ be a point, $D_1, \dots, D_c$ the components of D containing y , and Σ the irreducible component of $D_1 \cap \dots \cap D_c$ containing y . Let $f_1, \dots, f_c$ be local equations of $D_1, \dots, D_c$ at y and ℓ a local equation of L_a at y . Since D is snc, $df_1, \dots, df_c$ are linearly independent at y and $T_y \Sigma = \bigcap_{i=1}^c \ker df_i$. Since $L_a \cap \Sigma$ is a smooth divisor on Σ with local equation $\ell|_\Sigma$ at y , the restriction of $d\ell$ to $T_y \Sigma$ is nonzero. Thus $df_1, \dots, df_c, d\ell$ are linearly independent at y , and $D + L_a$ is snc at y . Since L_a is reduced and contains no component of D , $D + L_a$ is an snc divisor. $\square$

Lemma 7.2. *Let $g : Q \rightarrow S$ be a contraction with S affine, and let $(Q, B_Q + \mathbf{M})/S$ be a generalized klt pair such that*

$$(7.1) \quad -(K_Q + B_Q + \mathbf{M}_Q) \sim_{\mathbb{Q}, S} H_Q$$

for some $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor H_Q on Q which is ample over S . Then Q is of Fano type over S .

Proof. Let $\tau : Q' \rightarrow Q$ be a log resolution of (Q, B_Q) such that $\mathbf{M}$ descends to Q' , and write $K_{Q'} + B' + \mathbf{M}_{Q'} = \tau^*(K_Q + B_Q + \mathbf{M}_Q)$. By [Har77, Theorem II.7.17 and Proposition II.7.13], τ is the blow-up of a coherent ideal sheaf $\mathcal{I}$ on Q , and $\mathcal{I} \cdot \mathcal{O}_{Q'} = \mathcal{O}_{Q'}(-E)$ for some Cartier divisor $E \geq 0$ on Q' . Since S is affine and H_Q is ample over S , H_Q is ample (cf. [Har77, Theorem II.7.6]). By [Har77, Exercise II.7.14(b)], there exists a positive integer m such that mH_Q is Cartier and $A' := m\tau^*H_Q - E$ is ample over S .

Let $\pi : Q'' \rightarrow Q'$ be a log resolution of $(Q', \text{Supp } B' + E)$, and write $K_{Q''} + B'' = \pi^*(K_{Q'} + B')$ and $E'' := \pi^*E$. Since $\mathbf{M}$ descends to Q'' and $(Q, B_Q + \mathbf{M})$ is generalized klt, every coefficient of B'' is less than 1. Fix a rational number $0 < \epsilon < 1$ such that every coefficient of $B'' + \frac{\epsilon}{m}E''$ is less than 1. Since $\mathbf{M}_{Q'}$ is nef over S ,

$$L := \mathbf{M}_{Q'} + \frac{\epsilon}{m}A'$$

is ample over S . Since S is affine, there is an integer $k \geq 2$ such that kL is Cartier and $\mathcal{O}_{Q'}(kL)$ is generated by global sections (cf. [Har77, Theorem II.7.6]). Let $H \subset H^0(Q', \mathcal{O}_{Q'}(kL))$ be a finite-dimensional subspace generating $\mathcal{O}_{Q'}(kL)$, $a \in H$ a general element, and J the divisor of zeros of a . By Lemma 7.1 applied to Q'' , $\text{Supp}(B'' + E'')$ and π^*H , π^*J is smooth and contains no component of $\text{Supp}(B'' + E'')$, and $\text{Supp}(B'' + E'') + \pi^*J$ is snc. Let

$$C' := \frac{\epsilon}{m}E + \frac{1}{k}J \quad \text{and} \quad \Gamma'' := B'' + \frac{\epsilon}{m}E'' + \frac{1}{k}\pi^*J.$$

Then $C' \geq 0$, $K_{Q''} + \Gamma'' = \pi^*(K_{Q'} + B' + C')$, $\text{Supp } \Gamma''$ is snc, and every coefficient of Γ'' is less than 1. Moreover,

$$C' \sim_{\mathbb{Q}} \frac{\epsilon}{m}E + L = \mathbf{M}_{Q'} + \epsilon\tau^*H_Q.$$

Let $\Delta := \tau_*(B' + C') = B_Q + \tau_*C' \geq 0$. By (7.1),

$$K_{Q'} + B' + C' \sim_{\mathbb{Q}} \tau^*(K_Q + B_Q + \mathbf{M}_Q) + \epsilon\tau^*H_Q \sim_{\mathbb{Q}, S} -(1 - \epsilon)\tau^*H_Q.$$

Thus there exist a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor L_S on S , a positive integer r , and a nonzero rational function $\varphi \in \mathbb{C}(Q)$ such that

$$K_{Q'} + B' + C' + (1 - \epsilon)\tau^*H_Q = (g \circ \tau)^*L_S + \frac{1}{r}\text{div}(\varphi).$$

Pushing forward to Q , we get $K_Q + \Delta + (1 - \epsilon)H_Q = g^*L_S + \frac{1}{r}\text{div}(\varphi)$. Hence $K_Q + \Delta$ is $\mathbb{Q}$ -Cartier, $\tau^*(K_Q + \Delta) = K_{Q'} + B' + C'$, and $-(K_Q + \Delta) \sim_{\mathbb{Q}, S} (1 - \epsilon)H_Q$ is ample over S . Since $(\tau \circ \pi)^*(K_Q + \Delta) = K_{Q''} + \Gamma''$, where Q'' is smooth, $\text{Supp } \Gamma''$ is snc, and every coefficient of Γ'' is less than 1, the pair (Q, Δ) is klt [KM98, Corollary 2.31]. Thus Q is of Fano type over S . $\square$

Theorem 7.3 ([FM20, Theorem 1.2 and Conjecture 1.1]). *Let e and p be positive integers and $\Lambda \subset [0, 1] \cap \mathbb{Q}$ a DCC set whose accumulation points are rational. Then there exists a positive integer n depending only on e , p and Λ satisfying the following. Assume that $X \rightarrow Z$ is a contraction of normal quasi-projective varieties and $(X, B + \mathbf{M})/Z$ is a generalized lc pair of dimension e such that*

- (1) $B \in \Lambda$,
- (2) $\mathbf{M}$ descends to a birational model X' of X such that $\mathbf{M}_{X'}$ is nef and $p\mathbf{M}_{X'}$ is Cartier,
- (3) X is of Fano type over Z , and
- (4) $-(K_X + B + \mathbf{M}_X)$ is nef over Z .

Then for every point $z \in Z$, there exists a monotonic generalized n -complement $(X/Z \ni z, B^+ + \mathbf{M})$ of $(X/Z \ni z, B + \mathbf{M})$.

In [FM20], the nef part of a generalized pair is required to be nef on a birational model where it descends, as in Theorem 7.3(2). If Z is affine, then every complete curve on X' is contracted to a point by $X' \rightarrow Z$, so $\mathbf{M}_{X'}$ is nef if it is nef over Z .

Proposition 7.4. *Let d, p, q be positive integers and $\Lambda \subset [0, 1] \cap \mathbb{Q}$ a DCC set whose accumulation points are rational. Then there exists a positive integer N divisible by p and q and depending only on d, p, q and Λ satisfying the following.*

Let $f : X \rightarrow U$ be a projective morphism of normal quasi-projective varieties with $\dim X \leq d$, let $X \xrightarrow{v} S \xrightarrow{\nu} U$ be the Stein factorization of f , and let $u \in f(X)$ be a point. Let $(X, B + \mathbf{M})/S$ be a generalized lc pair such that

- (1) $B \in \Lambda$,
- (2) $\mathbf{M}$ descends to a birational model X' of X such that $p\mathbf{M}_{X'}$ is Cartier,
- (3) X is of Fano type over S , and
- (4) $-(K_X + B + \mathbf{M}_X)$ is nef over S .

Then there exist an open neighborhood U° of u in U , a nonzero rational function $a \in \mathbb{C}(X)$, and a $\mathbb{Q}$ -divisor B^+ on $X^\circ := f^{-1}(U^\circ)$ with $B^+ \geq B|_{X^\circ}$, such that $(X^\circ, B^+ + \mathbf{M})/U^\circ$ is generalized lc and

$$(7.2) \quad N(K_{X^\circ} + B^+ + \mathbf{M}_{X^\circ}) = \operatorname{div}_{X^\circ}(a) \quad \text{and} \quad NB^+ \geq N\lfloor B|_{X^\circ} \rfloor + \lfloor (N+1)\{B|_{X^\circ}\} \rfloor.$$

Proof. For each integer $1 \leq e \leq d$, let n_e be the positive integer given by Theorem 7.3 for e, p and Λ , and let

$$N := \operatorname{lcm}(p, q, n_1, \dots, n_d).$$

If $\dim X = 0$, then X is a point, $B = 0$ and $\mathbf{M}_X = 0$, and we may take $U^\circ := U$, $B^+ := 0$ and $a := 1$. Thus we may assume that $\dim X \geq 1$, and we let $e := \dim X$. Recall that v is a contraction, ν is finite [Har77, Corollary III.11.5], and S is a normal quasi-projective variety as $\mathcal{O}_S = v_*\mathcal{O}_X$.

Step 1. We reduce to the case where u is a closed point and U is affine, and construct complements near the points of $\nu^{-1}(u)$.

Let $u_0 \in \overline{\{u\}}$ be a closed point. Then $u_0 \in f(X)$ as $f(X)$ is closed, and every open subset of U containing u_0 contains u . Thus we may replace u by u_0 and assume that u is a closed point. Let U_1 be an affine open neighborhood of u in U , $X_1 := f^{-1}(U_1)$ and $S_1 := \nu^{-1}(U_1)$. Then $X_1 \rightarrow S_1 \rightarrow U_1$ is the Stein factorization of $X_1 \rightarrow U_1$, and $X_1 \rightarrow U_1$ with the generalized lc pair $(X_1, B|_{X_1} + \mathbf{M})/S_1$ satisfies the hypotheses of the proposition. Since every open subset of U_1 is open in U , the conclusion for $X_1 \rightarrow U_1$ implies the conclusion for f . Replacing U, S and X by U_1, S_1 and X_1 , we may assume that U is affine. Then S is affine as ν is finite, and $\nu^{-1}(u) = \{s_1, \dots, s_r\}$ is a nonempty finite set.

Since $\mathbf{M}$ is b-nef over S , possibly replacing X' by a higher birational model, we may assume that $\mathbf{M}_{X'}$ is nef over S . Since S is affine, $\mathbf{M}_{X'}$ is nef. By Theorem 7.3 applied to $v : X \rightarrow S$, for each $1 \leq i \leq r$ there exists a monotonic generalized n_e -complement $(X/S \ni s_i, B_i^+ + \mathbf{M})$ of $(X/S \ni s_i, B + \mathbf{M})$. Since $n_e \mid N$, there exist affine open neighborhoods S_i of s_i in S and nonzero rational functions $a_i \in \mathbb{C}(X)$ such that, with $X_{S_i} := v^{-1}(S_i)$,

$$(7.3) \quad (X_{S_i}, B_i^+ + \mathbf{M})/S_i \text{ is generalized lc,} \quad N(K_X + B_i^+ + \mathbf{M}_X) = \operatorname{div}(a_i) \text{ on } X_{S_i}, \quad B_i^+ \geq B.$$

Step 2. We construct a smooth birational model Y of X , a Cartier divisor F and a $\mathbb{Q}$ -divisor Ξ on Y such that every coefficient of Ξ is at most one over a neighborhood of $\nu^{-1}(u)$.

Since $p\mathbf{M}_{X'}$ is Cartier and $p \mid N$, $N\mathbf{M}_X$ is a Weil divisor. Let

$$(7.4) \quad \mathcal{J} := \mathcal{O}_X(-NK_X - \lceil NB \rceil - N\mathbf{M}_X) \subset \mathbb{C}(X),$$

where for a Weil divisor D on X we denote by $\mathcal{O}_X(D)$ the coherent subsheaf of the constant sheaf $\mathbb{C}(X)$ whose sections over an open subset V are the rational functions g with $\operatorname{div}(g) + D \geq 0$ on V . By (7.3), NB_i^+ is a Weil divisor and $NB_i^+ \geq NB$ on X_{S_i} , so $NB_i^+ \geq \lceil NB \rceil$ and $\operatorname{div}(a_i) \geq NK_X + \lceil NB \rceil + N\mathbf{M}_X$ on X_{S_i} . Thus $a_i \in H^0(X_{S_i}, \mathcal{J})$. Since S is affine, the coherent sheaf $v_*\mathcal{J}$ is generated by finitely many global sections $a'_1, \dots, a'_t \in H^0(X, \mathcal{J}) \subset \mathbb{C}(X)$, which

are not all zero as $a_1 \neq 0$. Since S_i is affine, we may write $a_i = \sum_{j=1}^t (c_{ij} \circ v) a'_j$ on X_{S_i} for some $c_{ij} \in \mathcal{O}_S(S_i)$.

Let $\mu : Y \rightarrow X$ be a projective birational morphism from a smooth variety Y such that $\mathbf{M}$ descends to Y and the rational map $X \dashrightarrow \mathbb{P}^{t-1}$ defined by $(a'_1 : \dots : a'_t)$ extends to a morphism on Y (cf. [Har77, Example II.7.17.3]). Near each point of Y , this morphism is given by the regular functions a'_j/a'_{j_0} for some j_0 , so the $\mathcal{O}_Y$ -submodule of $\mathbb{C}(Y) = \mathbb{C}(X)$ generated by $a'_1, \dots, a'_t$ is invertible. We write it as $\mathcal{O}_Y(-F)$ for a Cartier divisor F on Y . Let $H \subset H^0(Y, \mathcal{O}_Y(-F))$ be the span of $a'_1, \dots, a'_t$. Then H generates $\mathcal{O}_Y(-F)$, and for every nonzero $a \in H$ we have

$$(7.5) \quad \operatorname{div}_Y(a) = F + L_a, \quad L_a \geq 0,$$

where L_a is the divisor of zeros of a as a section of $\mathcal{O}_Y(-F)$. Let

$$(7.6) \quad \Xi := \frac{1}{N}F - K_Y - \mathbf{M}_Y.$$

Replacing Y by a log resolution of $(Y, \operatorname{Supp} \Xi)$ and $F, \mathbf{M}_Y$ by their pullbacks, we may assume that $\operatorname{Supp} \Xi$ is snc. The $\mathcal{O}_Y$ -submodule generated by $a'_1, \dots, a'_t$ is again $\mathcal{O}_Y(-F)$, and (7.5) and (7.6) still hold.

Let Q be a prime divisor on Y . Since $\mathcal{O}_Y(-F)$ is generated by $a'_1, \dots, a'_t$, we have $\operatorname{mult}_Q F = \min_j \operatorname{mult}_Q \operatorname{div}_Y(a'_j)$. Suppose that $v\mu(Q)$ meets S_i . Then $\operatorname{mult}_Q \operatorname{div}_Y(c_{ij} \circ v \circ \mu) \geq 0$ for every j with $c_{ij} \neq 0$, so $\operatorname{mult}_Q \operatorname{div}_Y(a_i) \geq \operatorname{mult}_Q F$. Let $\Xi_i := \frac{1}{N} \operatorname{div}_Y(a_i) - K_Y - \mathbf{M}_Y$. By (7.3), $K_Y + \Xi_i + \mathbf{M}_Y = \mu^*(K_X + B_i^+ + \mathbf{M}_X)$ over S_i . Since $\mathbf{M}$ descends to Y and $(X_{S_i}, B_i^+ + \mathbf{M})/S_i$ is generalized lc, $\operatorname{mult}_Q \Xi_i \leq 1$ by (2.1). Since $\Xi_i - \Xi = \frac{1}{N}(\operatorname{div}_Y(a_i) - F)$ by (7.6), we get

$$\operatorname{mult}_Q \Xi = \operatorname{mult}_Q \Xi_i - \frac{1}{N}(\operatorname{mult}_Q \operatorname{div}_Y(a_i) - \operatorname{mult}_Q F) \leq \operatorname{mult}_Q \Xi_i \leq 1.$$

Let $\Xi^{>1}$ be the sum of the components of Ξ with coefficient greater than one, and let $S^\circ := S \setminus v\mu(\operatorname{Supp} \Xi^{>1})$. Then S° is open as $v\mu$ is proper. Since $\operatorname{mult}_Q \Xi \leq 1$ for every prime divisor Q on Y such that $v\mu(Q)$ meets some S_i , we have $s_1, \dots, s_r \in S^\circ$. Every coefficient of Ξ on $Y^\circ := (v\mu)^{-1}(S^\circ)$ is at most one.

Step 3. We choose a and B^+ , and show that $(X^\circ, B^+ + \mathbf{M})/U^\circ$ is generalized lc.

By Lemma 7.1 applied to Y° , $\operatorname{Supp} \Xi|_{Y^\circ}$ and the restriction of H to Y° , for a general $a \in H$ the divisor $L_a|_{Y^\circ}$ is smooth, $L_a|_{Y^\circ}$ has no common component with $\Xi|_{Y^\circ}$, and $\operatorname{Supp}(\Xi + L_a)|_{Y^\circ}$ is snc. Fix such an a , and let

$$B^+ := \frac{1}{N} \operatorname{div}_X(a) - K_X - \mathbf{M}_X.$$

Then $N(K_X + B^+ + \mathbf{M}_X) = \operatorname{div}_X(a)$, and NB^+ is a Weil divisor. Since $a \in H^0(X, \mathcal{J})$, by (7.4) we have $\operatorname{div}_X(a) \geq NK_X + \lceil NB \rceil + N\mathbf{M}_X$, so $NB^+ \geq \lceil NB \rceil$ and $B^+ \geq B \geq 0$. For every prime divisor D on X , Lemma 3.1(1) applied to $b := \operatorname{mult}_D B$ and $c := \frac{1}{N} \lceil Nb \rceil$ gives

$$N \operatorname{mult}_D B^+ \geq \lceil Nb \rceil \geq N \lfloor b \rfloor + \lfloor (N+1)\{b\} \rfloor.$$

By (7.5) and (7.6),

$$(7.7) \quad \mu^*(K_X + B^+ + \mathbf{M}_X) = \frac{1}{N} \operatorname{div}_Y(a) = K_Y + \Xi + \frac{1}{N} L_a + \mathbf{M}_Y.$$

Let $U^\circ := U \setminus v(S \setminus S^\circ)$ and $X^\circ := f^{-1}(U^\circ)$. Since v is finite, U° is open [Har77, Exercise II.3.5(b)], and $u \in U^\circ$ as $v^{-1}(u) \subset S^\circ$. Then $B^+|_{X^\circ} \geq B|_{X^\circ}$, and $B^+|_{X^\circ}$ satisfies (7.2). It suffices to show that $(X^\circ, B^+ + \mathbf{M})/U^\circ$ is generalized lc. Since v is finite, every complete curve on Y whose image in U is a point has a point as its image in S . Thus $\mathbf{M}_Y$ is nef over U , and $\mathbf{M}$ is b-nef over U° . Moreover $X^\circ \subset v^{-1}(S^\circ)$, so $\mu^{-1}(X^\circ) \subset Y^\circ$. On Y° , the support of $\Xi + \frac{1}{N} L_a$ is snc, and every coefficient of $\Xi + \frac{1}{N} L_a$ is at most one, since every coefficient of Ξ is at most

one, L_a is reduced and has no common component with Ξ , and $\frac{1}{N} \leq 1$. Since $\mathbf{M}$ descends to Y , $(X^\circ, B^+ + \mathbf{M})/U^\circ$ is generalized lc by (7.7) and [KM98, Corollary 2.31]. $\square$

8. LOG CANONICITY UNDER PULLBACK

Proposition 8.1. *Let $Y \xrightarrow{g} T \xrightarrow{\pi} Z$ be contractions of normal quasi-projective varieties with $\dim T > 0$, and (Y, A) an lc pair such that A is a $\mathbb{Q}$ -divisor and $K_Y + A \sim_{\mathbb{Q}, T} 0$, so that $g : (Y, A) \rightarrow T$ is a crepant log structure. Let m be a positive integer, $\psi \in \mathbb{C}(Y)$ a nonzero rational function, and D_T a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor on T such that*

$$(8.1) \quad K_Y + A + \frac{1}{m} \operatorname{div}(\psi) = g^* D_T.$$

Let B_T be the discriminant part of g and $\mathbf{M}$ the moduli b -divisor of (8.1) (Definition 4.2), so that $D_T = K_T + B_T + \mathbf{M}_T$. Assume that $\mathbf{M}$ descends to a birational model T' of T and that $\mathbf{M}_{T'}$ is nef over Z . Then:

- (1) *Let $P \geq 0$ be a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor on T . Then $(T, B_T + P + \mathbf{M})/Z$ is a generalized pair, and $(Y, A + g^*P)$ is lc if and only if $(T, B_T + P + \mathbf{M})/Z$ is generalized lc.*
- (2) *Let n be a positive integer such that $m \mid n$, P a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor on T , L a Cartier divisor on Z , and $u \in \mathbb{C}(T)$ a nonzero rational function such that $n(D_T + P) = \pi^* L + \operatorname{div}(u)$. Then*

$$(8.2) \quad n(K_Y + A + g^*P) = (\pi \circ g)^* L + \operatorname{div}\left((u \circ g)\psi^{-n/m}\right).$$

- (3) *Let n be a positive integer such that $m \mid n$ and $z \in Z$ a point. If $(T/Z \ni z, B_T^+ + \mathbf{M})$ is a monotonic generalized n -complement of $(T/Z \ni z, B_T + \mathbf{M})$, then $B_T^+ - B_T$ is $\mathbb{Q}$ -Cartier over a neighborhood of z and, possibly after shrinking Z to a neighborhood of z , $(Y/Z \ni z, A + g^*(B_T^+ - B_T))$ is a monotonic n -complement of $(Y/Z \ni z, A)$.*

Proof. For a prime divisor Q on T , let t_Q be as in Definition 4.2, so that $B_T = \sum_Q (1 - t_Q)Q$. Since g is surjective, there is a prime divisor F on Y such that $g(F) = Q$. Since $(Y, A + t_Q g^*Q)$ is lc over the generic point of Q and $A \geq 0$, $t_Q \operatorname{mult}_F g^*Q \leq 1$, so $t_Q \leq 1$. Thus $B_T \geq 0$. Since $\mathbf{M}$ descends to T' , $\mathbf{M}_{T'}$ is nef over Z , and $K_T + B_T + \mathbf{M}_T = D_T$ is $\mathbb{Q}$ -Cartier, $(T, B_T + \mathbf{M})/Z$ is a generalized pair (Definition 2.7).

(1) Since $P \geq 0$ is $\mathbb{Q}$ -Cartier, $(T, B_T + P + \mathbf{M})/Z$ is a generalized pair. Let $\tau : T'' \rightarrow T$ be a birational model of T , and $\mu : Y'' \rightarrow Y$, $g'' : Y'' \rightarrow T''$, and $A_{Y''}$ as in Definition 4.2, so that $K_{Y''} + A_{Y''} = \mu^*(K_Y + A)$ and $(Y'', A_{Y''})$ is sub-lc. Let E be a prime divisor on T'' and t_E the largest rational number t such that $(Y'', A_{Y''} + t g''^*E)$ is sub-lc over the generic point of E , so that $B_{T''} = \sum_E (1 - t_E)E$ (Definition 4.2). Over the generic point of E we have $\tau^*P = (\operatorname{mult}_E \tau^*P)E$, hence $\mu^*g^*P = g''^*\tau^*P = (\operatorname{mult}_E \tau^*P)g''^*E$. Thus, for every rational number t , $(Y'', A_{Y''} + \mu^*g^*P + t g''^*E)$ is sub-lc over the generic point of E if and only if $t \leq t_E - \operatorname{mult}_E \tau^*P$. If τ factors through T' , then $\mathbf{M}$ descends to T'' and $K_{T''} + B_{T''} + \mathbf{M}_{T''} = \tau^*D_T$ (Definition 4.2), so

$$(8.3) \quad a(E, T, B_T + P + \mathbf{M}) = 1 - \operatorname{mult}_E(B_{T''} + \tau^*P) = t_E - \operatorname{mult}_E \tau^*P$$

by (2.1).

Suppose that $(Y, A + g^*P)$ is lc. Since $K_{Y''} + A_{Y''} + \mu^*g^*P = \mu^*(K_Y + A + g^*P)$, $(Y'', A_{Y''} + \mu^*g^*P)$ is sub-lc, so $t_E - \operatorname{mult}_E \tau^*P \geq 0$ for every T'' and E as above. Since every prime divisor over T is a prime divisor on some birational model T'' of T such that τ factors through T' , $(T, B_T + P + \mathbf{M})/Z$ is generalized lc by (8.3).

Conversely, suppose that $(T, B_T + P + \mathbf{M})/Z$ is generalized lc, and let F be a prime divisor over Y . If $g(\operatorname{center}_Y F) = T$, then $\operatorname{center}_Y F \not\subset \operatorname{Supp} g^*P$, so $a(F, Y, A + g^*P) = a(F, Y, A) \geq 0$. Otherwise, $g(\operatorname{center}_Y F)$ is a proper closed subset of T , and the restriction v of ord_F to $\mathbb{C}(T)$ is a nontrivial discrete valuation. The residue field of v is a subfield of $\mathbb{C}(F)$, and elements of $\mathbb{C}(F)$ algebraically independent over it lift to elements of $\mathbb{C}(Y)$ algebraically independent over $\mathbb{C}(T)$.

Thus the residue field of v has transcendence degree $\dim T - 1$ over $\mathbb{C}$. By [KM98, Lemma 2.45], there exist a birational model $\tau : T'' \rightarrow T$ of T and a prime divisor E on T'' such that $v = e \operatorname{ord}_E$ for some positive integer e ; we may assume that τ factors through T' . Let $\mu : Y'' \rightarrow Y$, g'' , $A_{Y''}$, and t_E be as above. Then $g''(\operatorname{center}_{Y''} F) = E$ as $v = e \operatorname{ord}_E$, so g''^*E has multiplicity e along F and $a(F, Y'', A_{Y''} + sg''^*E) = a(F, Y'', A_{Y''}) - es$ for every rational number s . Since $(Y'', A_{Y''} + t_E g''^*E)$ is sub-lc over the generic point of E , $a(F, Y'', A_{Y''}) \geq et_E$. Hence

$$\begin{aligned} a(F, Y, A + g^*P) &= a(F, Y'', A_{Y''} + \mu^*g^*P) = a(F, Y'', A_{Y''}) - e \operatorname{mult}_E \tau^*P \\ &\geq e(t_E - \operatorname{mult}_E \tau^*P) \geq 0 \end{aligned}$$

by (8.3). Thus $(Y, A + g^*P)$ is lc. This proves (1).

(2) Multiplying (8.1) by n and using $g^* \operatorname{div}(u) = \operatorname{div}(u \circ g)$, we get

$$n(K_Y + A + g^*P) = g^*(n(D_T + P)) - \frac{n}{m} \operatorname{div}(\psi) = (\pi \circ g)^*L + \operatorname{div}(u \circ g) - \operatorname{div}(\psi^{n/m}),$$

which is (8.2).

(3) Let $P := B_T^+ - B_T$. Possibly shrinking Z to a neighborhood of z , we may assume that $(T, B_T^+ + \mathbf{M})/Z$ is generalized lc and $n(K_T + B_T^+ + \mathbf{M}_T) \sim 0$ (Definition 2.7). Then $P \geq 0$, and P is $\mathbb{Q}$ -Cartier since D_T and $K_T + B_T^+ + \mathbf{M}_T$ are. By (1), $(Y, A + g^*P)$ is lc. Write $n(D_T + P) = \operatorname{div}(u)$ for a nonzero $u \in \mathbb{C}(T)$. By (2) with $L = 0$, $n(K_Y + A + g^*P) \sim 0$, and $n(A + g^*P)$ is a Weil divisor. Since $A + g^*P \geq A$, we have

$$n(A + g^*P) \geq [nA] \geq n[A] + [(n+1)\{A\}]$$

(cf. Lemma 3.1). Thus $(Y/Z \ni z, A + g^*P)$ is a monotonic n -complement of $(Y/Z \ni z, A)$ (Definition 2.4). $\square$

Let (X, B) be a dlt $\mathbb{Q}$ -pair and S an lc center of (X, B) . We write (S, Θ) for the adjunction pair of S (Definition 2.3); Θ is the different of B on S , obtained by iterating the different for divisors [Kol92, 16.5–16.6] along a chain of irreducible components of $[B]$ [Kol16, Definition 4].

Theorem 8.2 ([Kol16, Corollary 11]). *Let $f : (X, B) \rightarrow Z$ be a dlt crepant log structure, S an lc center of (X, B) , (S, Θ) the adjunction pair of S , and $S \xrightarrow{g} T \xrightarrow{\tau} Z$ the Stein factorization of $f|_S$. Then:*

- (1) $g : (S, \Theta) \rightarrow T$ is a dlt crepant log structure.
- (2) For every lc center V_T of g , $\tau(V_T)$ is an lc center of f . Moreover, every lc center of (S, Θ) which is minimal among the lc centers of (S, Θ) dominating V_T is minimal among the lc centers of (X, B) dominating $\tau(V_T)$.
- (3) For every lc center V of f , every irreducible component of $\tau^{-1}(V)$ is an lc center of g .

Theorem 8.3 ([Kaw07, Theorem]). *Let X be a normal variety, S a reduced divisor on X , and $B \geq 0$ an $\mathbb{R}$ -divisor on X such that S and $\operatorname{Supp} B$ have no common irreducible component and $K_X + S + B$ is $\mathbb{R}$ -Cartier. Let $\nu : S^\nu \rightarrow S$ be the normalization of S and B^ν the different of B on S^ν as in [Kaw07], so that $K_{S^\nu} + B^\nu = \nu^*((K_X + S + B)|_S)$. Then $(X, S + B)$ is lc near S if and only if (S^ν, B^ν) is lc.*

Let X be a normal variety and $\Delta \geq 0$ an $\mathbb{R}$ -divisor on X such that $K_X + \Delta$ is $\mathbb{R}$ -Cartier. The non-lc locus $\operatorname{Nlc}(X, \Delta)$ is the complement of the largest open subset of X on which (X, Δ) is lc. An lc center of (X, Δ) is $\operatorname{center}_X E$ for a prime divisor E over X such that $a(E, X, \Delta) = 0$ and $\operatorname{center}_X E \not\subset \operatorname{Nlc}(X, \Delta)$ (cf. [Fuj11, 4.4 and Definition 4.6]).

Theorem 8.4 ([Fuj17, Theorem 6.3]). *Let X be a normal variety and $\pi : X \rightarrow V$ a proper morphism to a variety such that $\pi_* \mathcal{O}_X = \mathcal{O}_V$. Let $\Delta \geq 0$ be an $\mathbb{R}$ -divisor on X such that $K_X + \Delta$ is $\mathbb{R}$ -Cartier and $-(K_X + \Delta)$ is semi-ample over V . Let $x \in V$ be a closed point such that $\operatorname{Nlc}(X, \Delta) \cap \pi^{-1}(x) \neq \emptyset$. Then $\operatorname{Nlc}(X, \Delta) \cap \pi^{-1}(x)$ is connected and intersects every lc center C of (X, Δ) such that $C \cap \pi^{-1}(x) \neq \emptyset$.*

Proposition 8.5. *Let $f : (X, B) \rightarrow Z$ be a dlt crepant log structure with $\dim Z > 0$, S an lc center of (X, B) dominating Z , and (S, Θ) the adjunction pair of S . Let $H \geq 0$ be a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor on Z . Then $(X, B + f^*H)$ is lc if and only if $(S, \Theta + (f|_S)^*H)$ is lc. In particular,*

$$(8.4) \quad \text{lct}(X, B; f^*H) = \text{lct}(S, \Theta; (f|_S)^*H),$$

where both sides are $+\infty$ if $H = 0$.

Proof. Assume first that S is an irreducible component of $[B]$. Since S dominates Z , $S \not\subset \text{Supp } f^*H$. By Definition 2.3, S is normal. Since f^*H is $\mathbb{Q}$ -Cartier, the different of $B - S + f^*H$ on S is $\Theta + (f|_S)^*H$ (cf. [Kol92, 16.5–16.6]), and

$$K_S + \Theta + (f|_S)^*H \sim_{\mathbb{Q}} (K_X + B + f^*H)|_S.$$

By Theorem 8.3, $(X, B + f^*H)$ is lc near S if and only if $(S, \Theta + (f|_S)^*H)$ is lc. It remains to show that $(X, B + f^*H)$ is lc if it is lc near S . Suppose not. Then $\text{Nlc}(X, B + f^*H)$ is nonempty and disjoint from S . Let $x \in Z$ be a closed point in $f(\text{Nlc}(X, B + f^*H))$. Let E be an lc place of (X, B) such that $\text{center}_X E = S$. Since $S \not\subset \text{Supp } f^*H$, $a(E, X, B + f^*H) = a(E, X, B) = 0$. Since $S \cap \text{Nlc}(X, B + f^*H) = \emptyset$, S is an lc center of $(X, B + f^*H)$. Since S dominates Z and $f|_S$ is proper, $S \cap f^{-1}(x) \neq \emptyset$. Since f is a contraction, $f_*\mathcal{O}_X = \mathcal{O}_Z$. Since $K_X + B \sim_{\mathbb{Q}, Z} 0$, $-(K_X + B + f^*H) \sim_{\mathbb{Q}, Z} 0$ is semi-ample over Z . By Theorem 8.4, $\text{Nlc}(X, B + f^*H) \cap f^{-1}(x)$ intersects S , a contradiction.

In general, we prove the first assertion by induction on $\dim X$. We may assume that S is not an irreducible component of $[B]$. By Definition 2.3, $S \subsetneq S'$ for some irreducible component S' of $[B]$. Let (S', Θ') be the adjunction pair of S' and $S' \xrightarrow{g'} Z' \xrightarrow{\tau'} Z$ the Stein factorization of $f|_{S'}$. By Theorem 8.2(1), $g' : (S', \Theta') \rightarrow Z'$ is a dlt crepant log structure. By [Kol16, Definitions 4 and 18], S is an lc center of (S', Θ') and its adjunction pair is (S, Θ) . Moreover, S' and S dominate Z' , $\dim Z' = \dim Z > 0$, $g'^*\tau'^*H = (f|_{S'})^*H$, and $(g'|_S)^*\tau'^*H = (f|_S)^*H$. By the case treated above, $(X, B + f^*H)$ is lc if and only if $(S', \Theta' + (f|_{S'})^*H)$ is lc. By induction applied to g' , S , and τ'^*H , the latter holds if and only if $(S, \Theta + (f|_S)^*H)$ is lc. This proves the first assertion.

Finally, since (X, B) and (S, Θ) are lc and log discrepancies are affine in t , the sets

$$\{t \in \mathbb{R}_{\geq 0} \mid (X, B + tf^*H) \text{ is lc}\} \quad \text{and} \quad \{t \in \mathbb{R}_{\geq 0} \mid (S, \Theta + t(f|_S)^*H) \text{ is lc}\}$$

are intervals containing 0. By the first assertion applied to tH for every rational $t \geq 0$, they have the same rational points. Thus they have the same supremum, which is (8.4). $\square$

9. PROOF OF THE MAIN THEOREMS

Proof of Theorem 1.1. Step 1. We fix the constants.

Let $d_0 := \max\{d, 2\}$ and $\Phi_1 := \Phi \cup \{1\}$. Let $\mathfrak{R}$ be the finite set given by Lemma 4.4 for d_0 and Φ_1 , and $\Gamma := \Phi(\mathfrak{R})$. Then $\Phi_1 \subset \Gamma$, and Γ is a DCC set whose only accumulation point is 1. Let $b_{\mathfrak{R}}$ be the smallest positive integer such that $b_{\mathfrak{R}}\mathfrak{R} \subset \mathbb{Z}$ and $b_{\mathfrak{R}}\Phi \subset \mathbb{Z}$. Let $m_1 \mid p_1$ be the positive integers given by Theorem 4.7 for d_0 and Γ , and let $\Lambda := \Phi(\mathfrak{R}[p_1])$, where $\mathfrak{R}[p_1]$ is defined in (4.6). By Proposition 4.8, Theorem 4.7 holds for d_0 and Γ with m_1 , p_1 , and Λ , and Λ is a DCC set whose only accumulation point is 1. Let N_0 be the positive even integer defined in (6.11) for d_0 , Φ_1 , $b_{\mathfrak{R}}$, and m_1 . Since $b_{\mathfrak{R}} \mid N_0$ by (6.11), $N_0\Phi \subset \mathbb{Z}$ and $N_0\mathfrak{R} \subset \mathbb{Z}$. Let E_0 , $b := m_1E_0$, and p_v be the integers defined in (5.4) for d_0 and m_1 ; then $m_1 \mid b \mid p_v$. Let

$$q_{\text{prop}} := \text{lcm}(N_0, b, p_1, p_v),$$

and let n_{prop} and n_{dom} be the positive integers given by Proposition 7.4 for $(d_0, p_v, q_{\text{prop}}, \Lambda)$ and for $(d_0, p_1, 1, \Lambda)$ respectively. Let $a(s, \Gamma)$ be as in Theorem 2.11 for $0 \leq s \leq d_0$, and let

$$(9.1) \quad N := p \cdot \text{lcm}(N_0, n_{\text{prop}}, n_{\text{dom}}, p_1, p_v, b, a(0, \Gamma), \dots, a(d_0, \Gamma)).$$

Then N depends only on d , p , and Φ , $p \mid N$, and N is a positive even multiple of N_0 . In particular, $N\Phi_1 \subset \mathbb{Z}$ and $N\mathfrak{R} \subset \mathbb{Z}$. We show that N satisfies the requirements of the theorem.

Step 2. We reduce to the case where z is a closed point, and we construct a $\mathbb{Q}$ -factorial dlt pair (Y, A) with contractions $Y \xrightarrow{g} T \xrightarrow{\pi} Z$.

Let $(X/Z \ni z, B)$ be an $\mathbb{R}$ -complementary pair of dimension d such that $B \in \Phi$. Possibly shrinking Z to a neighborhood of z , we may assume that $(X/Z, B^+)$ is an $\mathbb{R}$ -complement of $(X/Z, B)$ for some B^+ . Let z_0 be a closed point of $\overline{\{z\}}$. Then $(X/Z \ni z_0, B)$ is $\mathbb{R}$ -complementary, and every open neighborhood of z_0 contains z . Thus a monotonic N -complement of $(X/Z \ni z_0, B)$ is a monotonic N -complement of $(X/Z \ni z, B)$. Replacing z with z_0 , we may assume that z is a closed point.

By Proposition 3.5, possibly shrinking Z to an affine neighborhood of z , there exist a $\mathbb{Q}$ -factorial dlt pair (Y, A) with $A \in \Phi_1$ and a birational map $X \dashrightarrow Y$ over Z with the following properties. There are contractions $g : Y \rightarrow T$ and $\pi : T \rightarrow Z$ and a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor H on T ample over Z such that $-(K_Y + A) \sim_{\mathbb{Q}} g^*H$. If $(Y/Z \ni z, A)$ has a monotonic N -complement, then so does $(X/Z \ni z, B)$. Here $\dim Y = d$, and $g : (Y, A) \rightarrow T$ is a $\mathbb{Q}$ -factorial dlt crepant log structure (Definition 2.8). It suffices to show that $(Y/Z \ni z, A)$ has a monotonic N -complement.

The lc centers of g are finitely many closed subvarieties $V_1, \dots, V_r$ of T , and $\pi(V_i)$ is closed in Z for each i , as π is proper. Replacing Z with an affine neighborhood of z in $Z \setminus \bigcup_{z \notin \pi(V_i)} \pi(V_i)$, and Y and T with their preimages, we may assume that $z \in \pi(V)$ for every lc center V of g . Indeed, the lc centers of (Y, A) over the smaller base are the nonempty intersections of the lc centers of (Y, A) with the preimage of the smaller base. The properties of (Y, A) , g , π , and H listed above are preserved. Let $T_z := \pi^{-1}(z)$ and $Y_z := (\pi \circ g)^{-1}(z)$. Then every lc center of g meets T_z , and every lc center of (Y, A) meets Y_z .

There are two cases. Either every lc center of (Y, A) dominates T , which includes the case $[A] = 0$, or some lc center of g is a proper subset of T . In the second case, let $V \subsetneq T$ be an lc center of g which is minimal with respect to inclusion among the lc centers of g properly contained in T . Then V is minimal with respect to inclusion among all lc centers of g .

Step 3. We show that $(Y/Z \ni z, A)$ has a monotonic N -complement when every lc center of (Y, A) dominates T .

Assume first that $\dim T = 0$. Then Z is a point as π is surjective, Y is projective, and $K_Y + A \sim_{\mathbb{Q}} 0$. Since (Y, A) is a projective lc pair of dimension $d \leq d_0$ with $A \in \Phi_1 \subset \Gamma$, Theorem 2.11 gives $a(d, \Gamma)(K_Y + A) \sim 0$, so $N(K_Y + A) \sim 0$ by (9.1). Since NA is a Weil divisor and every coefficient of A is at most 1, Lemma 3.1(1) gives $NA \geq N[A] + \lfloor (N+1)\{A\} \rfloor$. Thus $(Y/Z \ni z, A)$ is a monotonic N -complement of itself.

Assume that $\dim T > 0$. Then $0 < \dim T \leq \dim Y = d \leq d_0$, $A \in \Gamma$, and $K_Y + A \sim_{\mathbb{Q}} g^*(-H)$. By Theorem 4.7(1) for d_0 and Γ , there exist a nonzero $\psi \in \mathbb{C}(Y)$ and a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor D_T on T such that

$$(9.2) \quad K_Y + A + \frac{1}{m_1} \operatorname{div}(\psi) = g^*D_T, \quad D_T = K_T + B_T + \mathbf{M}_T \sim_{\mathbb{Q}} -H.$$

Here B_T is the discriminant part of g , and $\mathbf{M}$ is the moduli b-divisor of (9.2) (Definition 4.2). By Theorem 4.7(2)–(4), applied with $S := Z$, and Proposition 4.8, $B_T \geq 0$ and $B_T \in \Lambda$. Moreover, $\mathbf{M}$ descends to a smooth variety T' projective and birational over T such that $p_1\mathbf{M}_{T'}$ is Cartier and nef over Z . In particular, $(T, B_T + \mathbf{M})/Z$ is a generalized pair (Definition 2.7).

We show that $(T, B_T + \mathbf{M})/Z$ is generalized klt. Let F be a prime divisor over T . Let $\tau : T'' \rightarrow T$ be a birational model of T which factors through T' and on which F is a prime divisor. Let $\mu'' : Y'' \rightarrow Y$, $g'' : Y'' \rightarrow T''$, $A_{Y''}$, t_F , and $B_{T''}$ be as in Definition 4.2. Then $K_{Y''} + A_{Y''} = \mu''^*(K_Y + A)$, $\operatorname{mult}_F B_{T''} = 1 - t_F$, and $K_{T''} + B_{T''} + \mathbf{M}_{T''} = \tau^*D_T$. By (2.1), $a(F, T, B_T + \mathbf{M}) = 1 - \operatorname{mult}_F B_{T''} = t_F$. Let $\nu : \hat{Y} \rightarrow Y''$ be a log resolution of $(Y'', \operatorname{Supp} A_{Y''} \cup \operatorname{Supp} g''^*F)$ over a neighborhood of the generic point of F . By [KM98, Corollary 2.31] (cf. the proof of Theorem 4.7(2)), t_F is the minimum of $a(F', Y'', A_{Y''}) / \operatorname{mult}_{F'} \nu^*g''^*F$ over the finitely many prime divisors $F' \subset \operatorname{Supp} \nu^*g''^*F$ on $\hat{Y}$ with $g''(\nu(F')) = F$. Suppose that $t_F = 0$. Then $a(F', Y'', A_{Y''}) = 0$ for some such F' . Regarded as a prime divisor over Y , F' is an

lc place of (Y, A) with $g''(\text{center}_{Y''} F') = F$. Thus $\text{center}_Y F'$ is an lc center of (Y, A) with $g(\text{center}_Y F') = \tau(F) \subsetneq T$, which contradicts our assumption. Hence $t_F > 0$ for every F , and $(T, B_T + \mathbf{M})/Z$ is generalized klt.

Since π is a contraction, Z is affine, and $-(K_T + B_T + \mathbf{M}_T) \sim_{\mathbb{Q}} H$ is ample over Z , Lemma 7.2 shows that T is of Fano type over Z . The Stein factorization of π is $T \xrightarrow{\pi} Z \xrightarrow{\text{id}} Z$, $\dim T \leq d_0$, $z \in \pi(T)$, $B_T \in \Lambda$, $p_1 \mathbf{M}_{T'}$ is Cartier, and $-(K_T + B_T + \mathbf{M}_T)$ is nef over Z . By Proposition 7.4 for $(d_0, p_1, 1, \Lambda)$ and Definition 2.7, possibly shrinking Z to a neighborhood of z , there exists a $\mathbb{Q}$ -divisor $B_T^+ \geq B_T$ on T such that $(T/Z \ni z, B_T^+ + \mathbf{M})$ is a monotonic generalized n_{dom} -complement of $(T/Z \ni z, B_T + \mathbf{M})$. Since $p_1 \mid N$ and $p_1 \mathbf{M}_{T'}$ is Cartier, $N \mathbf{M}_T$ is a Weil divisor, so $(T/Z \ni z, B_T^+ + \mathbf{M})$ is a monotonic generalized N -complement of $(T/Z \ni z, B_T + \mathbf{M})$ by Lemma 3.1(3). Possibly shrinking Z again, we may write $N(K_T + B_T^+ + \mathbf{M}_T) = \text{div}_T(u)$ for some nonzero $u \in \mathbb{C}(T)$ (Notation 2.2). Let $A^+ := A + g^*(B_T^+ - B_T)$. The hypotheses of Proposition 8.1 hold for $Y \xrightarrow{g} T \xrightarrow{\pi} Z$, (Y, A) , m_1 , ψ , D_T , and T' , and $m_1 \mid N$. By Proposition 8.1(2) with $L = 0$,

$$(9.3) \quad N(K_Y + A^+) = \text{div}_Y((u \circ g)\psi^{-N/m_1})$$

over a neighborhood of z , and by Proposition 8.1(3), $(Y/Z \ni z, A^+)$ is a monotonic N -complement of $(Y/Z \ni z, A)$.

Step 4. Assume that $V \subsetneq T$ is as in Step 2. We fix a source (W, Θ_W) of V and, if $\dim V > 0$, we construct a generalized klt pair $(V, B_V + \mathbf{M}^V)$ with $B_V \in \Lambda$ which is of Fano type over the Stein factorization of $V \rightarrow Z$, and an invariant rational b -canonical form θ_0 on W .

Set-up 6.6 applies with (d, Φ, U, q) replaced by (d_0, Φ_1, Z, g) : Z is affine, $\dim Y = d \leq d_0$, (Y, A) is $\mathbb{Q}$ -factorial dlt with $A \in \Phi_1$, (6.1) is (3.2), and $V \subsetneq T$ is a minimal lc center of g . We use the notation of Set-up 6.6. Thus Z_V is the union of the lc centers of (Y, A) with image V , and (W, Θ_W) is a source of V with Stein factorization $W \xrightarrow{h} P \xrightarrow{\rho} V$. The log resolution $\mu : \tilde{Y} \rightarrow Y$, the divisors P_1 , E , and L_N , and the closed subsets $\tilde{Z}_V$ and $\tilde{W}$ of $\tilde{Y}$ are as in (6.2) and (6.3). For an lc center C of (Y, A) with $g(C) = V$, (C, Θ_C) is its adjunction pair and $C \xrightarrow{f_C} P_C \xrightarrow{\rho_C} V$ is the Stein factorization of $g|_C$. In particular, $\Theta_W \in \Gamma$, $\dim W \leq d - 1$, and $K_W + \Theta_W \sim_{\mathbb{Q}} -h^* \rho^*(H|_V)$. Let $\mathcal{A} := \text{Bir}_V^c(W, \Theta_W)$ and $G := \text{Gal}(\mathbb{C}(P)/\mathbb{C}(V))$. By Theorem 6.12(1), V is normal, (W, Θ_W) is klt, ρ is the quotient of P by G , and every element of G is induced by an element of $\mathcal{A}$. Thus $H|_V$ is a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor on V which is ample over Z , and every $\gamma \in \mathcal{A}$ satisfies $h \circ \gamma = \sigma \circ h$ for the element $\sigma \in G$ it induces. Let ω_W be the rational top form on W fixed in Notation 2.2, so that $K_W = \text{Div}(\omega_W)$.

If $\dim V = 0$, then V and P are points and G is trivial. Assume for the rest of this step that $\dim V > 0$. Fix a rational top form ω_V on V , and let $K_V := \text{Div}(\omega_V)$ and $K_P := \text{Div}(\rho^* \omega_V)$. The morphism $h : (W, \Theta_W) \rightarrow P$ is a crepant log structure, h is a contraction of normal quasi-projective varieties, $0 < \dim P \leq \dim W \leq d_0$, $\Theta_W \in \Gamma$, and $K_W + \Theta_W \sim_{\mathbb{Q}} h^* L_0$ for the $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor $L_0 := -\rho^*(H|_V)$ on P . By Theorem 4.7(1) for d_0 and Γ , there exist a nonzero $\psi_W \in \mathbb{C}(W)$ and a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor $D_P \sim_{\mathbb{Q}} -\rho^*(H|_V)$ on P such that

$$(9.4) \quad K_W + \Theta_W + \frac{1}{m_1} \text{div}(\psi_W) = h^* D_P, \quad D_P = K_P + B_P + \mathbf{M}_P.$$

Here B_P is the discriminant part of h , and $\mathbf{M}$ is the moduli b -divisor of (9.4) (Definition 4.2). By Theorem 4.7(2)–(4), applied with $S := Z$, and Proposition 4.8, $B_P \geq 0$ and $B_P \in \Lambda$. Moreover, $\mathbf{M}$ descends to a smooth variety P' projective and birational over P such that $p_1 \mathbf{M}_{P'}$ is Cartier and nef over Z .

We apply Theorem 5.5 to $h : (W, \Theta_W) \rightarrow P$, m_1 , ψ_W , D_P , G , ρ , $S := Z$, ω_V , and ω_W . Its hypotheses hold. We have $d_0 \geq 2$, G acts faithfully on P with quotient ρ , every $\sigma \in G$ lifts to an element of $\mathcal{A}$, and $V \rightarrow Z$ is projective. Since h is a contraction, the generic fiber W_η of h is a normal geometrically integral variety over $\mathbb{C}(P)$. The pair $(W_\eta, \Theta_W|_{W_\eta})$ is geometrically klt.

Indeed, the base change to an algebraic closure of $\mathbb{C}(P)$ of a log resolution of (W, Θ_W) is a log resolution of the geometric generic fiber of h with the same coefficients, all less than 1 (cf. the proof of Lemma 4.6). The constants of Theorem 5.5 for d_0 and m_1 are E_0 , b , and p_v of Step 1. Theorem 5.5(1) gives a nonzero $\chi \in \mathbb{C}(P)$ with the following property. The rational b -canonical form $\theta_0 := (\chi \circ h)\psi_W^{E_0}\omega_W^{\otimes b}$ on W satisfies $\text{Div}(\theta_0) + b\Theta_W = h^*(bD_P + \text{div}(\chi))$, and $\gamma^*\theta_0 = \theta_0$ for every $\gamma \in \mathcal{A}$. Theorem 5.5(2)–(4) give a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor D_V on V , a $\mathbb{Q}$ -divisor $B_V \geq 0$ on V , and a b- $\mathbb{Q}$ -Cartier b-divisor $\mathbf{M}^V$ over V such that $(V, B_V + \mathbf{M}^V)/Z$ is generalized lc and

$$(9.5) \quad D_P + \frac{1}{b} \text{div}(\chi) = \rho^* D_V, \quad K_P + B_P = \rho^*(K_V + B_V), \quad D_V = K_V + B_V + \mathbf{M}_V^V.$$

Moreover, $p_v \mathbf{M}_{V'}^V$ is Cartier for every smooth variety V' projective and birational over V , and $\mathbf{M}_{V'}^V$ is nef over Z if $\mathbf{M}^V$ descends to V' . For a birational model $\tau : U \rightarrow V$ of V , let $\rho_U : P_U \rightarrow U$, B_{P_U} , and B_U be as in Theorem 5.5(2). Then (5.11) holds for every prime divisor F on U and every prime divisor Q on P_U over F . If $\mathbf{M}^V$ descends to U , then B_U is the $\mathbb{Q}$ -divisor defined by (2.1) for $(V, B_V + \mathbf{M}^V)$. By (9.5),

$$(9.6) \quad \text{Div}(\theta_0) = bh^*\rho^*D_V - b\Theta_W.$$

We show that $(V, B_V + \mathbf{M}^V)/Z$ is generalized klt. Let F be a prime divisor over V . Let $\tau : U \rightarrow V$ be a birational model of V to which $\mathbf{M}^V$ descends and on which F is a prime divisor, and let Q be a prime divisor on P_U over F , with ramification index e_Q . Let W'' be the normalization of the main component of $W \times_P P_U$, $h'' : W'' \rightarrow P_U$ the induced morphism, and $K_{W''} + \Theta_{W''}$ the pullback of $K_W + \Theta_W$. By Definition 4.2, $\text{mult}_Q B_{P_U} = 1 - t_Q$, where t_Q is the largest rational number t such that $(W'', \Theta_{W''} + th''^*Q)$ is sub-lc over the generic point of Q . As in Step 3, t_Q is the minimum of the finitely many quotients $a(F', W'', \Theta_{W''}) / \text{mult}_{F'} \nu^* h''^* Q$, where ν is a log resolution of $(W'', \text{Supp } \Theta_{W''} \cup \text{Supp } h''^* Q)$ over a neighborhood of the generic point of Q and $F' \subset \text{Supp } \nu^* h''^* Q$ runs over the prime divisors with $h''(\nu(F')) = Q$. Since (W, Θ_W) is klt, $a(F', W'', \Theta_{W''}) = a(F', W, \Theta_W) > 0$ for every such F' . Thus $t_Q > 0$, and $a(F, V, B_V + \mathbf{M}^V) = 1 - \text{mult}_F B_U = t_Q/e_Q > 0$ by (5.11).

We show that $B_V \in \Lambda$. Let F be a prime divisor on V and Q a prime divisor on P over F , with ramification index e . Since $B_P \in \Lambda = \Phi(\mathfrak{A}[p_1])$, we may write $\text{mult}_Q B_P = 1 - \frac{r}{j}$ with $r \in \mathfrak{A}[p_1]$ and j a positive integer. By (5.11) for $U = V$, $\text{mult}_F B_V = 1 - \frac{r}{ej} \in \Phi(\mathfrak{A}[p_1])$.

We show that $D_V \sim_{\mathbb{Q}} -H|_V$. By (9.5), $\rho^*(D_V + H|_V) \sim_{\mathbb{Q}} D_P + \rho^*(H|_V) \sim_{\mathbb{Q}} 0$. Let k be a positive integer and $\xi \in \mathbb{C}(P)$ a nonzero rational function such that $k\rho^*(D_V + H|_V) = \text{div}_P(\xi)$. As in the proof of Lemma 5.3, $\rho^* \text{div}_V(\prod_{\sigma \in G} \sigma^* \xi) = \sum_{\sigma \in G} \sigma^* \text{div}_P(\xi) = \rho^*(k|G|(D_V + H|_V))$, so $k|G|(D_V + H|_V) = \text{div}_V(\prod_{\sigma \in G} \sigma^* \xi)$ by the injectivity of ρ^* .

Finally, let $V \xrightarrow{v} S_V \xrightarrow{\nu} Z$ be the Stein factorization of $V \rightarrow Z$. Then v is a contraction, ν is finite, and S_V is normal and affine. Since Z is affine, $H|_V$ is ample, hence ample over S_V . Since $\mathbf{M}^V$ is b-nef over Z , it is b-nef over S_V . Thus $(V, B_V + \mathbf{M}^V)/S_V$ is a generalized klt pair, and $-(K_V + B_V + \mathbf{M}_V^V) = -D_V \sim_{\mathbb{Q}} H|_V$ is ample over S_V , which is (7.1). By Lemma 7.2, V is of Fano type over S_V .

Step 5. We construct an $\mathcal{A}$ -invariant rational $(-N)$ -canonical form s_W on W satisfying (6.12) for $U' = V$, and a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor $E_V \geq 0$ on V such that $(W, \Theta_W + h^*\rho^*E_V)$ is lc.

Assume first that $\dim V > 0$. We apply Proposition 7.4 for $(d_0, p_v, q_{\text{prop}}, \Lambda)$ to $V \rightarrow Z$, its Stein factorization $V \xrightarrow{v} S_V \xrightarrow{\nu} Z$, the point $z \in \pi(V)$, and $(V, B_V + \mathbf{M}^V)/S_V$. Its hypotheses hold by Step 4: $\dim V \leq d_0$, $(V, B_V + \mathbf{M}^V)/S_V$ is generalized lc, $B_V \in \Lambda$, $\mathbf{M}^V$ descends to a smooth variety V' with $p_v \mathbf{M}_{V'}^V$ Cartier, V is of Fano type over S_V , and $-(K_V + B_V + \mathbf{M}_V^V) \sim_{\mathbb{Q}} H|_V$ is nef over S_V . Thus there exist an open neighborhood Z° of z , a nonzero $u_0 \in \mathbb{C}(V)$, and a $\mathbb{Q}$ -divisor $B_V^+ \geq B_V$ on $V \cap \pi^{-1}(Z^\circ)$ such that $(V \cap \pi^{-1}(Z^\circ), B_V^+ + \mathbf{M}^V)/Z^\circ$ is generalized lc and $n_{\text{prop}}(K_V + B_V^+ + \mathbf{M}_V^V) = \text{div}_V(u_0)$ on $V \cap \pi^{-1}(Z^\circ)$. Let Z' be an affine neighborhood of z in Z° , and let Y', T', V', W', P' , and $\tilde{Y}'$ be the preimages of Z' . Every lc center of g meets T_z , so Set-up 6.6 applies to $Y' \rightarrow T' \rightarrow Z'$, $(Y', A|_{Y'})$, V' , and the source $(W', \Theta_{W'}|_{W'})$, and the

conclusions of Steps 2 and 4 restrict to Z' . Let $\mathcal{A}' := \text{Bir}_{V'}^c(W', \Theta_{W'}|_{W'})$. Restriction defines an injective homomorphism $\mathcal{A} \rightarrow \mathcal{A}'$ over $G = \text{Gal}(\mathbb{C}(P')/\mathbb{C}(V'))$. We show that it is surjective. Let $\gamma' \in \mathcal{A}'$ lie over $\sigma \in G$, and let $\gamma_\sigma \in \mathcal{A}$ lie over σ (Theorem 6.12(1)). Then $\gamma_\sigma^{-1} \circ \gamma'$ is a birational self-map of W with $h \circ \gamma_\sigma^{-1} \circ \gamma' = h$, and it is crepant over the dense open subset P' of P . By Lemma 6.10, $\gamma_\sigma^{-1} \circ \gamma' \in \text{Bir}_P^c(W, \Theta_W) \subset \mathcal{A}$. Thus γ' is the restriction of $\gamma_\sigma \circ (\gamma_\sigma^{-1} \circ \gamma') \in \mathcal{A}$. Replacing $Z, Y, T, V, W, P, \tilde{Y}$, and $\mathcal{A}$ with $Z', Y', T', V', W', P', \tilde{Y}'$, and $\mathcal{A}'$, we may assume that $Z^\circ = Z$. In particular, $\gamma^* \theta_0 = \theta_0$ for every $\gamma \in \mathcal{A}$ by Step 4. Let $E_V := B_V^+ - B_V \geq 0$ and $u := u_0^{N/n_{\text{prop}}}$. Then E_V is $\mathbb{Q}$ -Cartier, as D_V and $K_V + B_V^+ + \mathbf{M}_V^V$ are, and (9.5) gives

$$(9.7) \quad N(D_V + E_V) = \text{div}_V(u), \quad (V, B_V + E_V + \mathbf{M}^V)/Z \text{ is generalized lc.}$$

Let

$$(9.8) \quad s_W := (h^* \rho^* u) \theta_0^{\otimes(-N/b)}, \quad \text{so that} \quad \text{Div}(s_W) = N\Theta_W + Nh^* \rho^* E_V$$

by (9.6) and (9.7). Thus s_W is a nonzero rational $(-N)$ -canonical form on W with $\text{Div}(s_W) \geq N\Theta_W$, hence $\text{Div}(s_W) \geq \lceil N\Theta_W \rceil$ as $\text{Div}(s_W)$ is a Weil divisor, which is (6.12) for $U' = V$. Let $\gamma \in \mathcal{A}$ and $\sigma \in G$ with $h \circ \gamma = \sigma \circ h$. Then $\gamma^*(h^* \rho^* u) = h^* \sigma^* \rho^* u = h^* \rho^* u$ as $\rho \circ \sigma = \rho$, and $\gamma^* \theta_0 = \theta_0$ as shown above. Thus s_W is $\mathcal{A}$ -invariant.

We show that $(W, \Theta_W + h^* \rho^* E_V)$ is lc. Let $P \xrightarrow{v_P} S_P \rightarrow Z$ be the Stein factorization of $P \rightarrow Z$. Then $W \xrightarrow{h} P \xrightarrow{v_P} S_P$ are contractions of normal quasi-projective varieties, $\dim P > 0$, and (W, Θ_W) is lc with $K_W + \Theta_W \sim_{\mathbb{Q}, P} 0$. The identity $\text{Div}(\theta_0) + b\Theta_W = h^*(bD_P + \text{div}(\chi))$ reads

$$K_W + \Theta_W + \frac{1}{b} \text{div}((\chi \circ h) \psi_W^{E_0}) = h^* D_\beta, \quad D_\beta := D_P + \frac{1}{b} \text{div}(\chi) = \rho^* D_V.$$

Its moduli b-divisor $\mathbf{M}_\beta$ satisfies $\mathbf{M}_{\beta, P''} = \mathbf{M}_{P''} + \frac{1}{b} \text{div}_{P''}(\chi)$ for every birational model P'' of P (Theorem 5.5(2)), so $\mathbf{M}_\beta$ descends to P' and $\mathbf{M}_{\beta, P'}$ is nef over Z , hence over S_P . By Proposition 8.1(1), applied to $W \xrightarrow{h} P \xrightarrow{v_P} S_P$ and the $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor $\rho^* E_V \geq 0$ on P , it suffices to show that $(P, B_P + \rho^* E_V + \mathbf{M}_\beta)/S_P$ is generalized lc. Let Q be a prime divisor over P . Since $\mathbb{C}(P)/\mathbb{C}(V)$ is finite, the restriction of ord_Q to $\mathbb{C}(V)$ is a discrete valuation whose residue field has transcendence degree $\dim V - 1$ over $\mathbb{C}$. By [KM98, Lemma 2.45], it is a positive multiple of ord_F for a prime divisor F on a birational model $\tau: U \rightarrow V$ of V ; possibly replacing U with a higher model, we may assume that $\mathbf{M}^V$ descends to U . Then Q is a prime divisor on P_U over F (standard); let e_Q be its ramification index. Computing on a birational model of P_U to which $\mathbf{M}_\beta$ descends, Definition 4.2 and (2.1) give

$$a(Q, P, B_P + \rho^* E_V + \mathbf{M}_\beta) = 1 - \text{mult}_Q B_{P_U} - \text{mult}_Q \rho_U^* \tau^* E_V = e_Q(1 - \text{mult}_F(B_U + \tau^* E_V)),$$

where the second equality follows from (5.11) and $\text{mult}_Q \rho_U^* F = e_Q$. Since $\mathbf{M}^V$ descends to U and $K_U + B_U + \mathbf{M}_U^V = \tau^* D_V$ by (5.10), $B_U + \tau^* E_V$ is the $\mathbb{Q}$ -divisor defined by (2.1) for $(V, B_V + E_V + \mathbf{M}^V)$, so $1 - \text{mult}_F(B_U + \tau^* E_V) = a(F, V, B_V + E_V + \mathbf{M}^V) \geq 0$ by (9.7). Thus $(P, B_P + \rho^* E_V + \mathbf{M}_\beta)/S_P$ is generalized lc, and $(W, \Theta_W + h^* \rho^* E_V)$ is lc.

Assume now that $\dim V = 0$. Then W is projective, (W, Θ_W) is klt with $\Theta_W \in \Gamma$, $\dim W \leq d_0 - 1$, and $K_W + \Theta_W \sim_{\mathbb{Q}} 0$. By Theorem 2.11 and (9.1), $N(K_W + \Theta_W) = \text{div}(\varphi_W)$ for some nonzero $\varphi_W \in \mathbb{C}(W)$. Let $E_V := 0$ and $s_W := \varphi_W \omega_W^{\otimes(-N)}$. Then $\text{Div}(s_W) = N\Theta_W = N\Theta_W + Nh^* \rho^* E_V$, so s_W satisfies the equality in (9.8) and (6.12), and $(W, \Theta_W + h^* \rho^* E_V) = (W, \Theta_W)$ is lc. Since G is trivial, $\mathcal{A}$ is the kernel of $\mathcal{A} \rightarrow G$, so s_W is $\mathcal{A}$ -invariant by Theorem 6.12(4).

Step 6. We construct a monotonic N -complement $(Y/Z \ni z, A^+)$ of $(Y/Z \ni z, A)$.

By Steps 4 and 5, s_W is a nonzero $\mathcal{A}$ -invariant rational $(-N)$ -canonical form on W satisfying (6.12) for $U' = V$, and N is a positive multiple of N_0 . By Theorem 6.12(5), there is a section $s \in H^0(\tilde{Z}_V, \mathcal{O}_{\tilde{Y}}(L_N))$ such that $s|_{\tilde{W}} = s_W$. By Theorem 6.12(6), $s = \tilde{s}|_{\tilde{Z}_V}$ for some $\tilde{s} \in$

$H^0(\tilde{Y}, \mathcal{O}_{\tilde{Y}}(L_N))$. By Set-up 6.6, $\tilde{s} = \varphi \omega^{\otimes(-N)}$ is a rational $(-N)$ -canonical form on $\tilde{Y}$, hence on Y , for some nonzero $\varphi \in \mathbb{C}(Y)$, and $\text{Div}_{\tilde{Y}}(\tilde{s}) \geq NP_1 + \lfloor (N+1)E \rfloor$. Let

$$A^+ := \frac{1}{N} \text{Div}_Y(\tilde{s}) = \frac{1}{N} \text{div}_Y(\varphi) - K_Y \quad \text{and} \quad R := A^+ - A.$$

Then $N(K_Y + A^+) = \text{div}_Y(\varphi)$, as in (9.3), and $K_Y + A^+$ is $\mathbb{Q}$ -Cartier. Since N is even and $N\mathfrak{R} \subset \mathbb{Z}$, Lemma 6.7(2) for $C = Y$ gives $\text{Div}_Y(\tilde{s}) \geq \lceil NA \rceil$, that is, $NA^+ \geq \lceil NA \rceil$. Since $A \in \Phi_1 \subset \Phi(\mathfrak{R})$, (4.2) gives

$$NA^+ \geq \lceil NA \rceil = N\lfloor A \rfloor + \lfloor (N+1)\{A\} \rfloor.$$

In particular, $A^+ \geq A \geq 0$, and $R \geq 0$ is $\mathbb{Q}$ -Cartier. It remains to show that (Y, A^+) is lc over a neighborhood of z .

Let C be an lc center of (Y, A) with $g(C) = V$, $\tilde{C} \subset \tilde{Z}_V$ its stratum, $\mu_C := \mu|_{\tilde{C}}$, $D_1, \dots, D_k$ the components of P_1 containing $\tilde{C}$, and S_C and E_C as in Set-up 6.6. Let θ_C , $E'_C \geq 0$, and $\Theta_C^+ = \Theta_C + f_C^* E'_C$ be given by Theorem 6.12(7) for $U' = V$, $\beta = s_W$, and s . Then $\text{Div}(\theta_C) = -N\Theta_C^+$, and $s|_{\tilde{C}}$ corresponds to $\theta_C^{-1} \neq 0$ under (6.5). By the definition of the restriction map after (6.5), no component of $\text{Div}_{\tilde{Y}}(\tilde{s}) - N(D_1 + \dots + D_k)$ contains $\tilde{C}$, and the residue of $\tilde{s}$ on $\tilde{C}$ is θ_C^{-1} . By (6.7) for Y ,

$$\text{Div}_{\tilde{Y}}(\tilde{s}) - N(D_1 + \dots + D_k) = N\mu^* R + N(P_1 - D_1 - \dots - D_k) + NE.$$

Since $\tilde{C}$ meets $\mu^{-1}(Y^{\text{snc}})$ (Set-up 6.6), no μ -exceptional divisor contains $\tilde{C}$. Hence every prime divisor containing $\tilde{C}$ has nonnegative coefficient in each of $\mu^* R$, $P_1 - D_1 - \dots - D_k$, and E , the non-exceptional part of E being $\mu_*^{-1}\{A\}$ by (6.2). Thus $\tilde{C} \not\subset \text{Supp } \mu^* R$, so $C \not\subset \text{Supp } R$ as μ is an isomorphism over the generic point of C . By (6.4) and (6.7) for C ,

$$N\mu_C^*(R|_C) + N(S_C + E_C) = \text{Div}_{\tilde{C}}(\theta_C^{-1}) = N\mu_C^*(\Theta_C^+ - \Theta_C) + N(S_C + E_C).$$

Pushing forward by μ_C , we get $R|_C = \Theta_C^+ - \Theta_C = f_C^* E'_C$.

We show that (C, Θ_C^+) is lc. If $\dim V = 0$, then P_C is a point, $E'_C = 0$, and $(C, \Theta_C^+) = (C, \Theta_C)$ is dlt. Assume that $\dim V > 0$. Let $W_C \subset C$ be a source of V , and let $\phi : (W, \Theta_W) \dashrightarrow (W_C, \Theta_{W_C})$ be a crepant birational map over V , which exists by Definition 6.1. Let $p : W' \rightarrow W$ and $r : W' \rightarrow W_C$ be a common resolution of ϕ , and $K_{W'} + \Gamma' := p^*(K_W + \Theta_W) = r^*(K_{W_C} + \Theta_{W_C})$. By Theorem 6.12(7), $r^*(s|_{\tilde{W}_C}) = p^* s_W$. The divisors $\text{Div}(s|_{\tilde{W}_C}) - N\Theta_{W_C}$ and $\text{Div}(s_W) - N\Theta_W$ are $\mathbb{Q}$ -Cartier by (6.16) and (9.8), and the computation of (5.12) gives

$$r^*(\text{Div}(s|_{\tilde{W}_C}) - N\Theta_{W_C}) = \text{Div}_{W'}(p^* s_W) - N\Gamma' = p^*(\text{Div}(s_W) - N\Theta_W).$$

By (6.16) and (9.8), $r^*(f_C|_{W_C})^* E'_C = p^* h^* \rho^* E_V$. Hence

$$r^*(K_{W_C} + \Theta_{W_C} + (f_C|_{W_C})^* E'_C) = p^*(K_W + \Theta_W + h^* \rho^* E_V),$$

and $(W_C, \Theta_{W_C} + (f_C|_{W_C})^* E'_C)$ is lc by Step 5. If $W_C = C$, this is the assertion. Otherwise, $f_C : (C, \Theta_C) \rightarrow P_C$ is a dlt crepant log structure by Theorem 8.2(1), $\dim P_C = \dim V > 0$, and W_C is an lc center of (C, Θ_C) dominating P_C whose adjunction pair is (W_C, Θ_{W_C}) [Kol16, Definitions 4 and 18]. By Proposition 8.5 applied to f_C , W_C , and E'_C , $(C, \Theta_C + f_C^* E'_C)$ is lc.

We show that (Y, A^+) is lc near C . By Definition 2.3 and [Kol16, Definition 4], there exist lc centers $Y = W_0 \supset W_1 \supset \dots \supset W_k = C$ of (Y, A) such that W_i is a prime divisor on W_{i-1} and a component of $\lfloor \Theta_{W_{i-1}} \rfloor$ for each i . Here (W_i, Θ_{W_i}) is the adjunction pair of W_i , $\Theta_{W_0} := A$, and Θ_{W_i} is the different of $\Theta_{W_{i-1}} - W_i$ on W_i . Each W_i is normal and contains C , so $W_i \not\subset \text{Supp } R$, and $R|_{W_i} \geq 0$ is $\mathbb{Q}$ -Cartier, with $R|_{W_k} = f_C^* E'_C$. We show by descending induction on i that $(W_i, \Theta_{W_i} + R|_{W_i})$ is lc near C . For $i = k$ this was shown above. Let $i \geq 1$, and assume that $(W_i, \Theta_{W_i} + R|_{W_i})$ is lc near C . Let $Y^\circ := Y \setminus \text{Nlc}(W_i, \Theta_{W_i} + R|_{W_i})$, an open neighborhood of C in Y . On $W_{i-1} \cap Y^\circ$, the reduced divisor W_i and $\text{Supp}(\Theta_{W_{i-1}} - W_i + R|_{W_{i-1}})$ have no common component, as W_i has coefficient 1 in $\Theta_{W_{i-1}}$ and $W_i \not\subset \text{Supp } R \supset \text{Supp}(R|_{W_{i-1}})$. Moreover, $K_{W_{i-1}} + \Theta_{W_{i-1}} + R|_{W_{i-1}}$ is $\mathbb{Q}$ -Cartier, and the different of $\Theta_{W_{i-1}} - W_i + R|_{W_{i-1}}$ on W_i is

$\Theta_{W_i} + R|_{W_i}$ [Kol92, 16.5–16.6], which is lc on $W_i \cap Y^\circ$. By Theorem 8.3, $(W_{i-1}, \Theta_{W_{i-1}} + R|_{W_{i-1}})$ is lc near $W_i \cap Y^\circ$, hence near C . For $i = 0$ we get that (Y, A^+) is lc near C . Since C was arbitrary, (Y, A^+) is lc near Z_V .

Finally, $\pi \circ g : Y \rightarrow Z$ is a contraction, $-(K_Y + A^+) \sim_{\mathbb{Q}} 0$ is semi-ample over Z , and z is a closed point. Let F_W be an lc place of (Y, A) with center $F_W = W$. Since $W \not\subset \text{Supp } R$, $a(F_W, Y, A^+) = a(F_W, Y, A) = 0$, and $W \cap \text{Nlc}(Y, A^+) = \emptyset$ as (Y, A^+) is lc near $Z_V \supset W$. Thus W is an lc center of (Y, A^+) , and $W \cap Y_z \neq \emptyset$ as $z \in \pi(V)$. If $\text{Nlc}(Y, A^+) \cap Y_z \neq \emptyset$, then $\text{Nlc}(Y, A^+) \cap Y_z \cap W \neq \emptyset$ by Theorem 8.4, a contradiction. Thus $\text{Nlc}(Y, A^+) \cap Y_z = \emptyset$. Since $\pi \circ g$ is proper, $(\pi \circ g)(\text{Nlc}(Y, A^+))$ is a closed subset of Z not containing z . Replacing Z with its complement, (Y, A^+) is lc. Therefore $(Y/Z \ni z, A^+)$ is a monotonic N -complement of $(Y/Z \ni z, A)$.

Step 7. We finish the proof.

By Steps 3 and 6, in both cases of Step 2, $(Y/Z \ni z, A)$ has a monotonic N -complement. By Proposition 3.5 and Lemma 3.2, $(X/Z \ni z, B)$ has a monotonic N -complement. Since $p \mid N$ and N depends only on d, p , and Φ by Step 1, this finishes the proof. $\square$

Proof of Corollary 1.2. It follows from Theorem 1.1, Theorem 2.9, and [HLS26, arXiv version 2, Theorem 8.25]. $\square$

REFERENCES

- [Bir11] C. Birkar, *On existence of log minimal models II*, J. Reine Angew. Math. **658** (2011), 99–113.
- [Bir19] C. Birkar, *Anti-pluricanonical systems on Fano varieties*, Ann. of Math. (2), **190** (2019), 345–463.
- [BCHM10] C. Birkar, P. Cascini, C. D. Hacon, and J. McKernan, *Existence of minimal models for varieties of log general type*, J. Amer. Math. Soc. **23** (2010), no. 2, 405–468.
- [CHL24] G. Chen, J. Han, and J. Liu, *On effective log Iitaka fibrations and existence of complements*, Int. Math. Res. Not. IMRN **2024** (2024), no. 10, 8329–8349.
- [FM20] S. Filipazzi and J. Moraga, *Strong (δ, n) -complements for semi-stable morphisms*, Doc. Math. **25** (2020), 1953–1996.
- [Fuj11] O. Fujino, *Fundamental theorems for the log minimal model program*, Publ. Res. Inst. Math. Sci. **47** (2011), no. 3, 727–789.
- [Fuj14] O. Fujino, *Fundamental theorems for semi log canonical pairs*, Algebr. Geom. **1** (2014), no. 2, 194–228.
- [Fuj17] O. Fujino, *Injectivity theorems*, in *Higher dimensional algebraic geometry: In honour of Professor Yujiro Kawamata’s sixtieth birthday*, Adv. Stud. Pure Math. **74**, Mathematical Society of Japan, Tokyo, 2017, 131–157.
- [FG14a] O. Fujino and Y. Gongyo, *Log pluricanonical representations and the abundance conjecture*, Compos. Math. **150** (2014), no. 4, 593–620.
- [FG14b] O. Fujino and Y. Gongyo, *On the moduli b-divisors of lc-trivial fibrations*, Ann. Inst. Fourier (Grenoble) **64** (2014), no. 4, 1721–1735.
- [SGA1] A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental (SGA 1)*, Documents Mathématiques (Paris) **3**, Société Mathématique de France, Paris, 2003.
- [HMX14] C. D. Hacon, J. McKernan, and C. Xu, *ACC for log canonical thresholds*, Ann. of Math. (2) **180** (2014), no. 2, 523–571.
- [HLS26] J. Han, J. Liu, and V. V. Shokurov, *ACC for minimal log discrepancies of exceptional singularities*, Peking Math. J. **9** (2026), no. 2, 225–257.
- [Har77] R. Hartshorne, *Algebraic geometry*, Graduate Texts in Math. **52**, Springer-Verlag, New York–Heidelberg, 1977.
- [Kaw07] M. Kawakita, *Inversion of adjunction on log canonicity*, Invent. Math. **167** (2007), no. 1, 129–133.
- [Kle98] S. L. Kleiman, *Bertini and his two fundamental theorems*, Rend. Circ. Mat. Palermo (2) Suppl. (1998), no. 55, 9–37.
- [Kol16] J. Kollár, *Sources of log canonical centers*, in *Minimal models and extremal rays*, Adv. Stud. Pure Math. **70**, Mathematical Society of Japan, Tokyo, 2016, 29–48.
- [KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Math. **134**, Cambridge University Press, Cambridge, 1998.
- [Kol92] J. Kollár et al., *Flips and abundance for algebraic threefolds*, Astérisque **211**, Soc. Math. France, Paris, 1992.
- [OAI26a] OpenAI, *Log abundance in characteristic zero*, [dated source](#).
- [OAI26b] OpenAI, *Uniform Pluricanonical Iitaka Fibrations*, [dated source](#).

- [OAI26c] OpenAI, *Uniform log Iitaka fibrations and bounded moduli denominators*, [dated source](#).
- [OAI26d] OpenAI, *Uniform indices for semi-log-canonical log Calabi–Yau pairs*, [dated source](#).
- [Sho92] V. V. Shokurov, *Three-fold log flips*, with an appendix in English by Y. Kawamata, *Izv. Ross. Akad. Nauk Ser. Mat.* **56** (1992), no. 1, 105–203.
- [Sho20] V. V. Shokurov, *Existence and boundedness of n -complements*, arXiv:2012.06495.